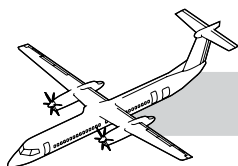


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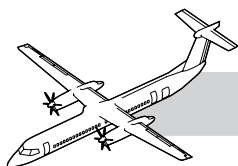
EQUIPMENT AND FURNISHING

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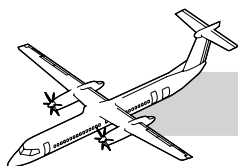
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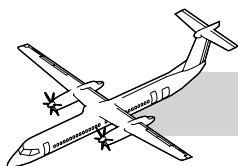
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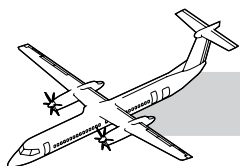
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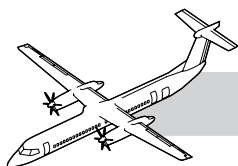
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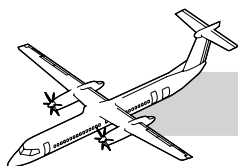


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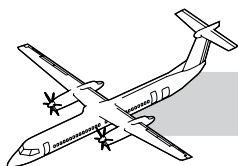


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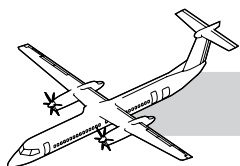
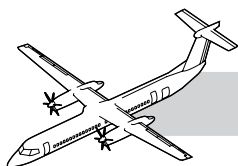


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CHAPTER 25

EQUIPMENT AND FURNISHING



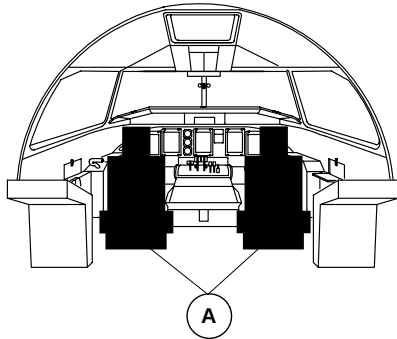
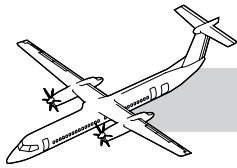
25-00-00 INTRODUCTION

The flight compartment has seating for the flight crew and a flight observer. It also has readily accessible normal and emergency equipment. The passenger compartment furnishings are provided to ensure the comfort and safety of the passengers and cabin crew. The emergency equipment located in the flight compartment and the passenger compartment is used by the flight crew during emergency situations.

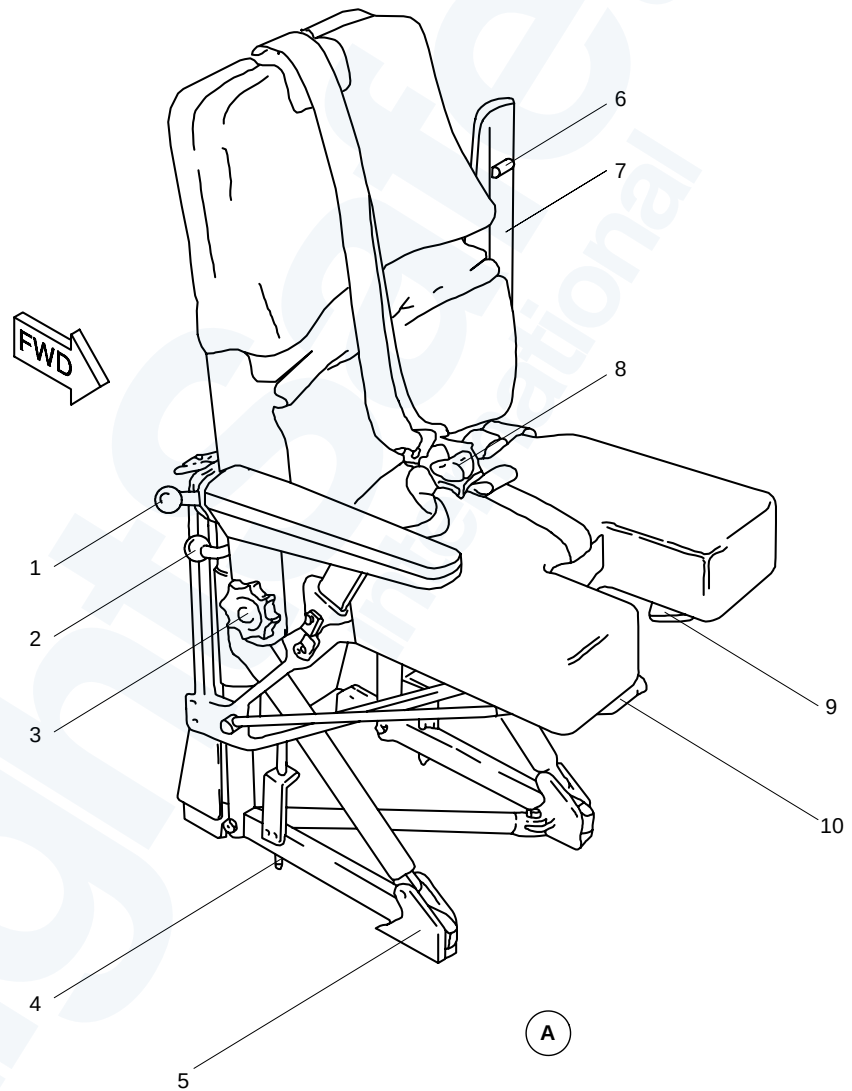
The galleys on board the aircraft supply a place to store and prepare food and beverages for the passengers and flight crew.

The lavatory compartment supplies the crew and passengers with washroom facilities.

The forward and aft baggage compartments have sufficient space for the safe storage of the passengers' baggage and cargo.



NOTE
Pilot's seat shown.
Co-pilot's seat similar.

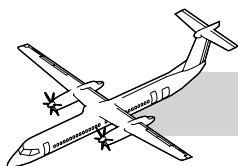


LEGEND

1. Inertia Reel Lock
2. Recline Adjustment Lever
3. Lumbar Support Adjustment
4. Track Locking Pin
5. Plate Assembly
6. Arm Position Adjustment
7. Armrest
8. Five Point Safety Harness
9. Height Control Lock Lever
10. Lock Release Lever (Fore and Aft).

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Figure 25-1. Pilot and Co-pilot Seats



GENERAL

The equipment and furnishings chapter is divided into the following sub-chapters:

- Flight compartment furnishings
- Passenger compartment furnishings
- Galleys
- Lavatories
- Baggage compartments
- Emergency equipment
- Insulation.

25-10-00 FLIGHT COMPARTMENT FURNISHINGS

GENERAL

The flight compartment has seats for the pilot, the co-pilot and an observer. Insulation blankets are installed where necessary in the flight compartment and can be removed for structural inspection. Access panels in the compartment give access to the aircraft flight equipment. The floor area is covered with a foam backed vinyl-coated covering.

The flight compartment furnishings include these components:

- Pilot and co-pilot seats
- Observer seat and
- Crew storage.

COMPONENT DESCRIPTION

Pilot and Co-pilot Seats

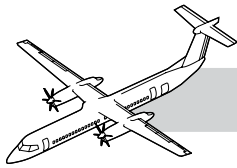
Refer to Figure 25-1. Pilot and Co-pilot Seats.

The pilot's and co-pilot's seats are mechanically adjustable (fore/aft and vertically) and are installed on track rails in the flight compartment. The seats have an aluminum alloy frame structure equipped with reclining backs and fold up padded armrests. Each seat has a removable seat cushion, contoured back cushion with lumbar support, coat hanger, head rest and a document holder.

The base frame of each seat is attached to four plate assemblies which roll on two machined tracks attached to the flight compartment floor. A roller is installed in each plate assembly to allow the seat to move freely along the tracks. Each seat is adjustable fore and aft along the floor tracks by means of a lock release lever located on the inboard side of each seat. The seat is held in place by four locking pins which engage in holes located along the length of the floor tracks. Pulling the lock release lever up unlocks the seat by disengaging the locking pins allowing the seat to move freely along the tracks.

The height control lock lever, located on the outboard side of each seat, raises or lowers the seat with the help of a gas spring. The recline adjustment lever, located on the right side of each seat, reclines the back rest from a range of 7 to 34 degrees.

The seat also has controls for the adjustment of the lumbar support. Turning the control knob on the right side of the seat adjusts the support IN and OUT. Turning the knob on the left side adjusts the support UP and DOWN. The padded armrests fold up to make it easier to enter and exit the seat. The armrests can be individually adjusted UP or DOWN by a knob located at the forward end of each armrest.



NOTES

The seats have a safety harness with a crotch strap, lap belts and a shoulder harness. An inertia reel controls the movement of the shoulder harness which is spring loaded to the retracted position. Selection of the inertia reel to lock secures the harness in position. Selection of the inertia reel to unlock lets the harness move freely until a force of 2 Gs or greater is applied, automatically locking the inertia reel. The inertia reel lock levers are located on the inboard side of the seats. This allows each flight crew member to access the other crew member's lock lever in case of incapacitation.

MAINTENANCE PRACTICES

The following are abbreviated descriptions of the maintenance practices and are intended for training purposes only. For a more detailed description of the practices, refer to the tasks in the Bombardier AMM PSM 1-84-2.

Pilot and Co-Pilot Seat Restraint System Detailed Inspection

- Inspect the pilot and co-pilot seat restraint system for cuts, tears, frayed webbing and/or stitching.
- Also that the inertia reel attached to the safety harness operates correctly.

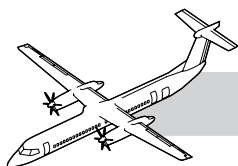
General Visual Inspection of the Pilot and Co-Pilot Seats

- Inspect the pilot and co-pilot seats for general condition.

TROUBLESHOOTING

Pilot and Co-Pilot Seats Operational Test

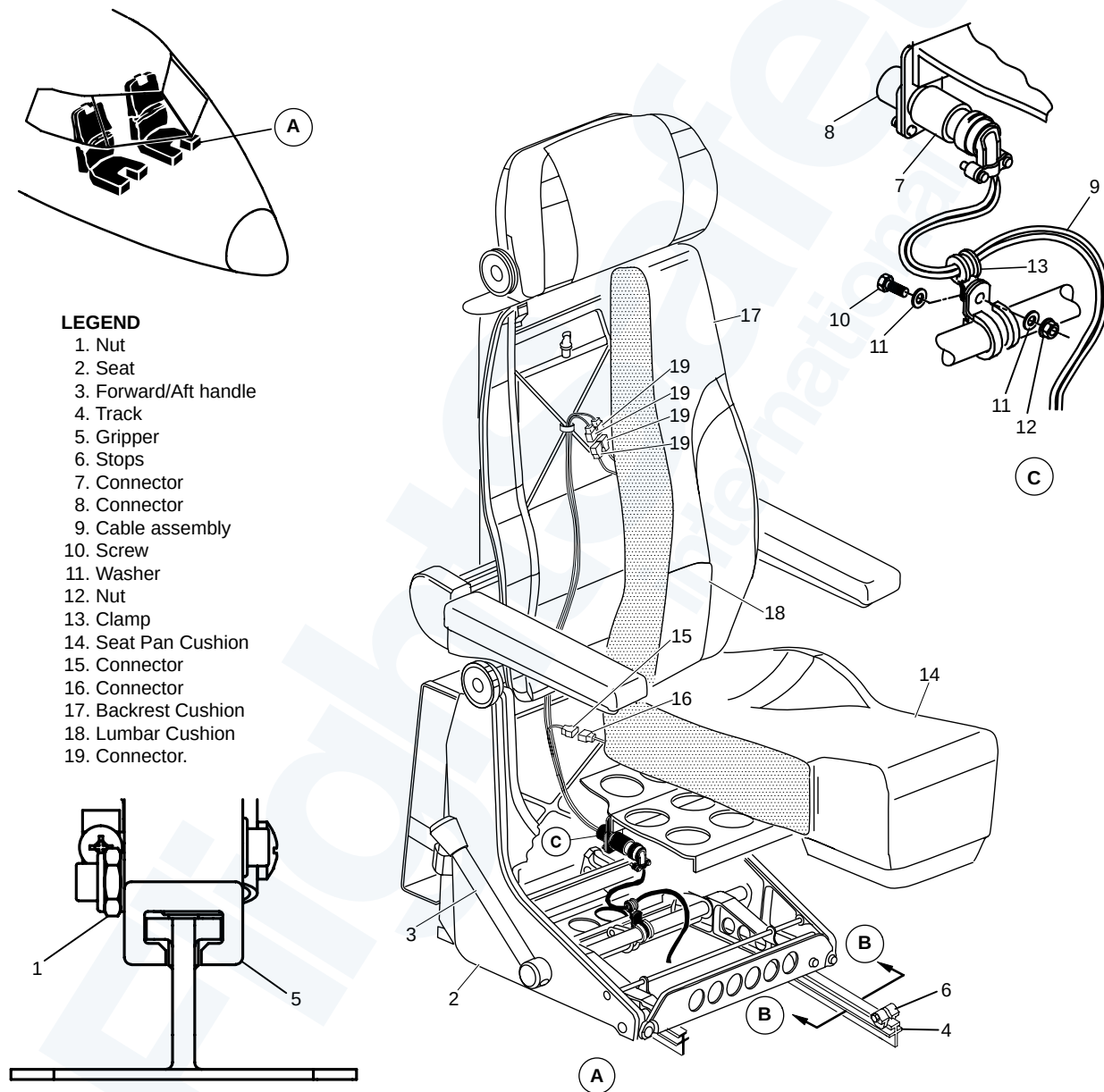
- This operational task is to confirm the correct operation of all the adjustments on the pilots and co-pilots seat.
- Each adjustment should allow free movement of the seat. When released the seat should remain locked in that position.



Heated Seat Pan

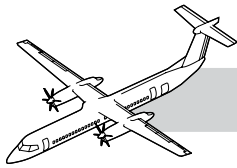
Refer to Figure 25-2. Heated Seat Pan.

On aircraft with the heated pilot seats (825CH01541): The seat cushions have heater pads to warm the seat pan. Each seat is heated by a 32 W heater pad.



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Figure 25-2. Heated Seat Pan



CONTROLS AND INDICATIONS

Pilot and Co-Pilot Seat Heating Switches

Refer to Figure 25-3. Heated Seat Pan Switch.

Annunciator switches located near the pilot and co-pilot seats in the flight compartment give manual control of the heating.

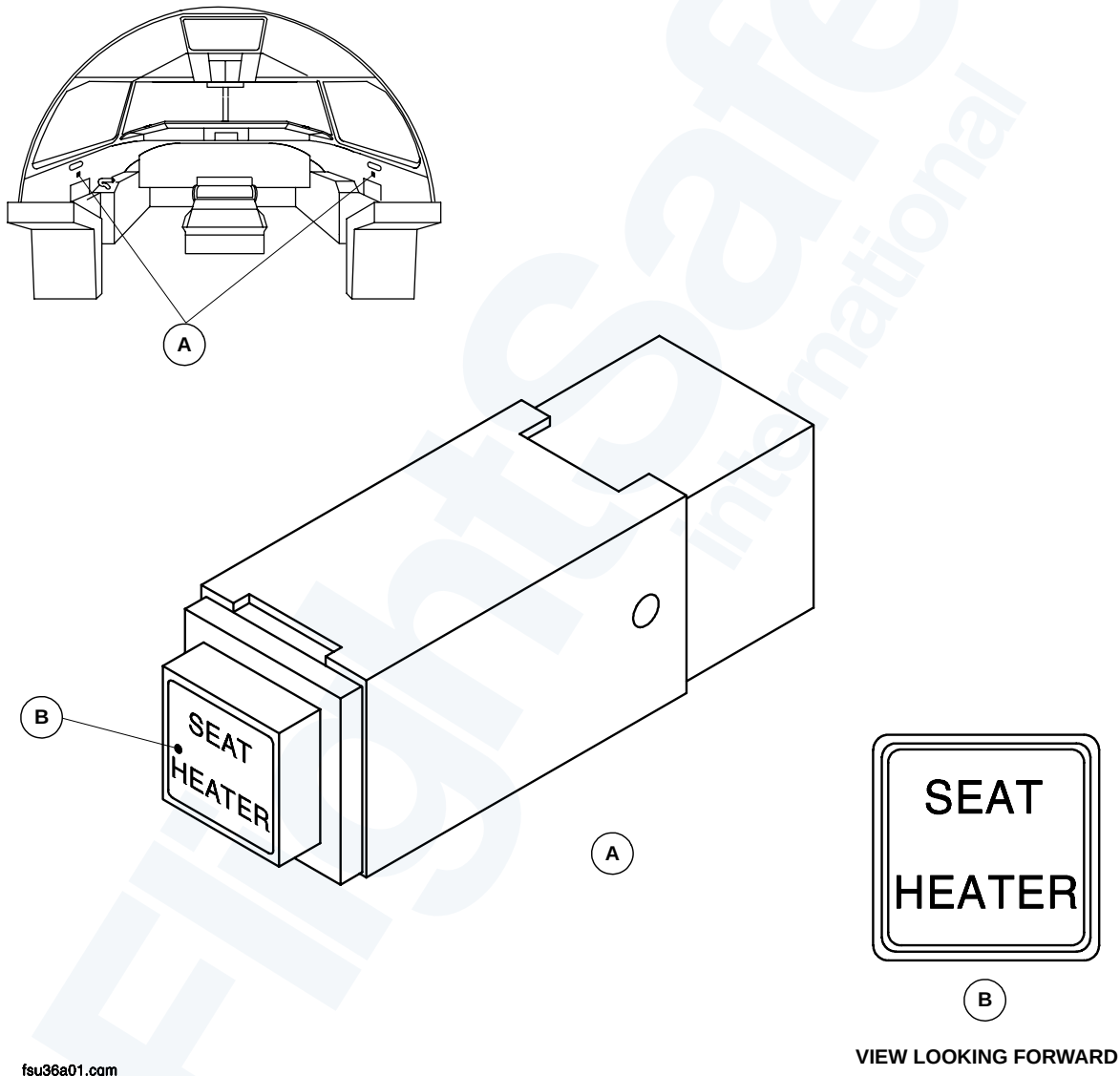
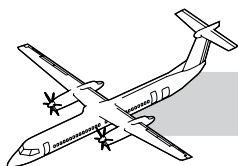


Figure 25-3. Heated Seat Pan Switch



OPERATION

Refer to Figure 25-4. Pilot and Co-pilot Seat Heater Schematic.

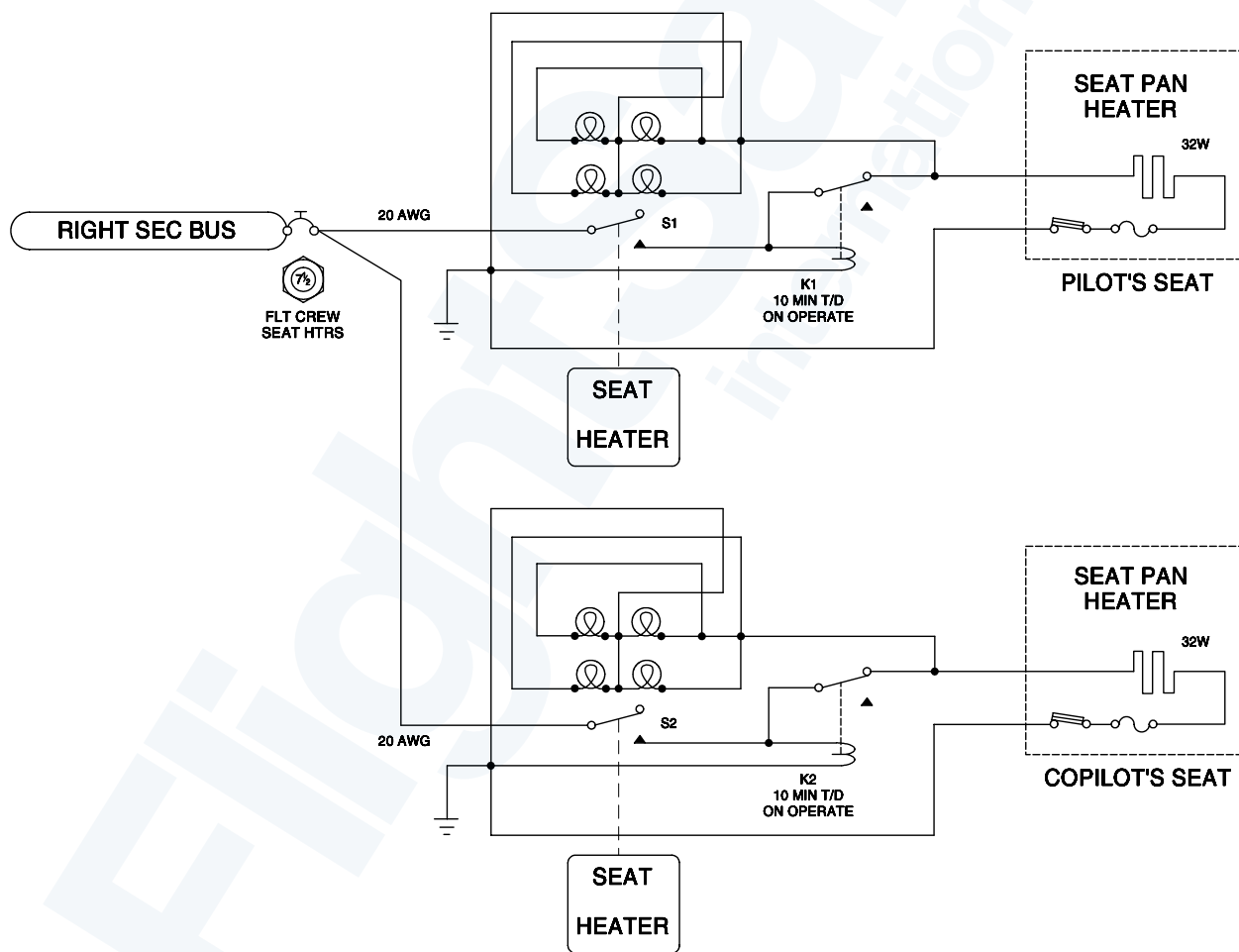
Pilot and Co-Pilot Seat Heater

Electrical power is supplied from the 28 VDC right secondary bus through related flight compartment annunciator switches and relays to energize the seat heater pads.

When a SEAT HEATER annunciator switch is pushed, the 28 VDC right secondary bus is used to energize the related relay and heater pad.

The SEAT HEATER lamps in the annunciator switch come on to show that electrical power is supplied to the heater. The relay has time delay circuit to de-energize the heater pads after 10 minutes.

The heater pad has a thermal switch for overheat protection. A fuse protects for over current.



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Figure 25-4. Pilot and Co-pilot Seat Heater Schematic

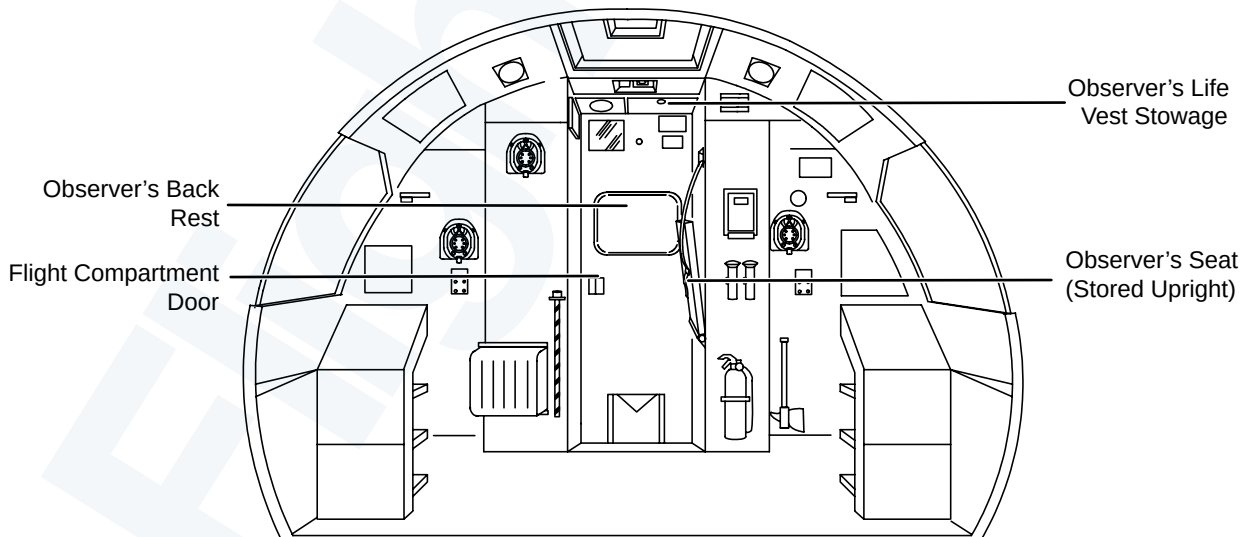
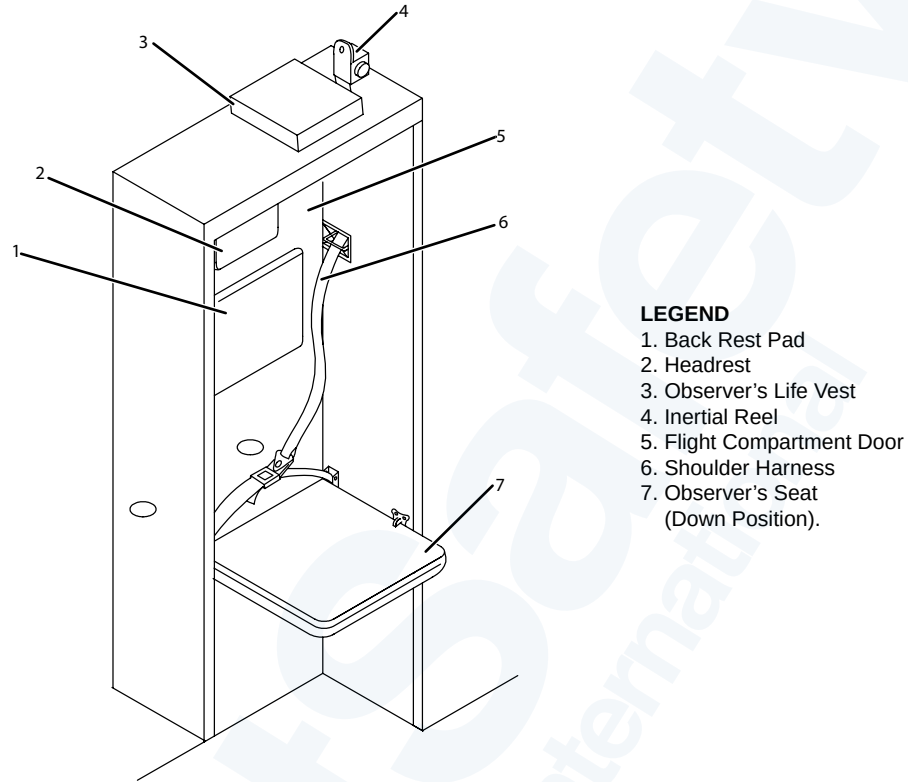
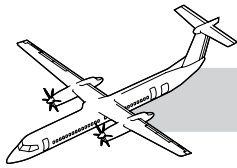
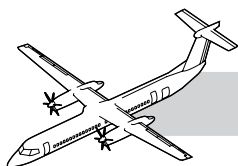


Figure 25-5. Observer Seat Location



Observer Seat

NOTES

Refer to:

- Figure 25-5. Observer Seat Location.
- Figure 25-6. Observer Seat - Detail.

The observer's seat has a machined steel frame, with a nylon covered padded seat cushion and a restraint system. The seat is located forward of the flight compartment door and is attached to the avionics rack by two hinge assemblies. The seat frame (with the bottom cushion) can be folded up and attached to the left side wall by a strap when the seat is not in use. When folded down, the seat is supported and held in place by latches on the inboard side of the lavatory wall. A padded back and head rest cushion is installed on the flight compartment door.

The seat has a single shoulder harness installed on an inertia reel and a lap belt with a metal to metal latching device. The flight compartment door can open when the observer's seat is in use.

Observer Seat Operational Test

- This operational task is to confirm the correct operation of lowering the seat and that the seat will latch in the level position.
- When the LIFT handle is pulled the seat goes back to the vertical position.

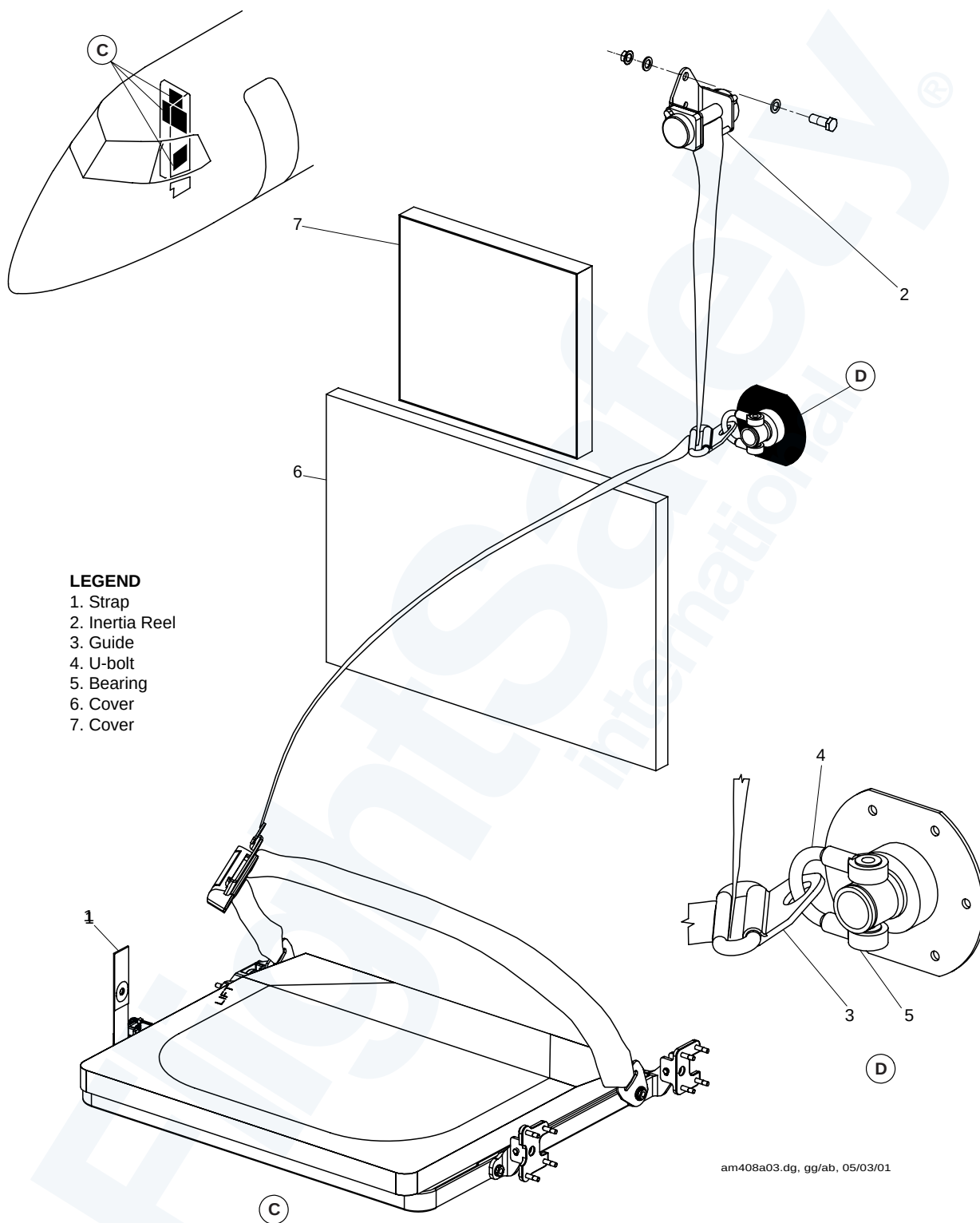
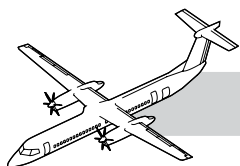
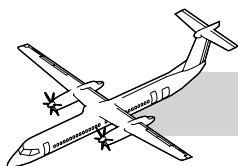
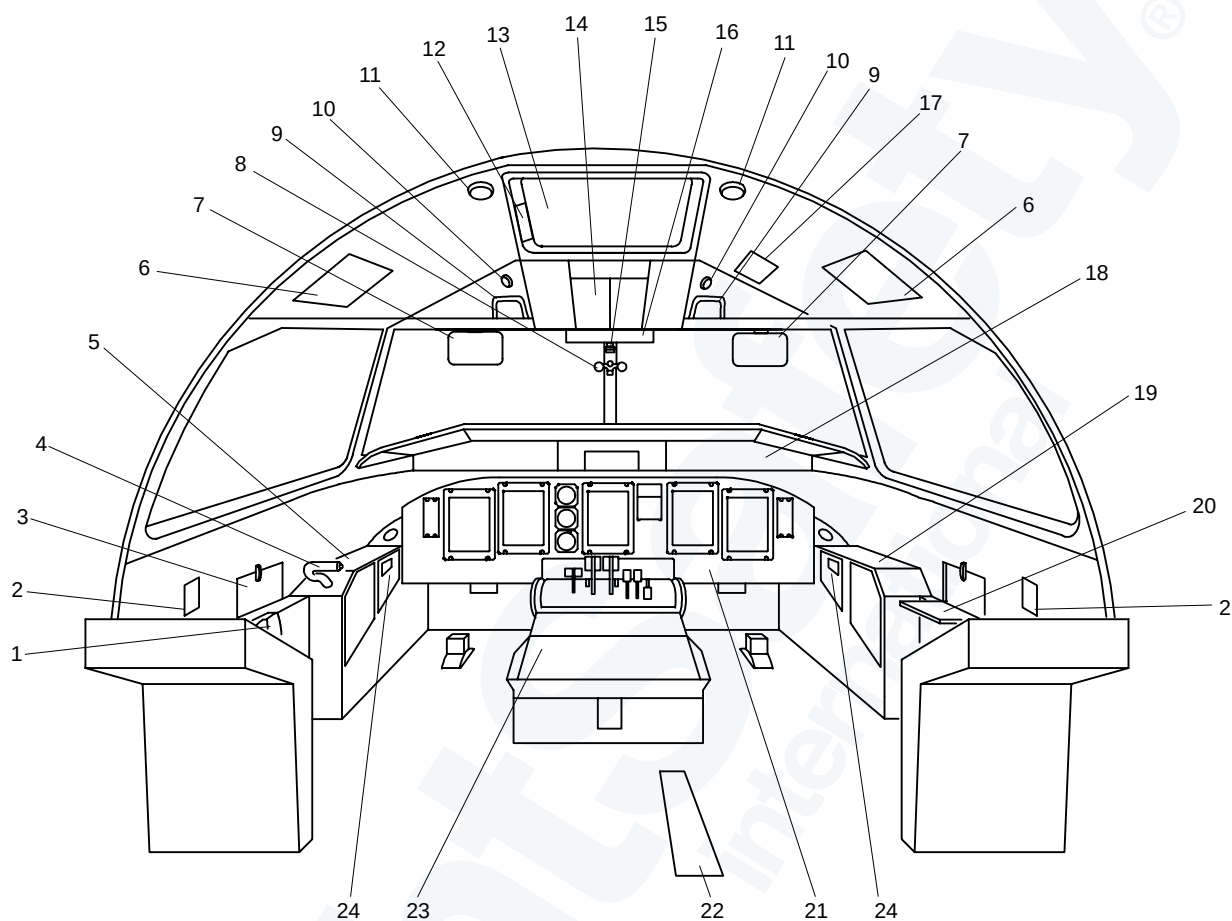
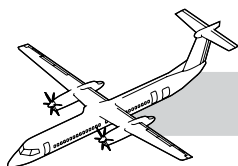


Figure 25-6. Observer Seat - Detail



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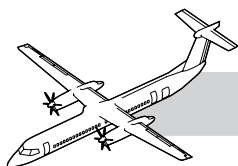


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LEGEND

- | | |
|-----------------------------------|---|
| 1. Chart Holder | 13. Emergency Exit |
| 2. Flow Control Levers | 14. Overhead Console |
| 3. Pilot's Map Table (Closed) | 15. Standby Compass |
| 4. Steering Hand Control | 16. Caution & Warning Panel |
| 5. Pilot's Side Panel | 17. Landing Gear Alternate Release Door |
| 6. Life Vest Stowage | 18. Glareshield |
| 7. Sun Visor | 19. Co-pilot's Side Panel |
| 8. Eye Lever Indicator | 20. Co-pilot's Map Table (Open) |
| 9. Assist Handle | 21. Instrument Panel |
| 10. Utility Light | 22. Landing Gear Alternate Extend Door |
| 11. Dome Light | 23. Centre Console |
| 12. Emergency Escape Rope Storage | 24. Smoke Goggles |

Figure 25-7. Flight Compartment Equipment - Detail (1 of 2)



Crew Storage

NOTES

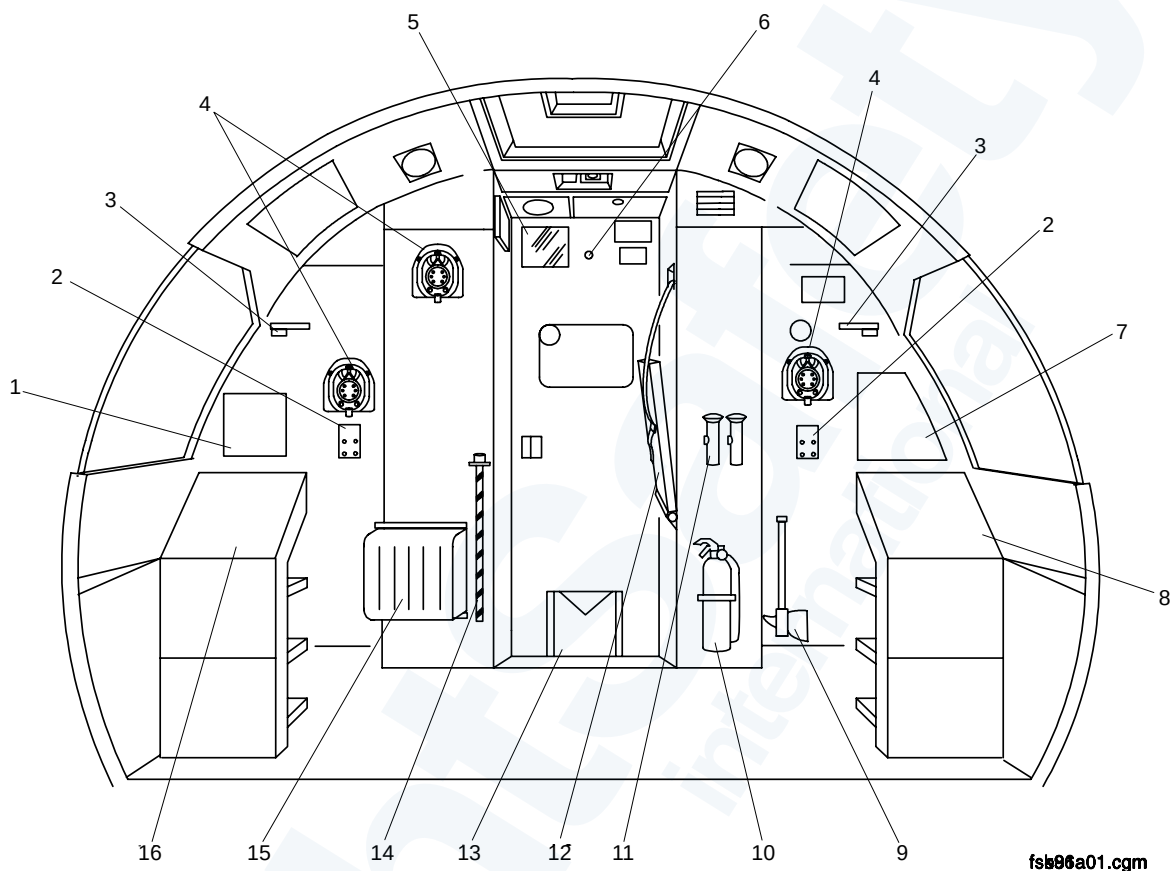
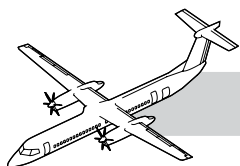
Refer to:

- Figure 25-7. Flight Compartment Equipment - Detail (1 of 2).
- Figure 25-8. Flight Compartment Equipment - Detail (2 of 2).

The flight compartment has these crew aids:

- Cup holders
- Map pouch
- Foot rests
- Assist handles
- Microphone
- Chart holder
- Individual air gasper
- Air conditioning grilles
- Console map light
- Dome light
- Reading light
- Flashlight
- Life vest
- Smoke goggles
- Oxygen mask.

The overhead escape hatch serves as an emergency exit.

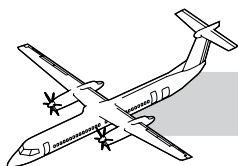


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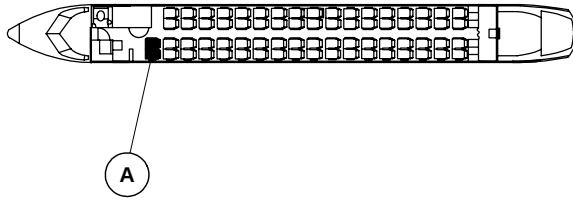
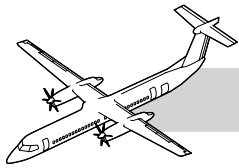
LEGEND

- | | |
|--|--|
| 1. Variable Frequency AC Circuit Breaker Panel | 10. Fire Extinguisher |
| 2. Headset Jacks | 11. Flashlights |
| 3. Circuit Panel Light | 12. Observer's Seat |
| 4. Oxygen Mask | 13. Weight and Balance Manual |
| 5. Mirror | 14. Landing Gear Emergency Extension Handpump Handle |
| 6. Viewer | 15. Protective Breathing Equipment (PBE) |
| 7. Avionics Circuit Breaker Panel | 16. Right DC Circuit Breaker Panel |
| 8. Left DC Circuit Breaker Panel | |
| 9. Fire Axe | |

Figure 25-8. Flight Compartment Equipment - Detail (2 of 2)

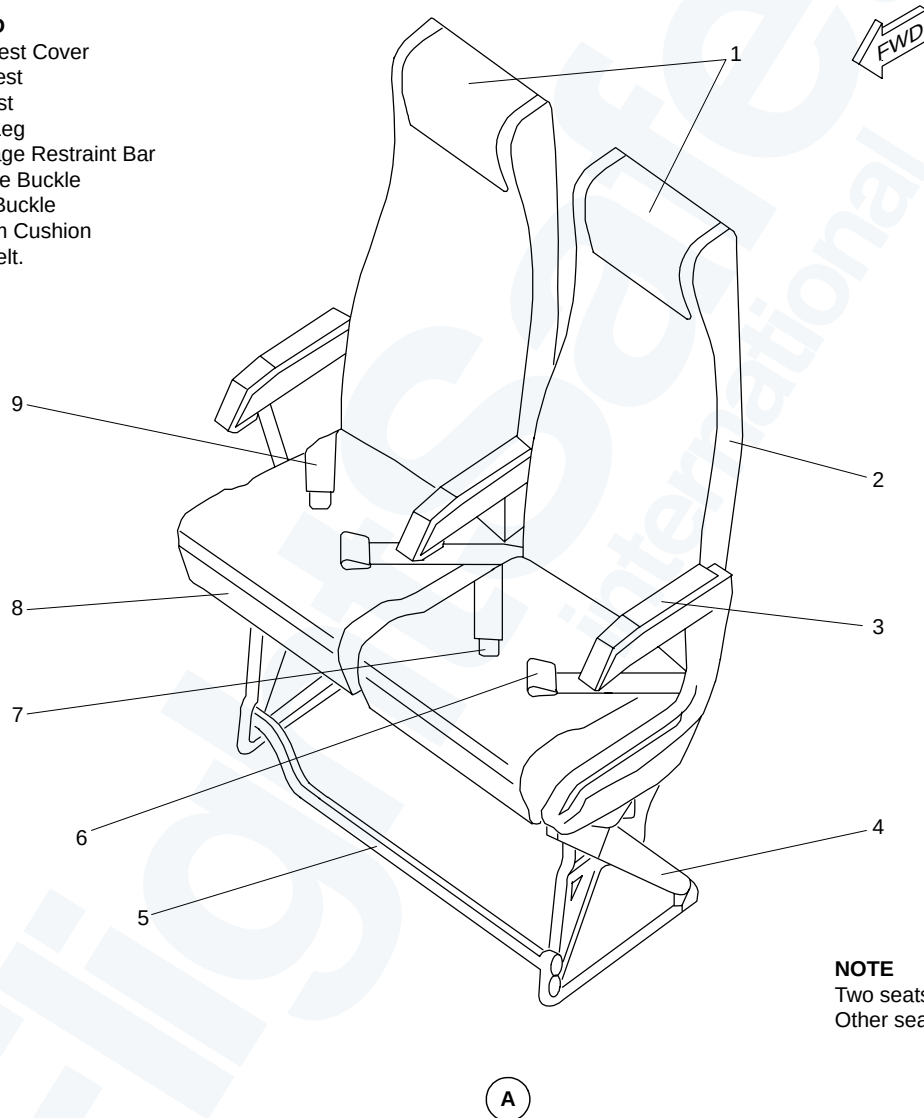


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LEGEND

1. Headrest Cover
2. Backrest
3. Armrest
4. Seat Leg
5. Baggage Restraint Bar
6. Female Buckle
7. Male Buckle
8. Bottom Cushion
9. Lap Belt.

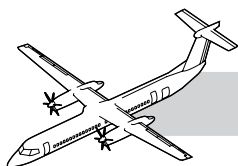


NOTE

Two seats shown.
Other seats similar.

Figure 25-9. Passenger Seat

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25-20-00 PASSENGER COMPARTMENT FURNISHINGS

GENERAL

The passenger compartment furnishings are provided to ensure the comfort and safety of the passengers and cabin crew.

The passenger compartment furnishings include these components:

- Passenger seats
- Flight attendant seats
- Overhead storage
- Wardrobes and partitions
- Passenger service units.

COMPONENT DESCRIPTION

Passenger Seats

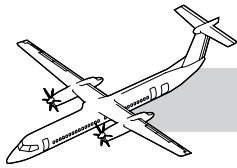
Refer to Figure 25-9. Passenger Seat.

The passenger cabin is designed to accommodate different seating configurations. The baseline configuration has up to 68 passengers, seated four abreast, at a mixed pitch of 33, 34, 35 and 36 in. (838, 864, 889 and 914 mm). The other configurations have 70 seats at 33 in. (838 mm) pitch and 78 seats at 30 in. (762 mm) pitch.

The standard passenger seats for the aircraft are non-reclining, and have fold-down meal trays, armrests, under-seat baggage restraints and removable ashtrays. Plug-in meal trays are supplied for seats without a fold-down tray in front of them. The surfaces of the seat backs are padded and smooth so as not to create an injury hazard to the passengers.

The passenger seats are made of an anodized aluminum frame, with plastic foam flotation cushions with dress covers. The cushions of each seat are removable and are certified for use as a flotation device in the event of the aircraft ditching in water. The cushions are attached to the seats by Velcro tape. The foam cushions, liners and seat dress covers satisfy the FAA and CAA flammability requirements.

A two-section lap belt is installed on each passenger seat and is connected to the seat frame by bolts. Each belt has a male and female buckle and a locking device which allows adjustment of the seat belt length.



Portable Meal Tray

Refer to Figure 25-10. Portable Meal Tray Location.

Fold-away meal trays are installed in the seat backrest, except for the front row of seats, where provision is made for a portable tray to be plugged into the front section of the arm rest.

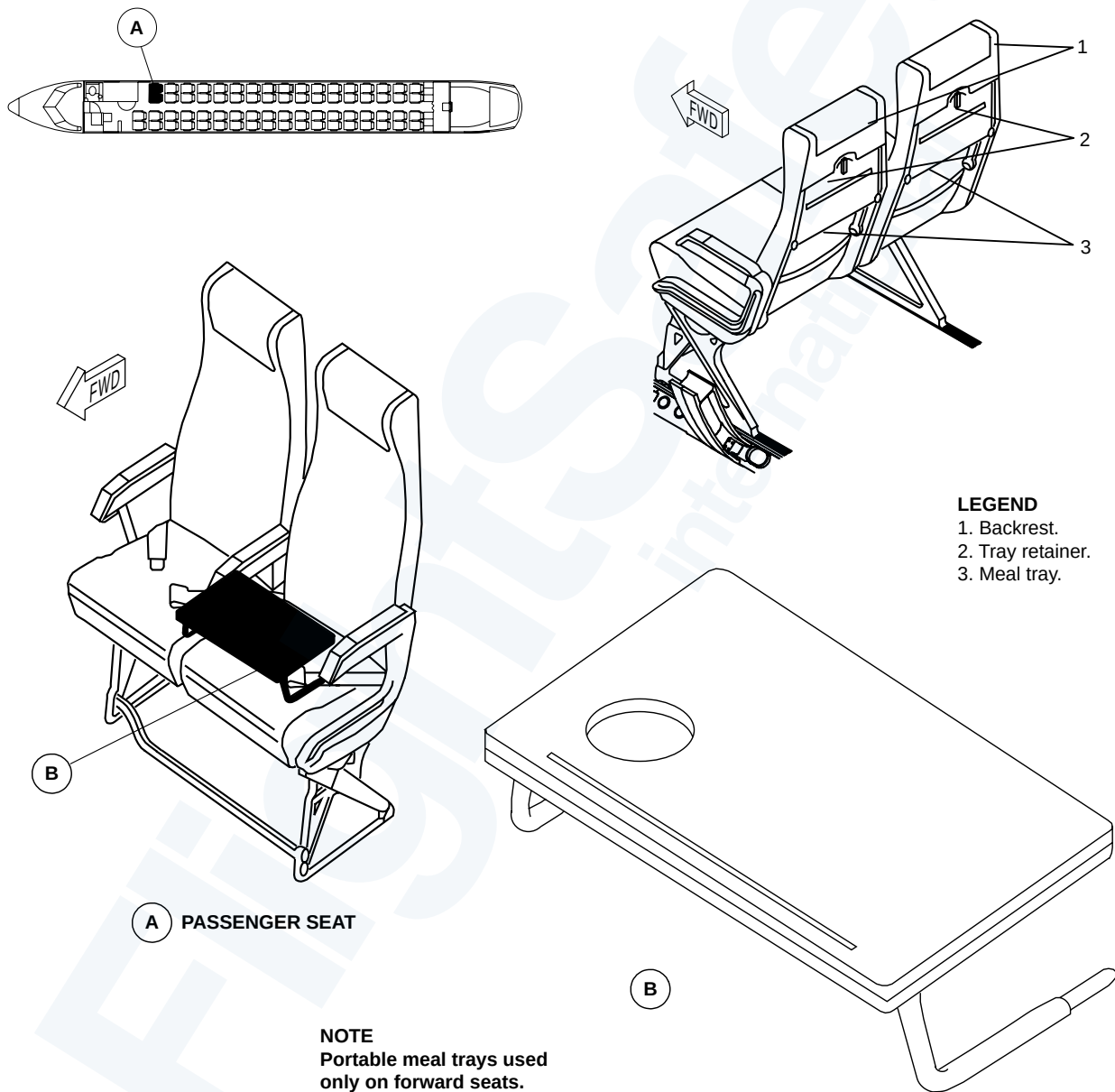
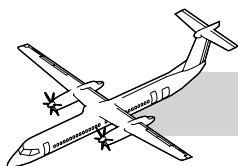


Figure 25-10. Portable Meal Tray Location



Passenger Seats

Refer to Figure 25-11. Passenger Seats.

The seats are installed on inboard and outboard seat tracks and are attached to the tracks by fittings and studs with anti-rattle locks. The seat leg studs fit into their related notches in the floor tracks. The seat frame structure, seat legs and

track attachments can withstand a force of 16 Gs. The seats can be moved along the tracks in one inch increments to give spacing flexibility. The fittings which attach the seats to the tracks are quick disconnect to allow for easy installation and removal. Seat track covers are installed to protect the tracks from dirt and water spillage.

WARNING

WHEN INSTALLING THE FRONT ROW SEATS, CARE MUST BE TAKEN TO ENSURE THAT THE FRONT ROW SEATS ARE THE CORRECT PART NUMBER AND ARE LOCATED IN THEIR APPROVED DESIGN POSITION. WHEN REPLACING PASSENGER SEAT BELTS, NOTE THAT THE FRONT ROW SEATS HAVE UNIQUE Y-STYLE BELTS. IT IS MANDATORY THAT THESE BELTS ARE REPLACED ONLY WITH THE CORRECT PART NUMBER OF BELT. REFER TO THE ILLUSTRATED PARTS CATALOGUE FOR CORRECT PART NUMBERS.

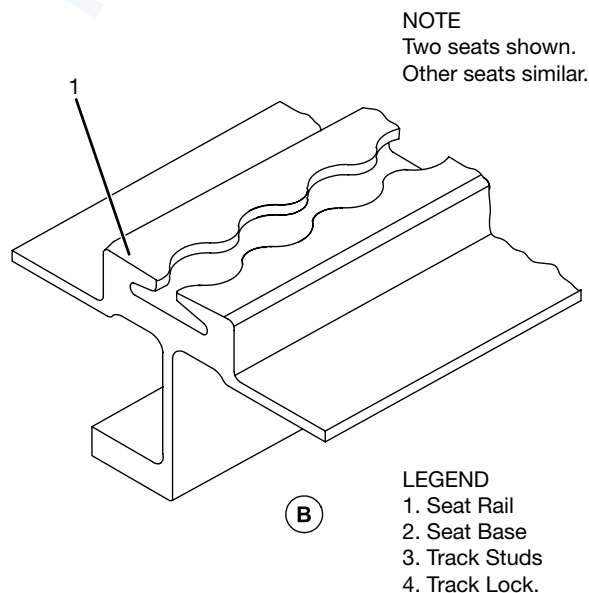
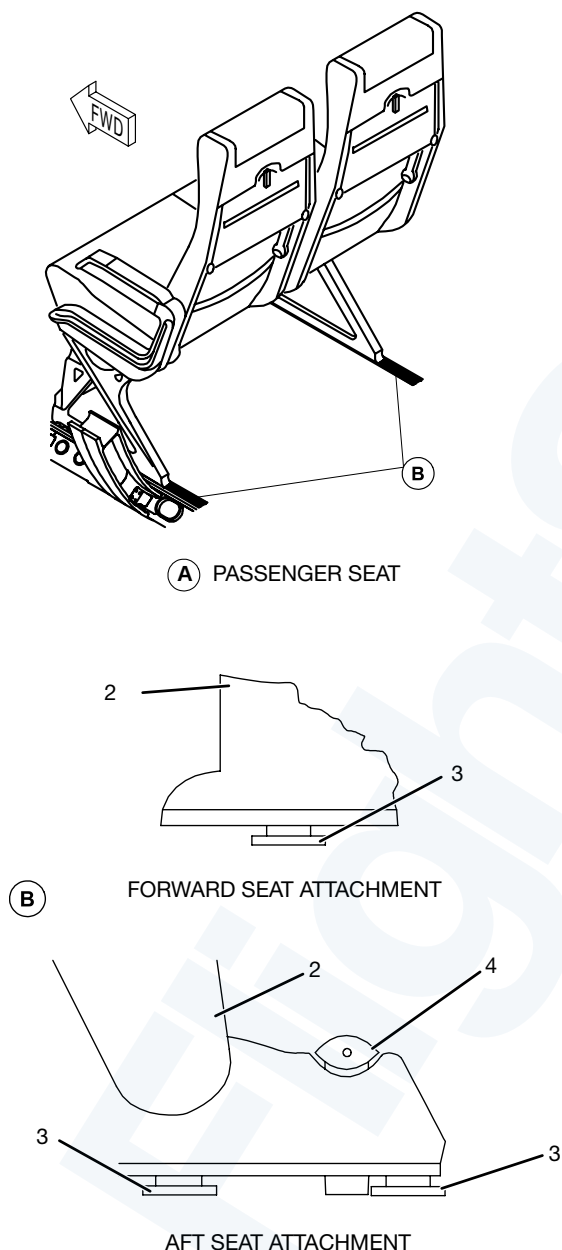
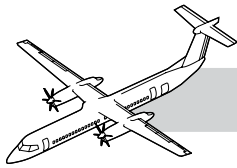


Figure 25-11. Passenger Seats



Flight Attendant Seats

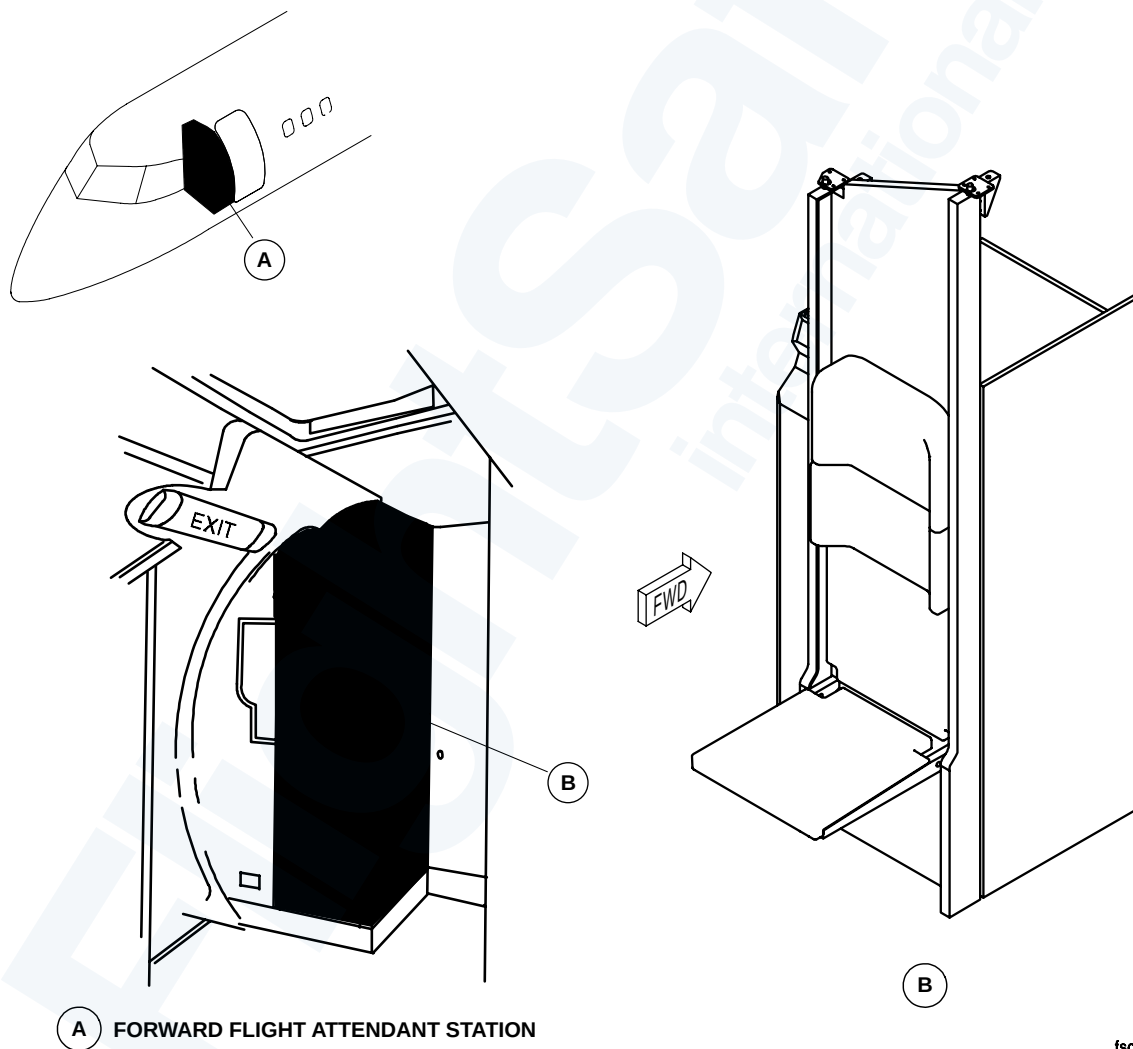
Refer to Figure 25-12. Forward Flight Attendant Seat.

The forward flight attendant's seat is aft facing and forms the aft wall of the wardrobe adjacent to the air-stair door. The seat back is 19 in. (482.6 mm) wide and is part of the wardrobe bulkhead. It has a contoured energy absorbing back cushion and headrest which supports the arms, shoulders, head and spine. The seat portion has a fold-down seat cushion, which is spring-loaded to stow automatically when

the seat is not occupied. The seat cushions are secured to the seat pan by Velcro tape. The flight attendant faces rearward when sitting and has a direct view of the cabin area.

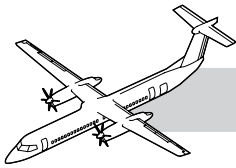
The seat is also equipped with a restraint harness consisting of shoulder restraints and a lap belt. The shoulder restraint is connected to an inertia reel fitted below the head cushion. When released, the harness is retracted by inertia reels after the seat is stowed.

Stowage for a life jacket and flashlight is provided below the seat (refer to 25-60-00).



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Figure 25-12. Forward Flight Attendant Seat



Refer to Figure 25-13. Flight Attendant Seats.

The aft flight attendant's seat is installed on the G3 galley bulkhead behind the two aft doors (aft passenger door and aft service door). If the G3 galley is not installed, the seat is attached to the forward wall of the baggage compartment.

When in use, the seat is located close to the center-line of the aircraft to give a direct view of the aft cabin area.

When stowed, the seat is latched to the left of the G3 galley. When the latch is released, the seat assembly rotates 180 degrees to latch to the galley counter-top. Lowering the seat cushion engages a pin into a receptacle on the floor. The

upper latch mechanism must be fully engaged in the counter-top fitting before the seat pan can be lowered. A handle on the right side of the seat frame releases both latches, enabling the seat to return to its stowed position. The seat may not be deployed unless the aisle carts in the G3 galley are secured behind the seat.

A third flight attendant's seat is optional and if installed, is located adjacent to the aft flight attendant's seat.

The aft flight attendant's seat has stowage room for emergency equipment within the seat base (refer to 25-60-00).

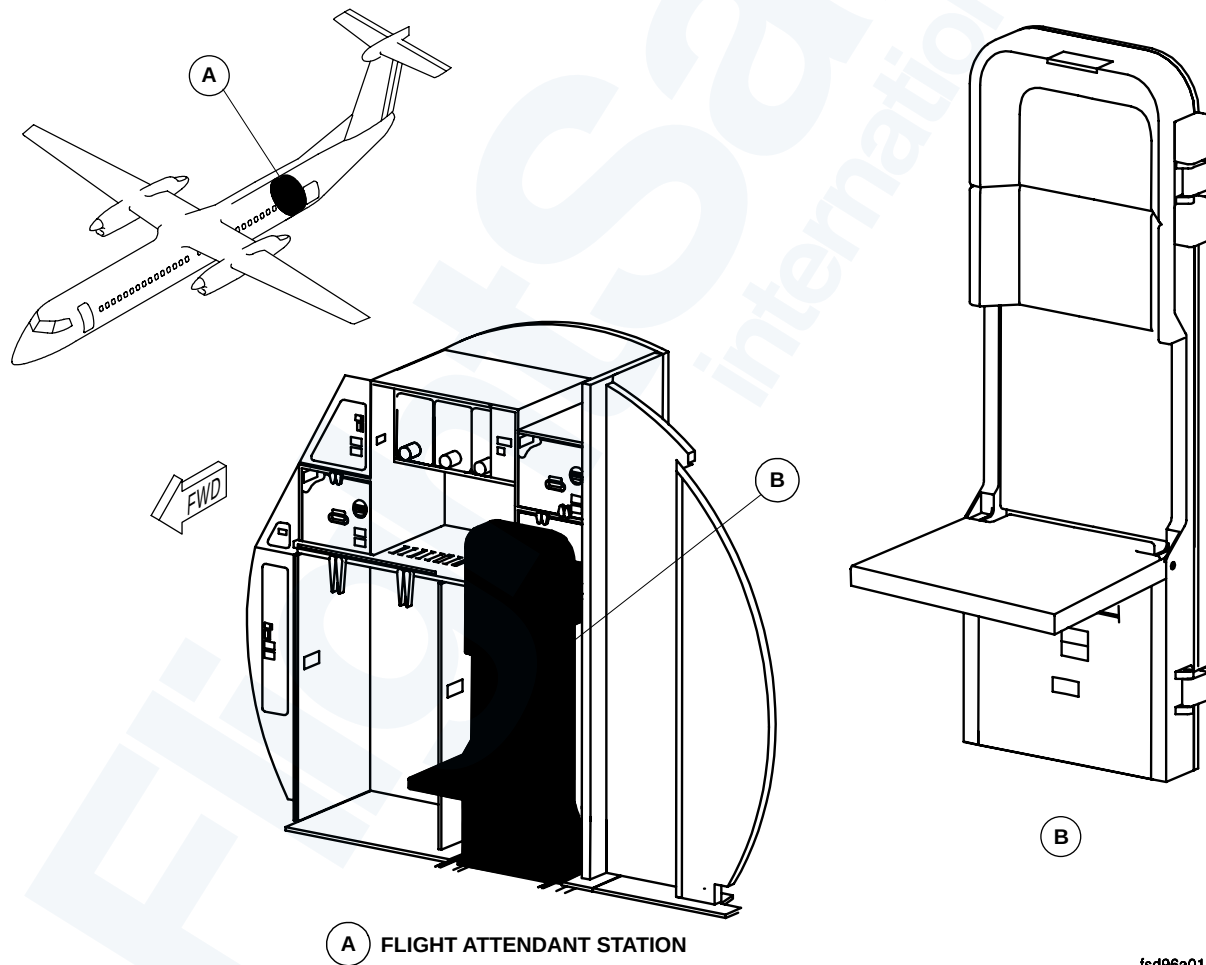
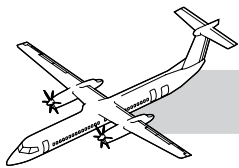


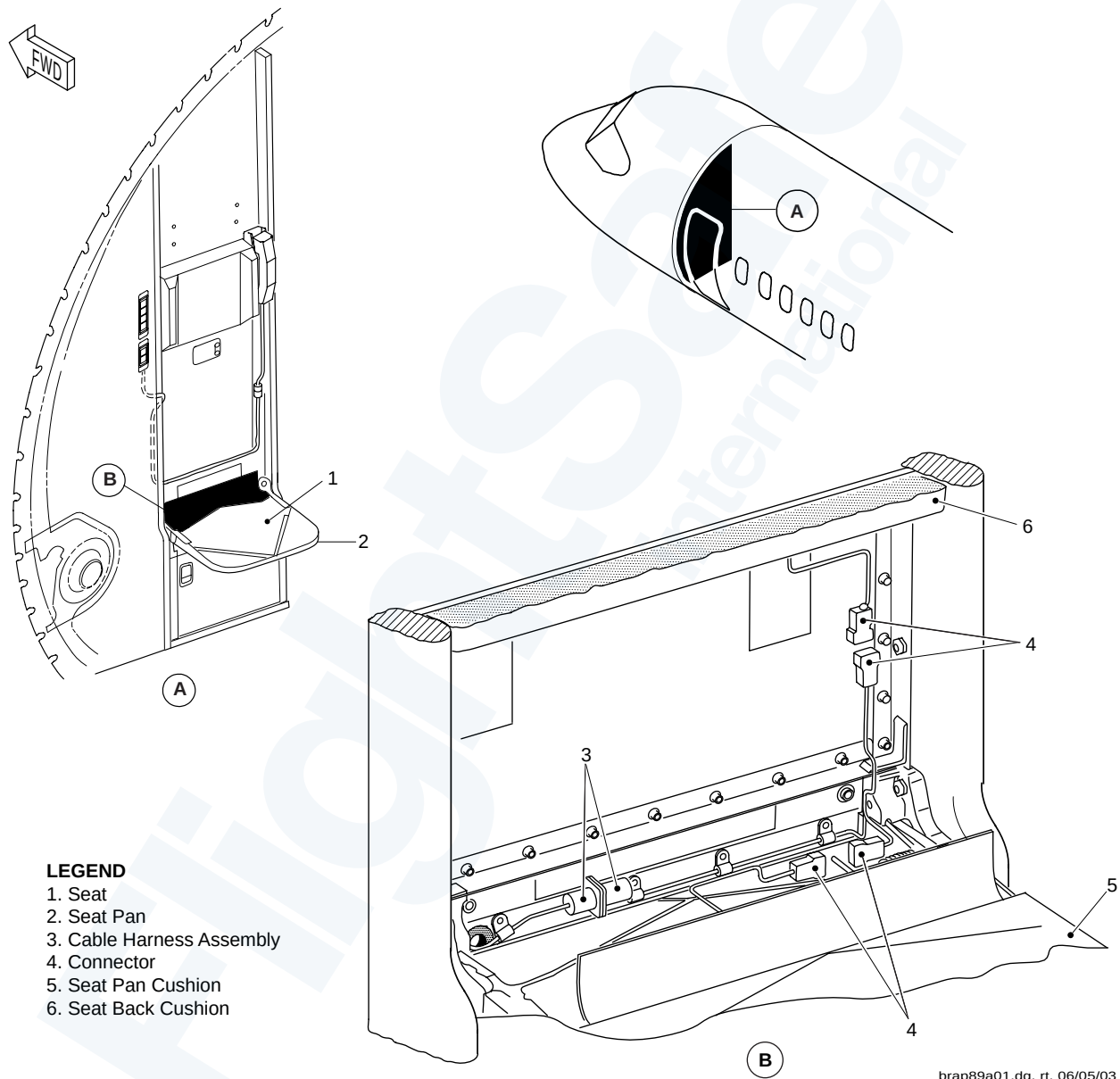
Figure 25-13. Flight Attendant Seats



Forward Flight Attendant Heated Seat

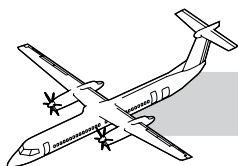
Refer to Figure 25-14. Forward Flight Attendant Seat - Detail.

On aircraft with the heated flight attendant's seats 825CH01244: Each seat is heated by two 32 W heater pads to warm the seat pan and seat back.



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Figure 25-14. Forward Flight Attendant Seat - Detail



Aft Flight Attendant Seat Heater

Refer to Figure 25-15. Aft Flight Attendant Seat.

The aft SEAT HEATER annunciator switch is attached to the aft face of the left aft draft bulkhead adjacent to the aft flight attendant's handset.

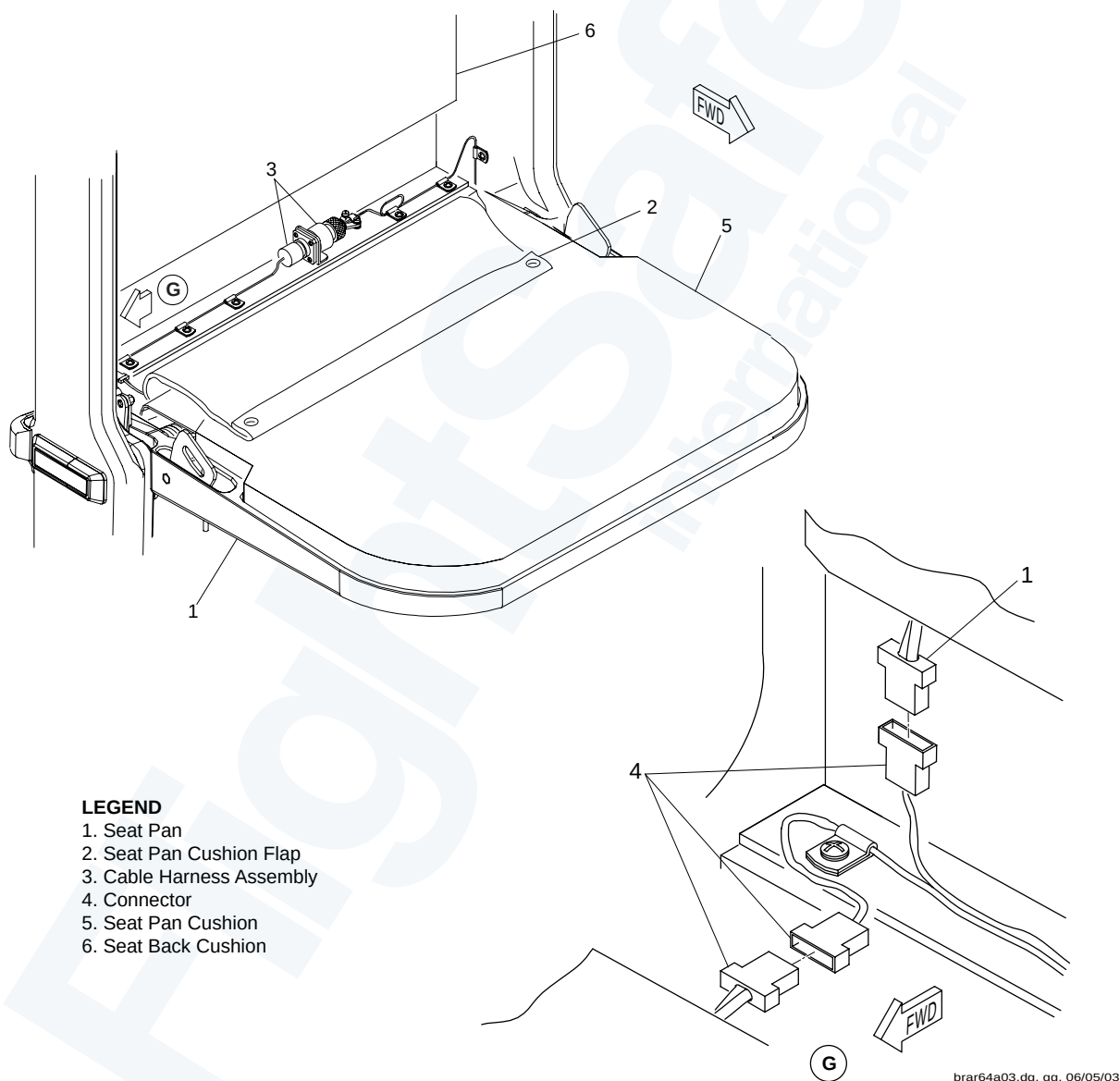


Figure 25-15. Aft Flight Attendant Seat

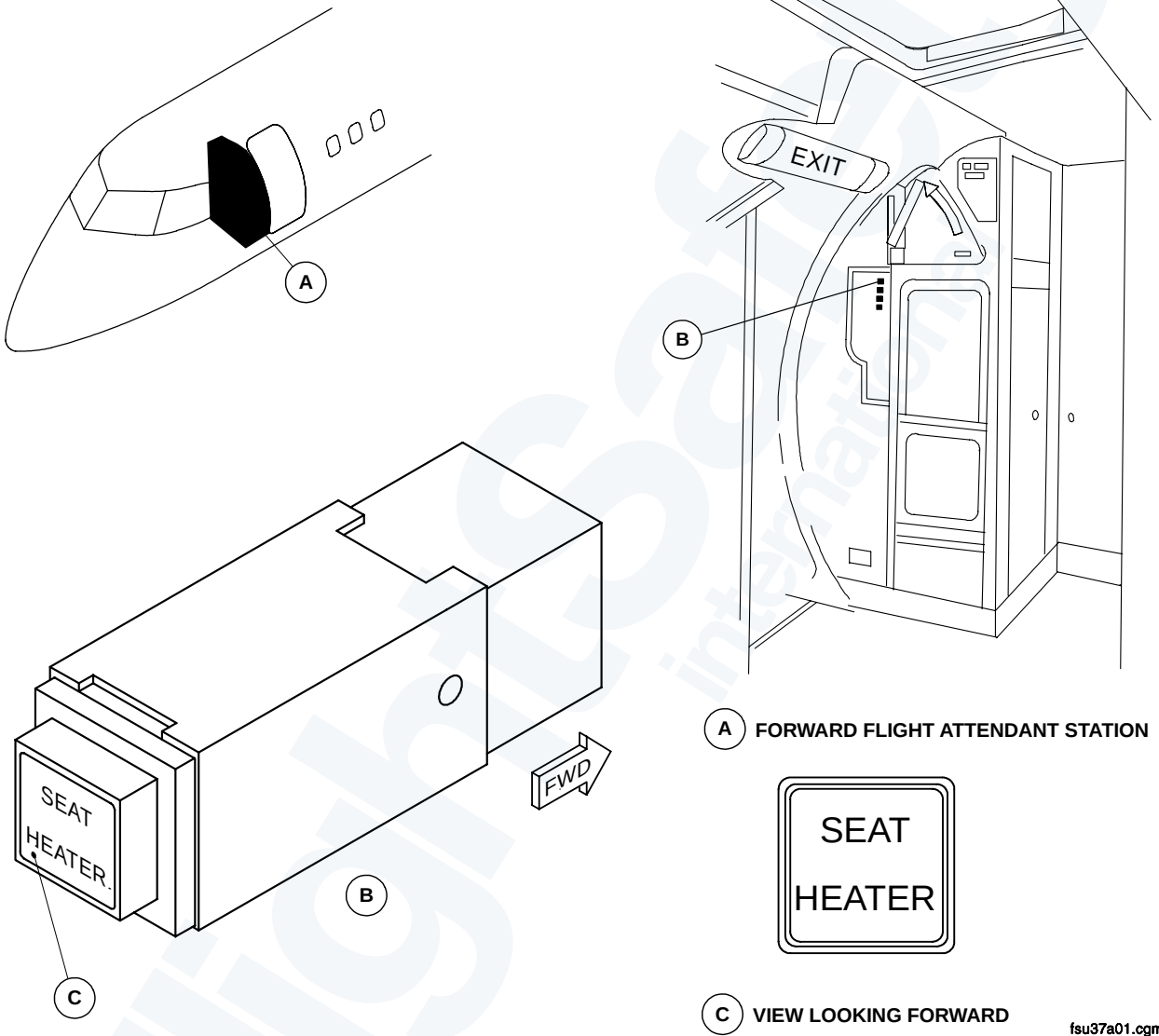
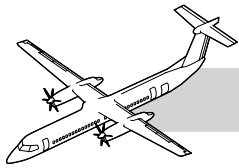
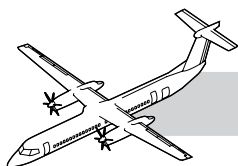


Figure 25-16. Forward Flight Attendant Heated Seat Switch Location



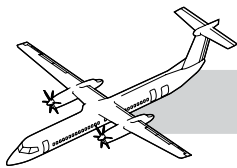
CONTROLS AND INDICATIONS

NOTES

Forward Flight Attendant Seat Heater Switch

Refer to Figure 25-16. Forward Flight Attendant Heated Seat Switch Location.

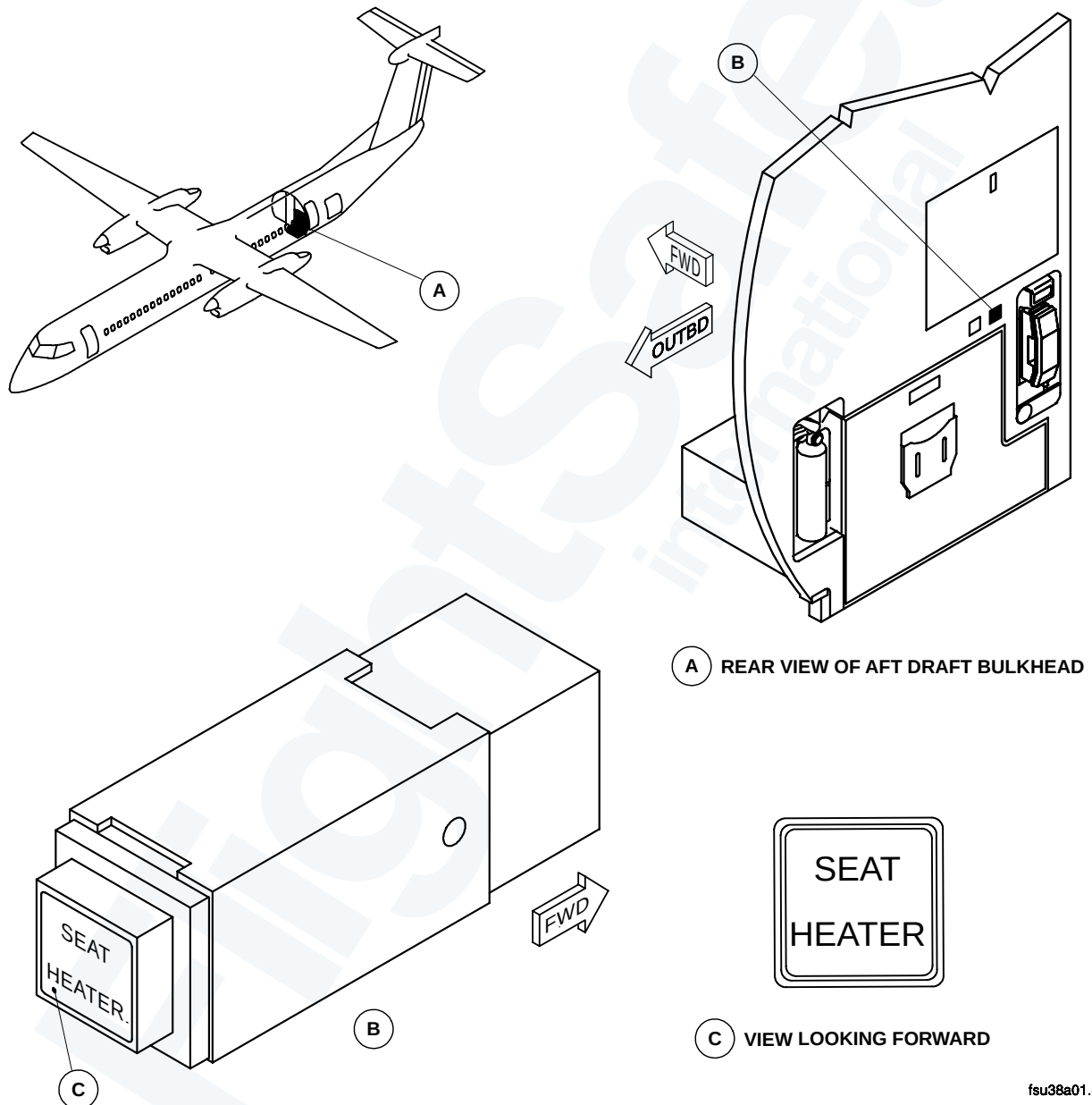
The annunciator switch located near the flight attendant seat in the passenger compartment give manual control of the heating.



Aft Flight Attendant Seat Heater Switch

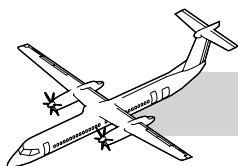
Refer to Figure 25-17. Aft Flight Seat Heater Switch Location.

The aft SEAT HEATER annunciator switch is attached to the aft face of the left aft draft bulkhead adjacent to the aft flight attendant's handset.



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Figure 25-17. Aft Flight Seat Heater Switch Location



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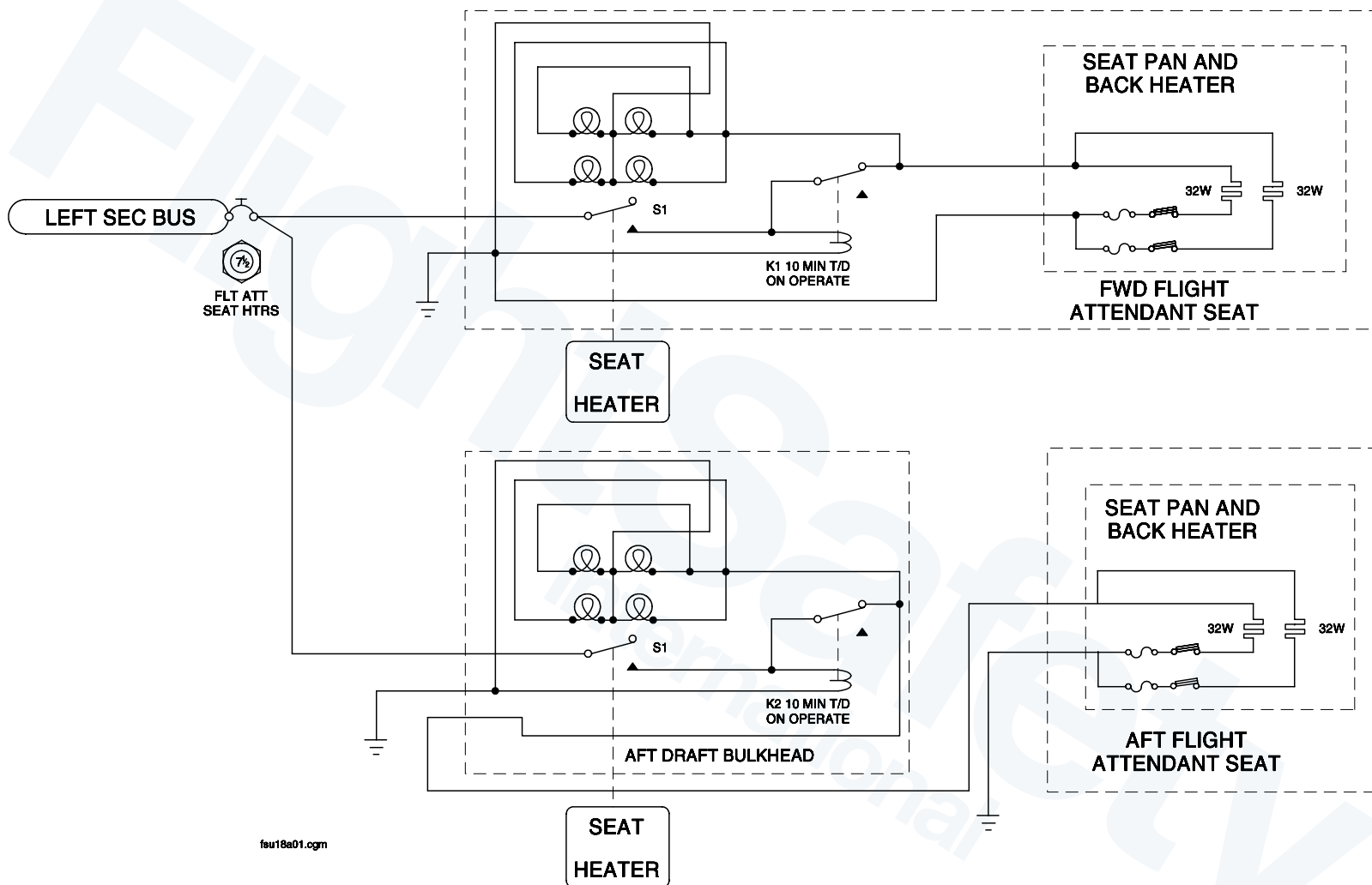
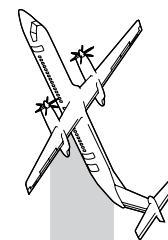
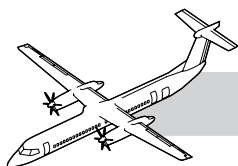


Figure 25-18. Flight Attendant Seat Heater Schematic



OPERATION

NOTES

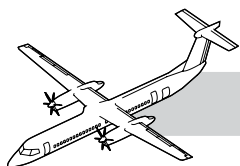
Refer to Figure 25-18. Flight Attendant Seat Heater Schematic.

Electrical power is supplied from the 28 VDC left secondary bus through related passenger compartment annunciator switches and relays to energize the seat heater pads.

When a SEAT HEATER annunciator switch is pushed, the 28 VDC left secondary bus is used to energize the related relay and heater pads.

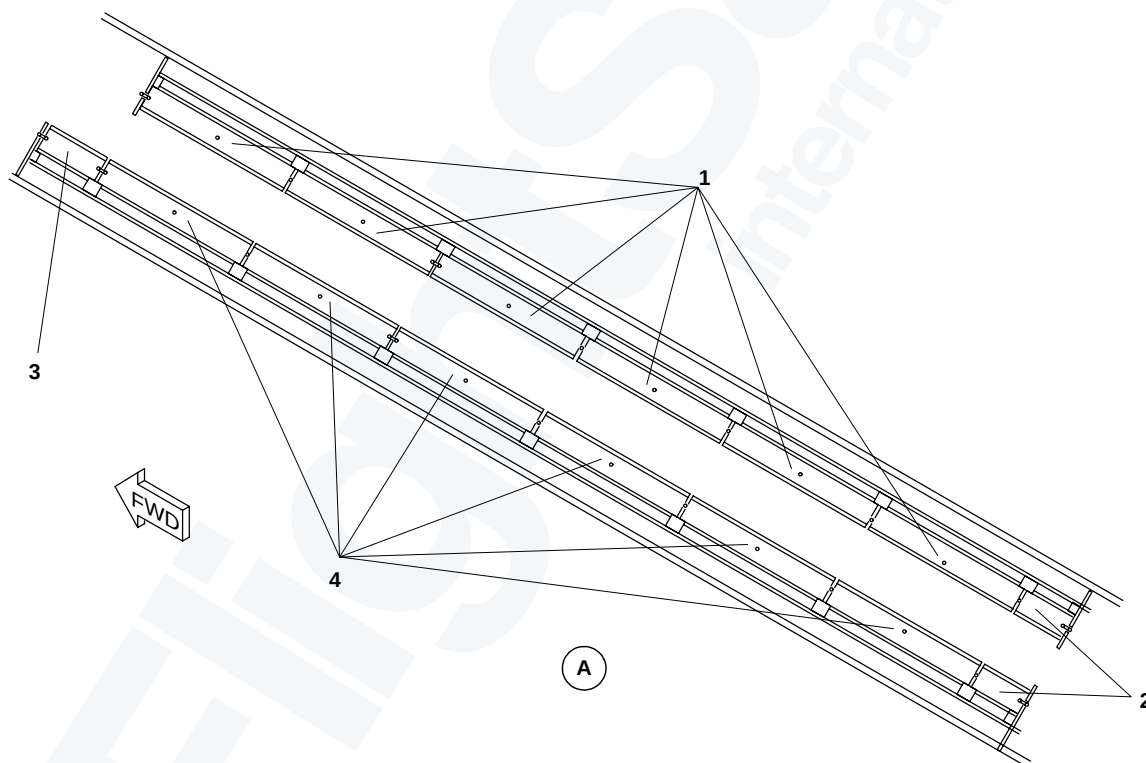
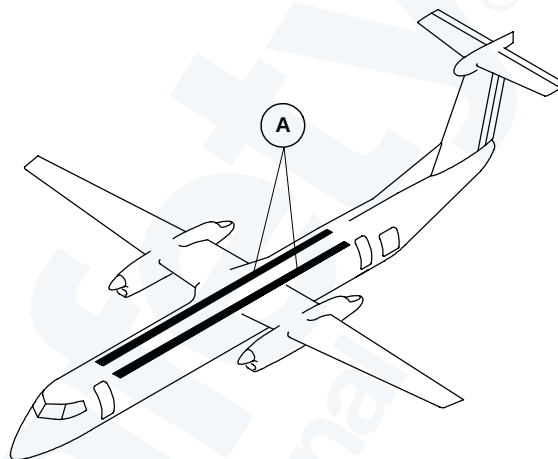
The SEAT HEATER lamps in the annunciator switch come on to show that electrical power is supplied to the heaters. The relay has time delay circuit to de-energize the heater pads after 10 minutes.

The heater pad has a thermal switch for overheat protection. A fuse protects for over current.



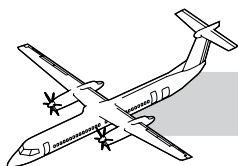
LEGEND

- 1. Bin Assembly (12 x 2 bins of 47 in each)
- 2. Aft Bin (2 bins x 15 in each)
- 3. Forward Bin (38.5).



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Figure 25-19. Overhead Storage Bins



Overhead Storage

NOTES

Refer to Figure 25-19. Overhead Storage Bins.

The overhead bins are installed in the passenger compartment above the seats. They extend the full length of the passenger compartment on both sides of the aisle. They are used for carry-on luggage and portable emergency equipment storage.

The bins are made from a fiberglass composite material with a honeycomb core. The interior of the bin is painted while the exterior has a plastic laminate decorative finish. The bins are attached to the aircraft structure with support brackets and bin hangers through anti-vibration mounts. Spacers are installed to keep an equal distance between each bin.

Each bin has a self-latching door which opens upward and pivots at the front and rear of each bin compartment. The door is attached to the bin with gas spring actuators. The door is held in the closed position by the latch. When the latch is released, the actuators open the door at a controlled rate.

The door latch assembly has a spring-loaded handle which hinges on a latch pin. A set screw is located on the inside and underneath the handle and can be adjusted to prevent the handle from rattling and vibrating. Lifting the handle firmly upwards gives access to the set screw.

There are 12 individual assemblies (six on each side of the aircraft), each assembly having two separate bin compartments. This gives a total of 24 bins (12 on each side of the aircraft). These bins are 47 in. (1.19m) in length and have a maximum weight capacity of 60 lbs. (27 kg) There are two 15 in. (381 mm) long aft bins located on the left and right side, forward of the aft draft bulkhead, which have a maximum capacity of 19 lbs. (8.6 kg) There is one 38.5 in. (978 mm) long forward bin on the left side with a maximum capacity of 49 lbs. (22 kg) This gives a total of 27 individual bins.

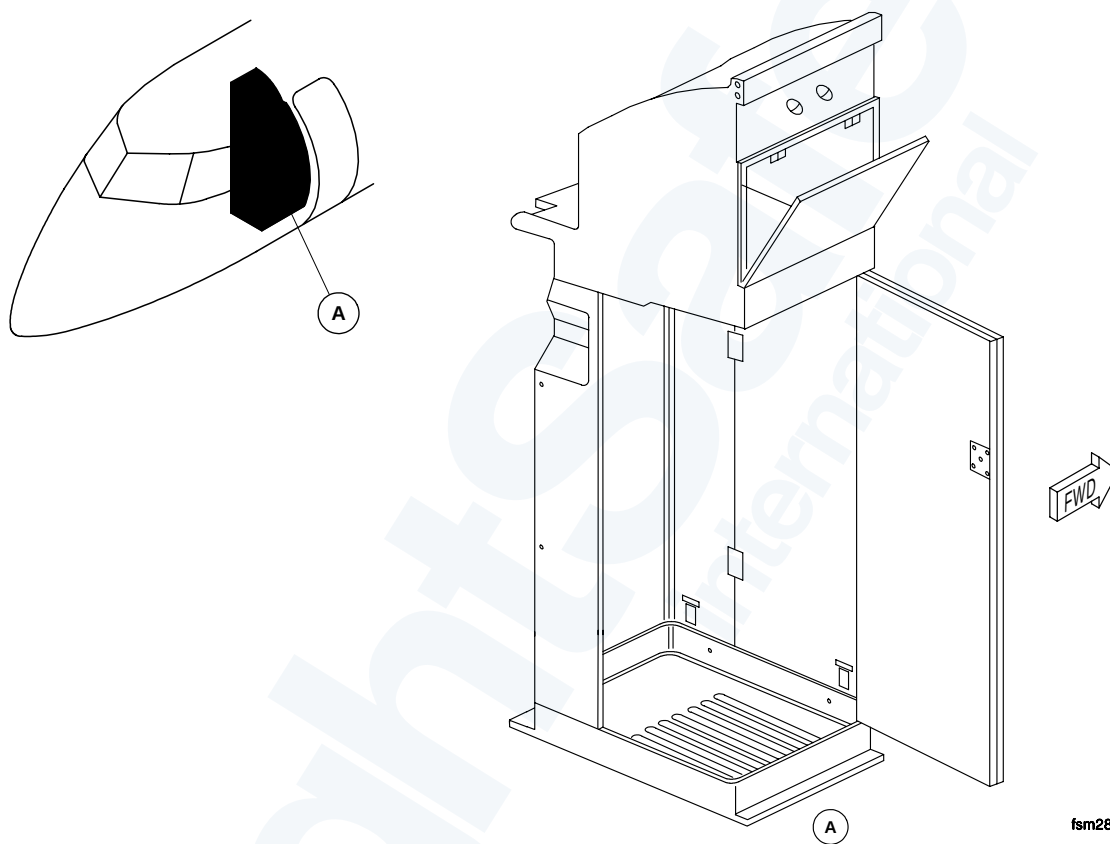
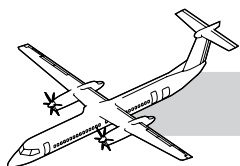
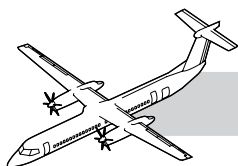


Figure 25-20. Forward Wardrobe Locator



WARDROBES AND PARTITIONS

NOTES

Forward Wardrobe

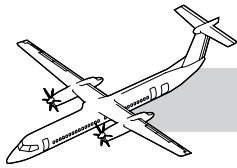
Refer to Figure 25-20. Forward Wardrobe Locator.

The forward wardrobe is located on the forward left side of the passenger compartment between the flight attendant's seat and the avionics equipment rack. The wardrobe is used for garment hanger bags and briefcase size bags. The space above the wardrobe is for avionics equipment.

The wardrobe base is attached to the floor with screws and to the ceiling structure with fittings. The wardrobe has a self contained shelf unit with a hanger bar, honeycomb composite panels with quick disconnect latches, a floor pan and a door. The door pivots at the top and bottom on two Teflon bushing and pin assemblies and opens into the aisle.

Panels within the closet compartment can easily be removed to give access to electrical equipment and utility lights. The wardrobe overhead light is controlled by a door micro-switch powered from the 28 VDC left essential bus through a circuit breaker K2.

The wardrobe enclosure is formed by the upper shelf unit, the floor pan and quick release sidewall panels. The floor pan is attached to the floor structure and the quick release sidewall panels connect the upper shelf to the floor pan. The wardrobe side panels are made of a honeycomb core and have a composite laminate surface. The flight attendant seat back forms part of the aft, inboard wardrobe compartment.



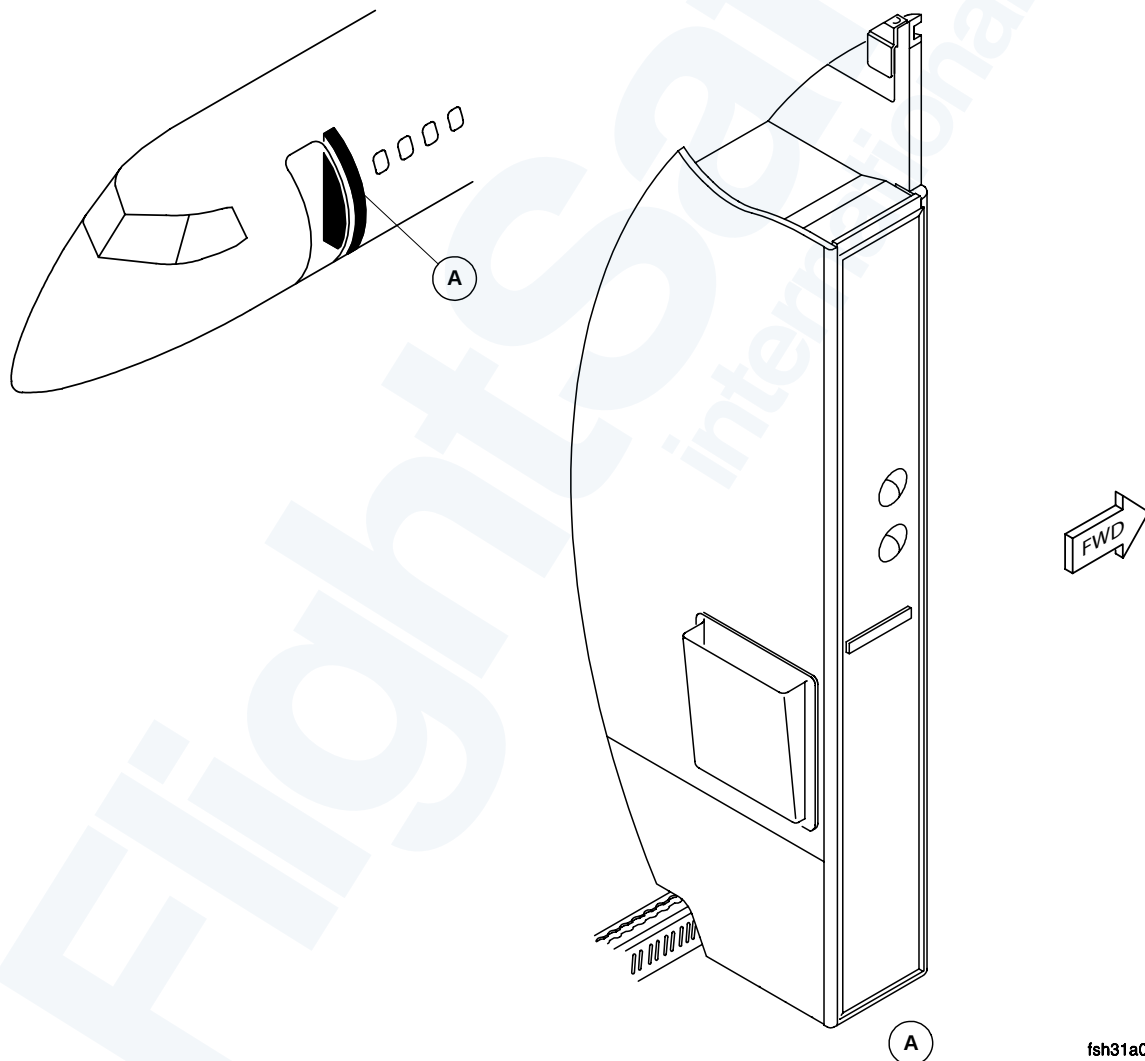
Forward Draft Bulkhead

Refer to Figure 25-21. Forward Draft Bulkhead.

The forward Draft Bulkhead forms a partition between the forward passenger door and the adjacent passenger seats. It shields the passengers from cold drafts when the forward passenger door is open. It also supplies a forward storage area for the passenger compartment emergency equipment (refer to 25-60-00).

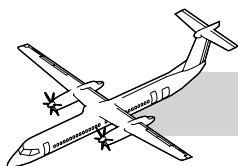
The bulkhead is made from a fiberglass/honeycomb material covered with a decorative plastic laminate. Its edges are trimmed with aluminum and a kick strip is attached at the floor line. The bulkhead has an edge-molding on the aft side to make a cover for the forward passenger door support mechanism. The bottom of the bulkhead is installed with two pins which engage with vibration isolators in the floor panels.

The top of the bulkhead is attached to the forward left hand overhead stowage bin with two screws. A document holder is installed on the aft side of the bulkhead.



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Figure 25-21. Forward Draft Bulkhead



Aft Draft Bulkhead

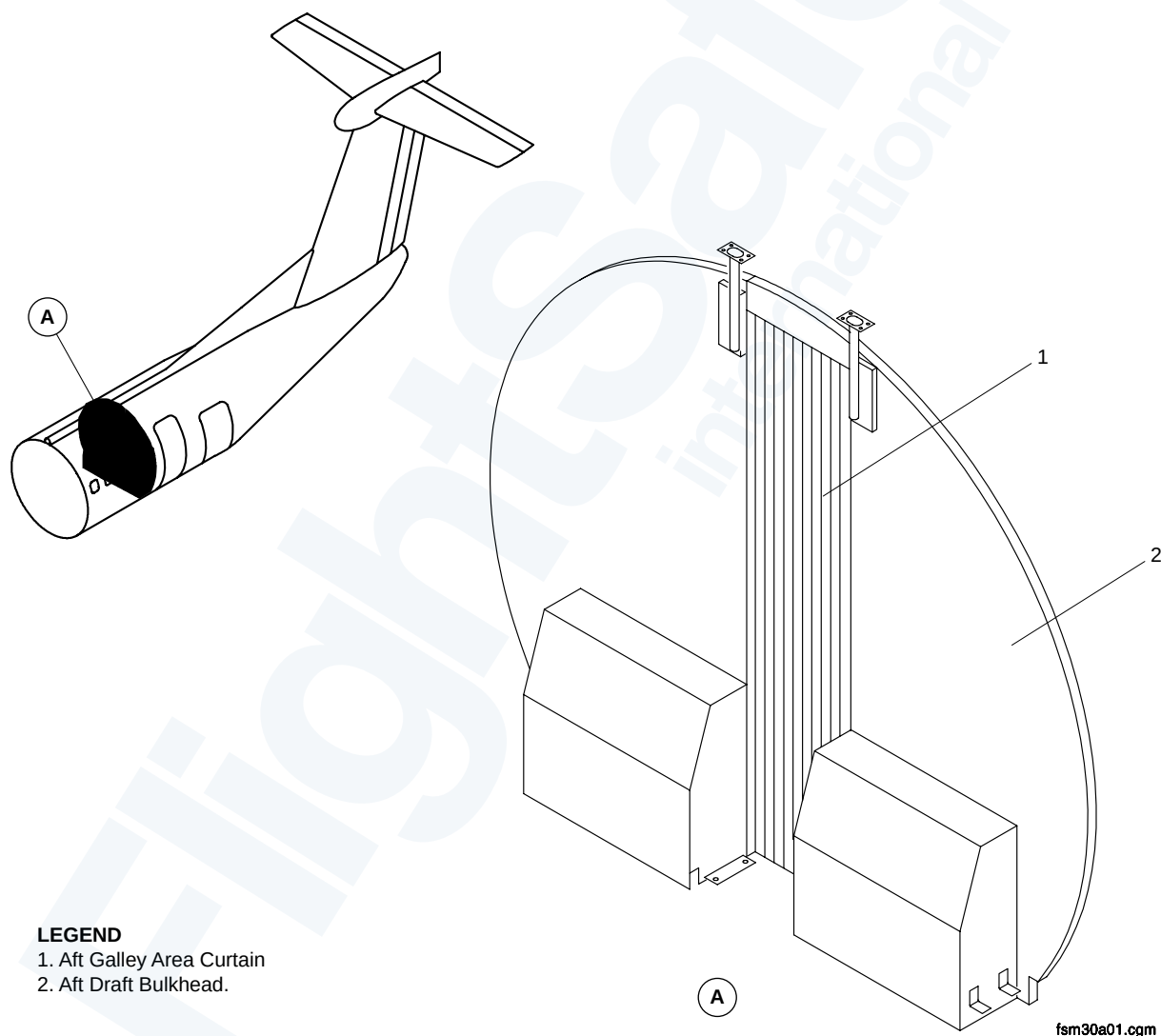
Refer to Figure 25-22. Aft Draft Bulkhead Locator.

The two aft bulkheads are located forward of the aft service door and the aft passenger door and immediately behind the rear passenger seats. They form a partition between the G3 galley area and the adjacent passenger seats. They shield the passengers from cold drafts when either of the two doors are opened.

The bulkheads are made from a fiberglass composite material and extend from the floor to ceiling outboard of the aisle. Each draft bulkhead contains emergency equipment stowed in drawer compartments, accessible from the aisle (refer to 25-60-00). A cabin inter-phone handset is installed on the lower rear part of the left bulkhead for use by the flight attendant in the aft cabin.

NOTE

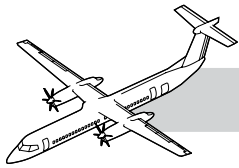
Optional for other configurations of stowage drawers.



LEGEND

- 1. Aft Galley Area Curtain
- 2. Aft Draft Bulkhead.

Figure 25-22. Aft Draft Bulkhead Locator

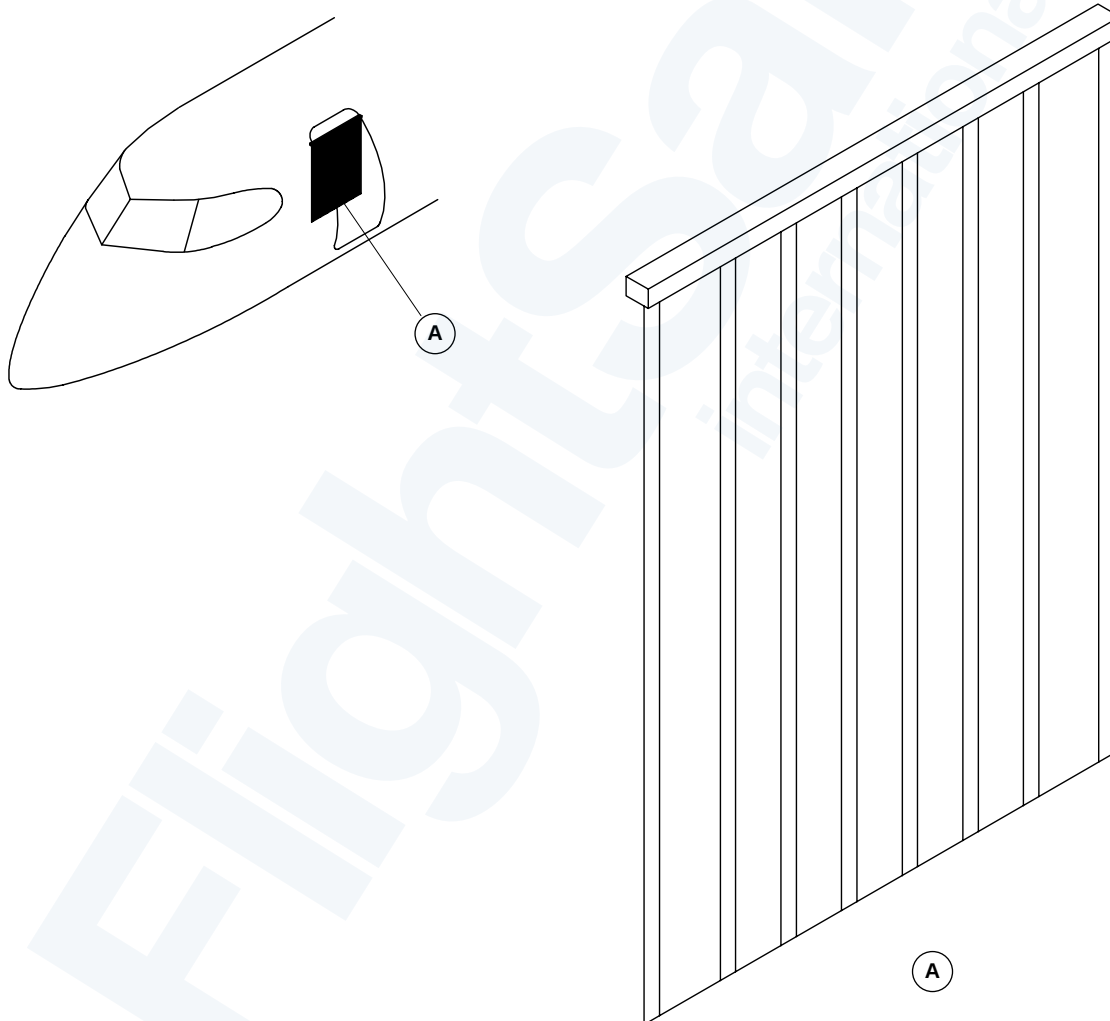


Airstair Door Weather Curtain

Refer to Figure 25-23. Airstair Door Weather Curtain.

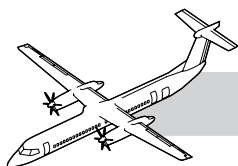
The airstair door weather curtain is installed above the passenger door and attached to a curtain rail. It is made of a weatherproof fabric. The curtain rail is part of the door header bumper panel assembly.

The curtain has hooks that are attached to gliders in the rail. The curtain is pulled down to cover the forward passenger door and the airstairs. A tie back strap is supplied to hold the curtain in the open position when not in use.



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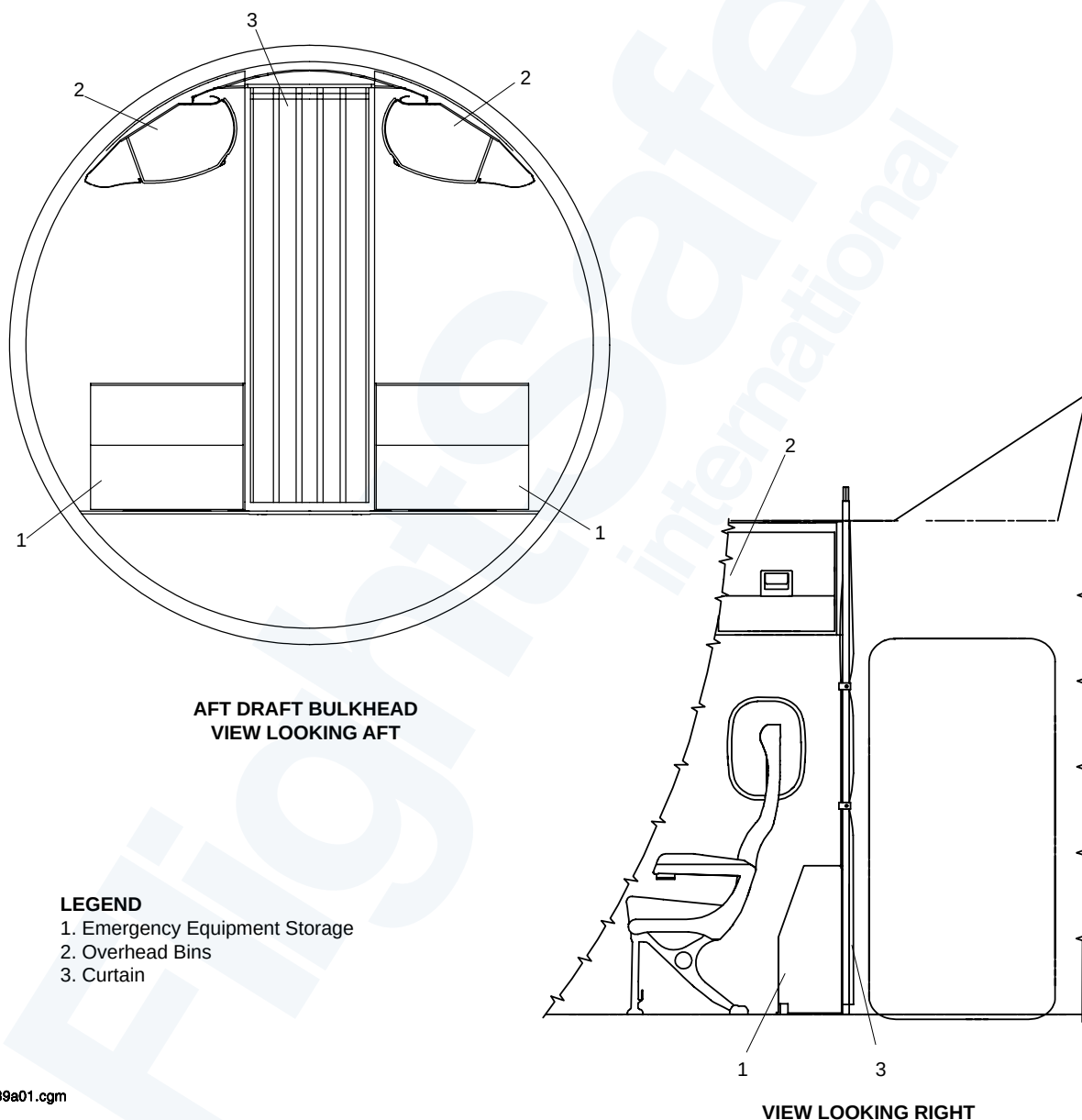
Figure 25-23. Airstair Door Weather Curtain



Aft Galley Area Curtain

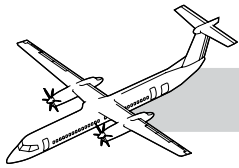
Refer to Figure 25-24. Aft Galley Area Curtain.

A curtain is suspended from the ceiling, in between the two aft draft bulkheads. It is stowed with Velcro straps. It supplies a degree of privacy for the flight attendant while on duty and acts as a draft screen when the rear doors are opened.



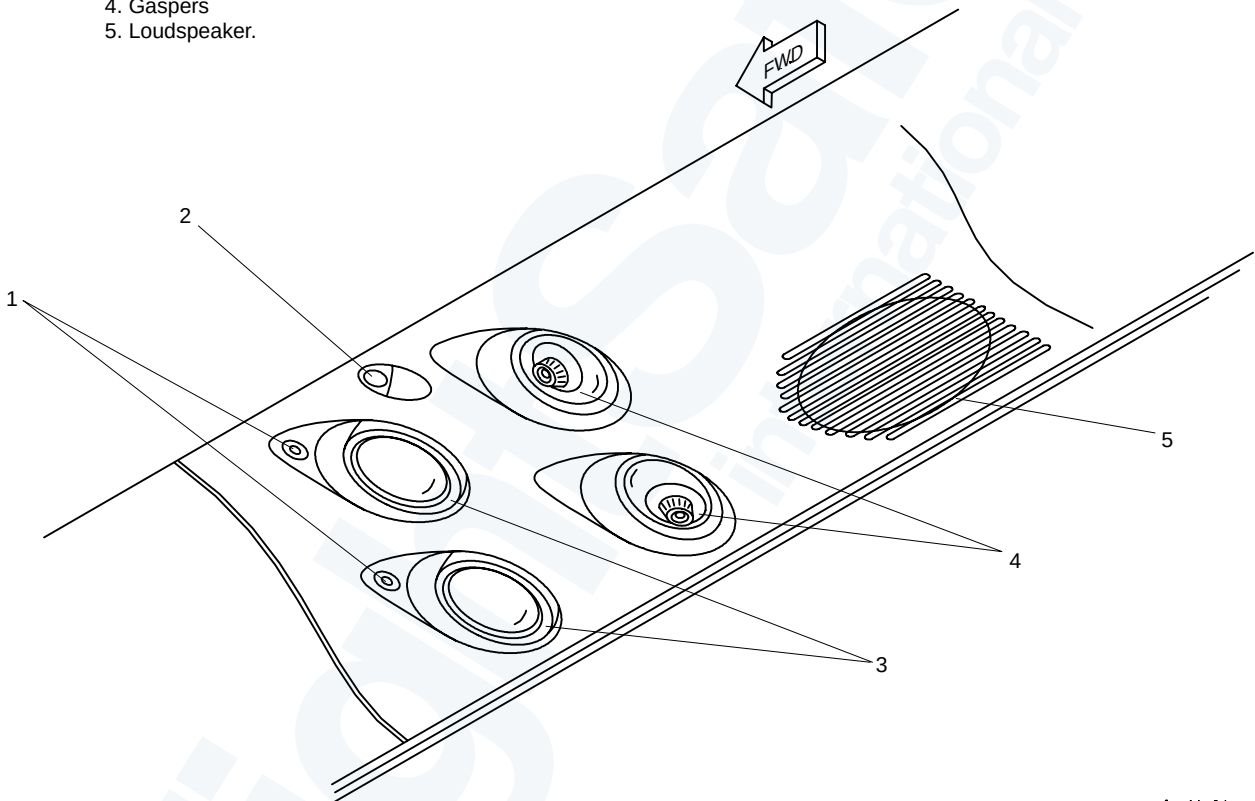
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Figure 25-24. Aft Galley Area Curtain



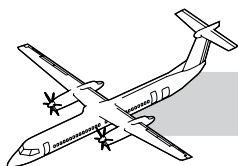
LEGEND

1. Reading Light Switch
2. Flight Attendant Call Button
3. Reading Lights
4. Gaspers
5. Loudspeaker.



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Figure 25-25. Passenger Service Units



Passenger Service Units (PSU)

Refer to:

- Figure 25-25. Passenger Service Units.
- Figure 25-26. Passenger Service Units.

The units have passenger service panels that contain:

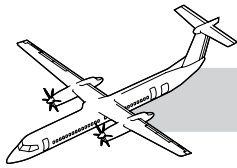
- Two adjustable reading lamps and operating switches
- Two adjustable air outlets (gaspers)
- Flight attendant call switch and light indicator
- NO SMOKING sign
- FASTEN SEAT BELT sign
- Passenger address loudspeakers.

The PSU are movable in one in. (24.5 mm) increments to allow for variable seat pitches. To move PSU, open all PSU using Removal Tool 9506145-501 and remove filler panels at both ends. Slide PSU to new location. Install proper sized filler panel at each end and raise PSU into position and secure with Installation Tool 9506145-502.

To replace lamp, press in and rotate bezel clockwise until bezel falls free. Press in and rotate lamp counterclockwise and remove lamp. Install new lamp by pressing in and rotating lamp clockwise to secure. Then rotate bezel counterclockwise and press in until secure.



Figure 25-26. Passenger Service Units



MAINTENANCE PRACTICES

The following are abbreviated descriptions of the maintenance practices and are intended for training purposes only. For a more detailed description of the practices, refer to the tasks in the Bombardier AMM PSM 1-84-2.

Removal of the Passenger Seats

- The seats are installed in the passenger compartment on rails. Remove the covers from the seat track, then remove hardware retaining the seat to the track. Align the Seat studs with the holes in the track then lift the seat from the track.

Removal of the Sidewall Panels

- The sidewall panels are installed between X 39.50 and X 665.50. Remove the infill strips from each side of the sidewall panel. Remove the attaching hardware, carefully raise the sidewall panel to disengage the attachment angle from the sidewall/dado rail.

Removal of the Passenger Window Reveal Panel

- Put the removal end of the latch tool in the space at the bottom of the groove in the reveal assembly. Push the latch tool down to release the lower clip. Remove the latch tool from the reveal assembly. Carefully move the lower part of the reveal assembly inboard and down to disengage the upper clip from the sidewall panel.

Removal of the Dado Panels

- For the removal of the dado panels installed between stations X 101.18 and X 665.50. Disconnect the Velcro hook tape from the Velcro loop tape. Move the dado panel down to disengage the dado panel from the sidewall panel.

Removal of the Ceiling Panels

- For the removal of the ceiling panels installed in the passenger compartment between X-24.30 and X700.75. Loosen the hardware that attaches the panel to the support rail. Lower the ceiling panel and disconnect the spring clips from the support rail. Disconnect the electrical connector from the aircraft interface connector if installed. Remove the ceiling panel

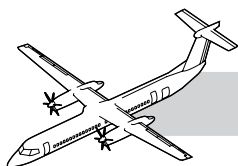
Removal of the Passenger Service Units

- For the removal of the short or long passenger service units (PSU) installed in the passenger compartment between X 32.60 and X 665.34 per aircraft configuration. With the hook, carefully hook the silicone tubing pull from between the PSU and the infill panels and overhead bin assembly. Let the silicone tubing hang from the panels.

NOTE

Careful not to cause damage to the silicone tube. If damage occurs, then replace the section of the damaged tube.

- Carefully lower the PSU from the infill panels, and the overhead bin assembly. The lanyards will hold the lowered PSU.
- Remove the flexible hose from the gasper air duct on the overhead bin assembly.
- Disconnect electrical connectors from the aircraft electrical plugs and install protective covers to the aircraft electrical plugs.
- Disconnect the PSU lanyard from the clips of the overhead bin and carefully remove the PSU from the overhead bin assembly.



Removal of the Passenger Service Unit Printed Circuit Board

NOTES

- Remove the PSU printed circuit board by disconnecting the circuit board connectors J1, J2, J3 and J4. Remove the circuit board attaching hardware. Remove the circuit board.

TROUBLESHOOTING

Operational Test of the Passenger Seats

- Push the recline button and push the seat rearward. Make sure the seat reclines noiselessly. Make sure that the seat stays in the reclined position when the recline button is released.
- Push the recline button with no weight on the seat back and make sure that the seat comes back to the vertical position.
- Make sure the meal tray goes down to the correct height and stops. Make sure the latch for the meal tray operates correctly and does not catch the tray.
- Make sure the moveable armrests do not catch the tray, and that they are free of damage.
- Make sure that the seat is correctly attached to the floor.
- Make sure that the lap belts engage and disengage correctly.

Operational Check of the Passenger Service Units

- For the operational test of the passenger service units (PSU). This test checks the operation of the following:
 - PSU panel gaspers
 - PSU and information signs
 - Cabin Reading Lights (Next Generations)
 - Information Signs (Next Generation)

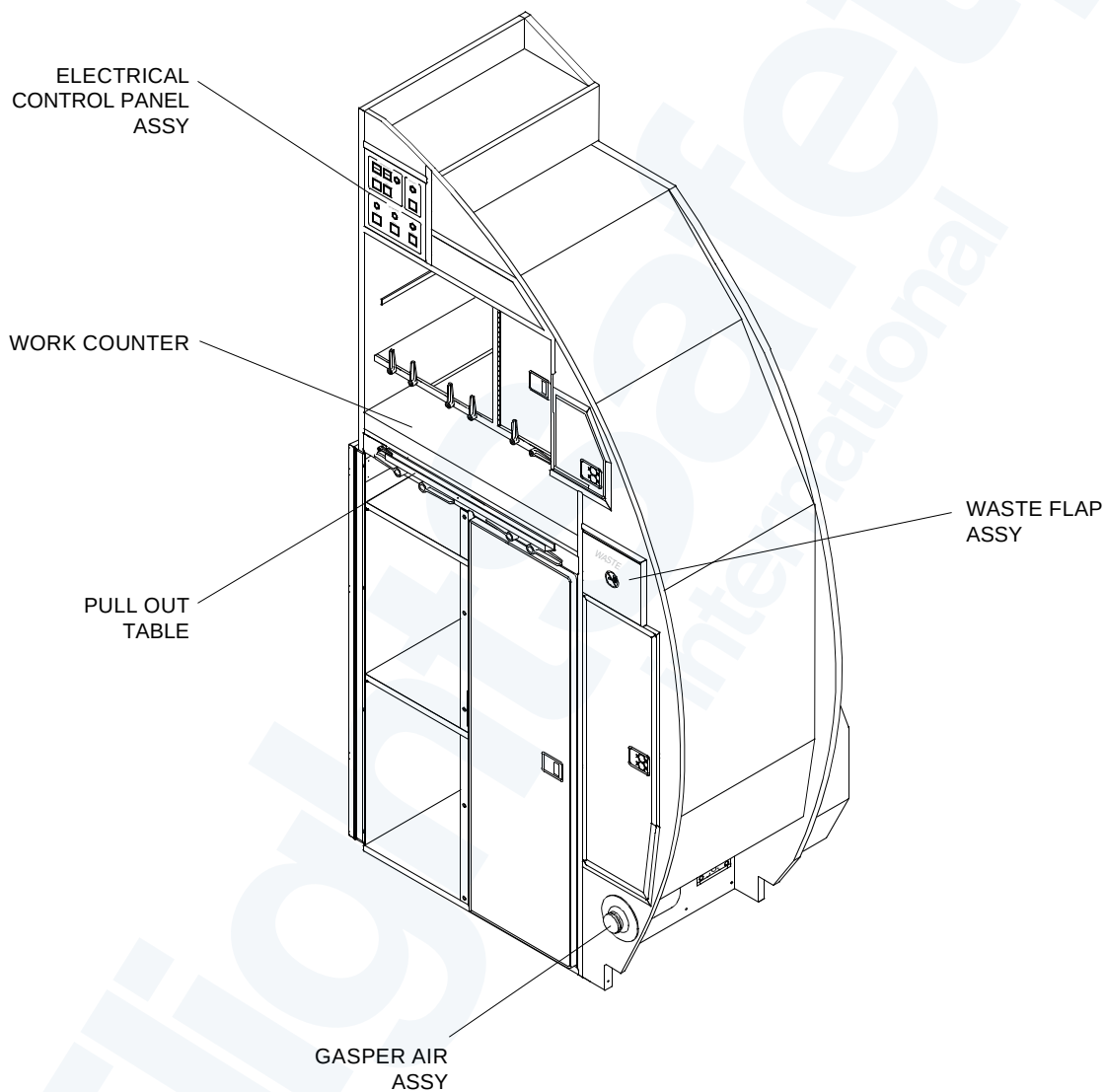
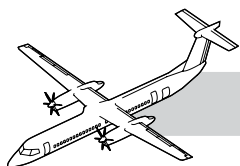
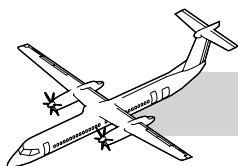


Figure 25-27. G1 Galley Assembly



25-30-00 GALLEYS

NOTES

GENERAL

The galleys allow for the storage of food and beverages (hot and cold) and for the storage of waste material. The galleys also supply adequate space and the required equipment for the preparation and serving of the food and beverages.

The galleys are supplied with AC and DC electrical power for the ovens and water heaters.

Several optional galley installations are available. The G1/G2 option has two galleys located on either side of the center aisle, forward of the aft draft bulkhead. The G3 option has one galley built in to the forward side of the aft baggage compartment. The galleys are supplied with hot and cold water for drinking and washing purposes. The water heaters are supplied with AC electrical power (refer to 38-00-00).

COMPONENT DESCRIPTION

Refer to:

- Figure 25-27. G1 Galley Assembly.
- Figure 25-28. G2 Galley Assembly.
- Figure 25-29. G1/G2/G3/G4/G6 Galley Configuration..

Galley (G1)

The aft right hand galley has a removable waste container, miscellaneous stowage, emergency equipment and provisions for three standard oven units and two ATLAS standard half carts. The unit has a durable work surface.

Aft Left Hand Galley (G2)

The aft left hand galley has three heated beverage dispensers (hot jugs), a removable waste container, miscellaneous stowage, emergency equipment and provisions for two ATLAS standard half carts. The hot jugs are powered by the DC electrical system. The unit has a durable work surface.

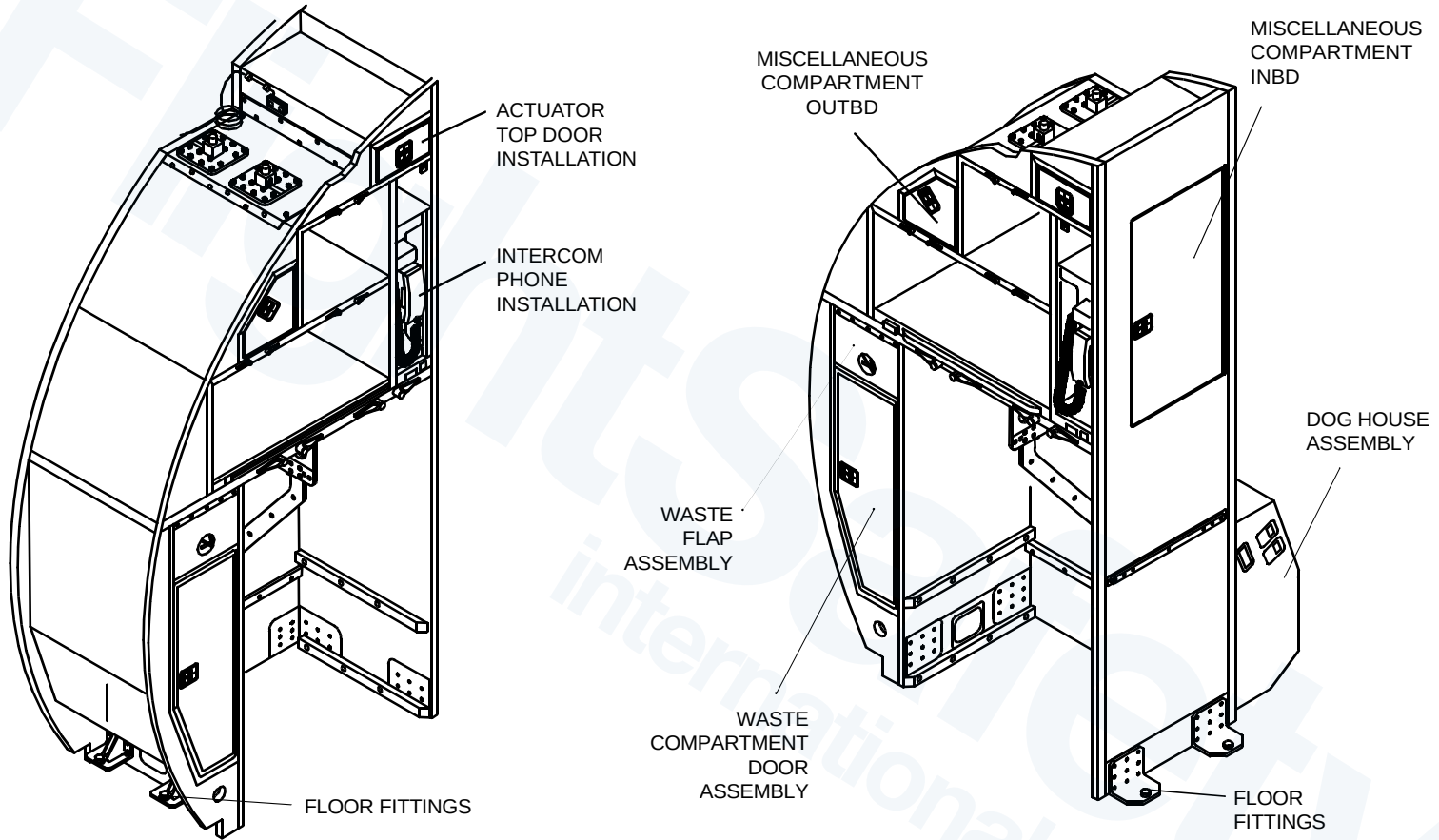
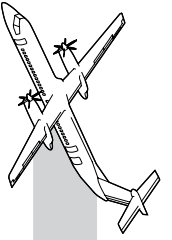
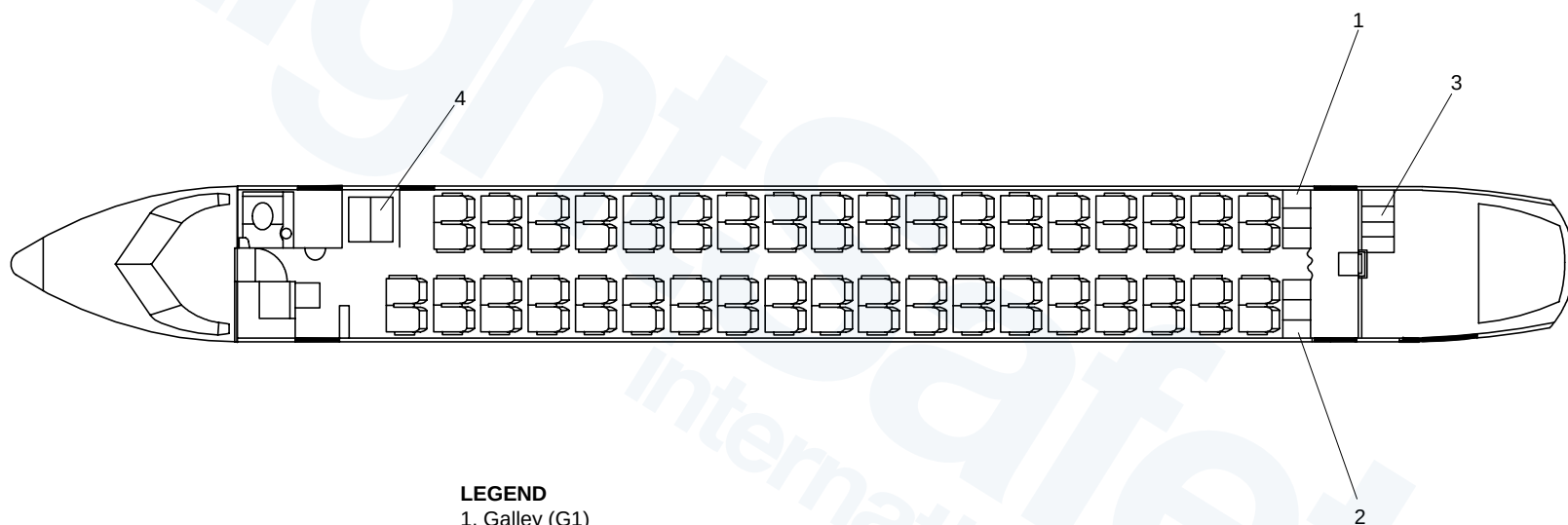
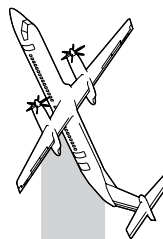


Figure 25-28. G2 Galley Assembly

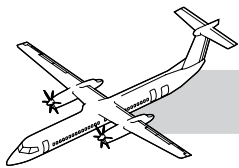


LEGEND

- 1. Galley (G1)
- 2. Galley (G2) or (G4)
- 3. Galley (G3)
- 4. Galley (G6)

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Figure 25-29. G1/G2/G3/G4/G6 Galley Configuration



Aft Galley (G3)

Refer to:

- Figure 25-29. G1/G2/G3/G4/G6 Galley Configuration.
- Figure 25-30. G3 Galley Assembly.

The G3 galley is located behind the aft service door. It contains three DC powered hot jugs, a removable waste container, miscellaneous stowage, three ovens and four Atlas standard half carts.

The carts are retained in the galley by 1/4 turn fasteners and have a pull out work shelf. The aft flight attendant's seat is part of the G3 galley.

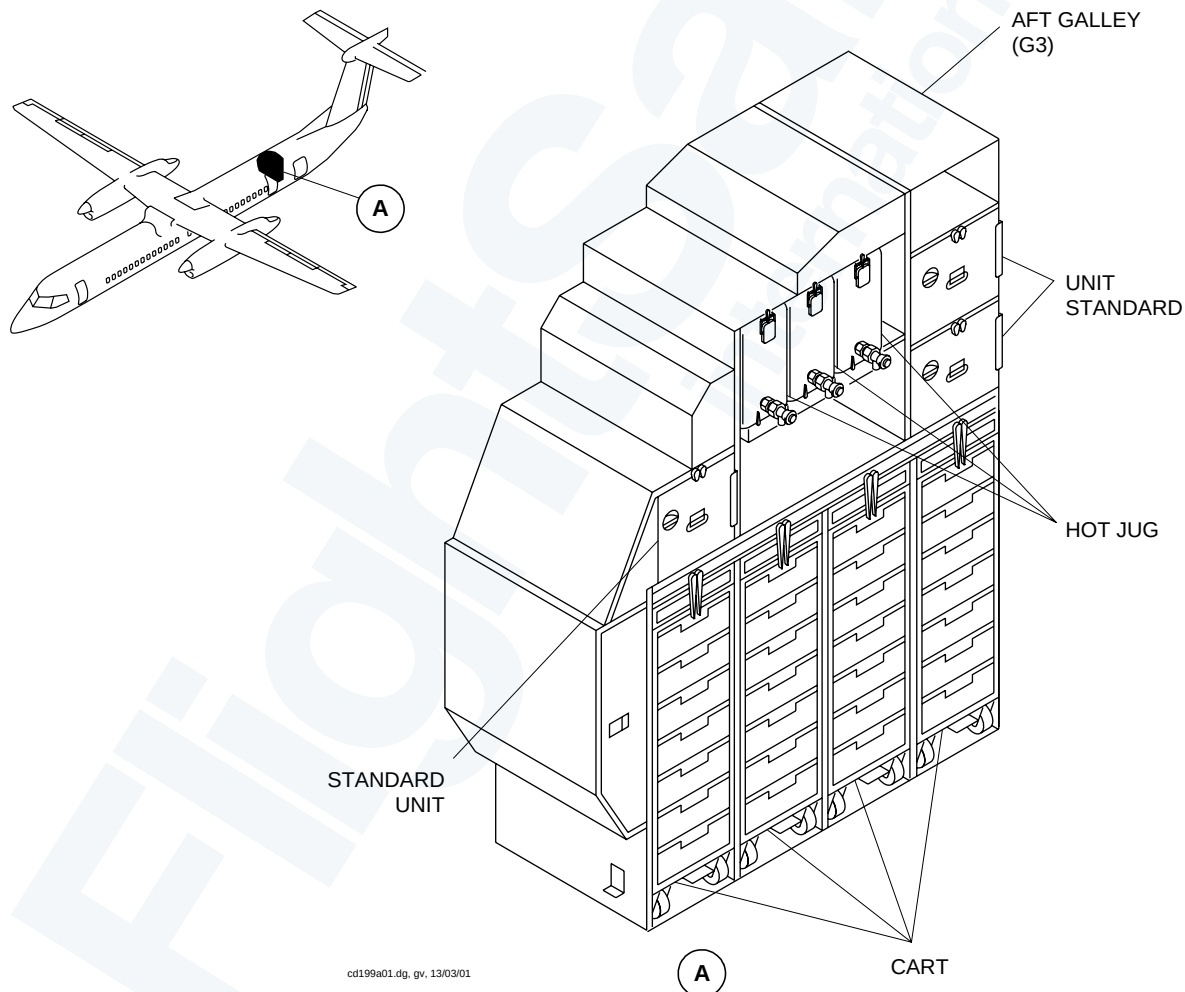
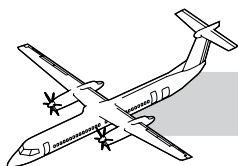


Figure 25-30. G3 Galley Assembly

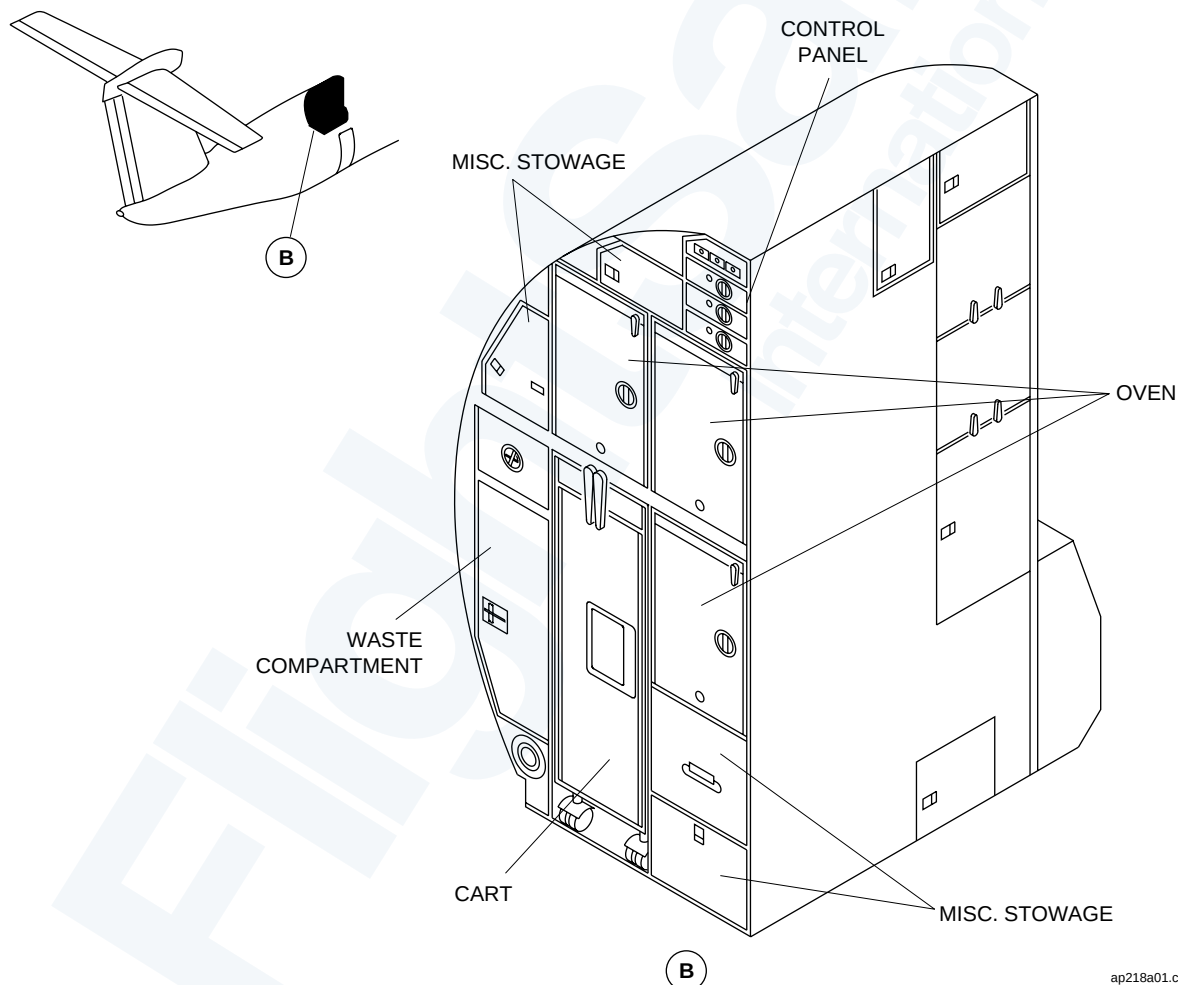


Aft Galley (G4)

Refer to:

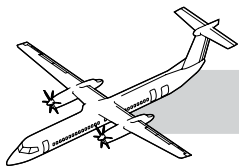
- Figure 25-29. G1/G2/G3/G4/G6 Galley Configuration.
- Figure 25-31. G4 Galley Assembly.

The G4 galley is located on the left side, forward of the aft passenger door. It contains one removable waste container, miscellaneous stowage, three ovens and one cart. An optional G4 galley has one removable waste container, miscellaneous stowage, three standard service units and two carts. The ovens are powered by the DC electrical system. The cart is retained in the galley by 1/4 turn fasteners.



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Figure 25-31. G4 Galley Assembly

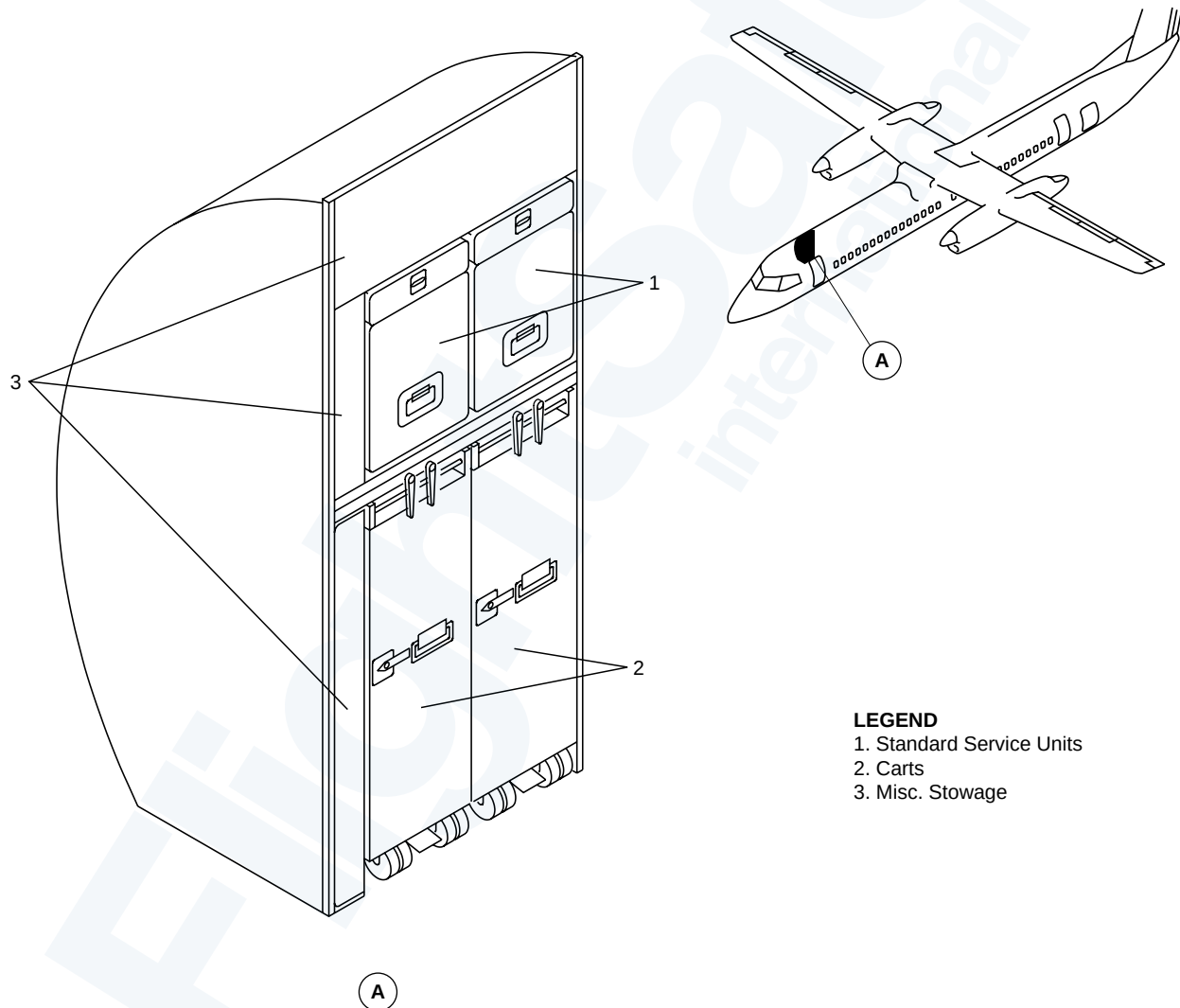


Aft Galley (G6)

Refer to:

- Figure 25-29. G1/G2/G3/G4/G6 Galley Configuration.
- Figure 25-32. G6 FWD Galley.

The G6 galley is located on the right side, forward of the emergency door. It contains two carts and two stowage containers. The cart is retained in the galley by 1/4 turn fasteners.

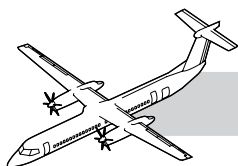


LEGEND

- 1. Standard Service Units
- 2. Carts
- 3. Misc. Stowage

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Figure 25-32. G6 FWD Galley



CONTROLS AND INDICATIONS

DC Power Hot Jug

Refer to Figure 25-33. DC Power Hot Jug Panel.

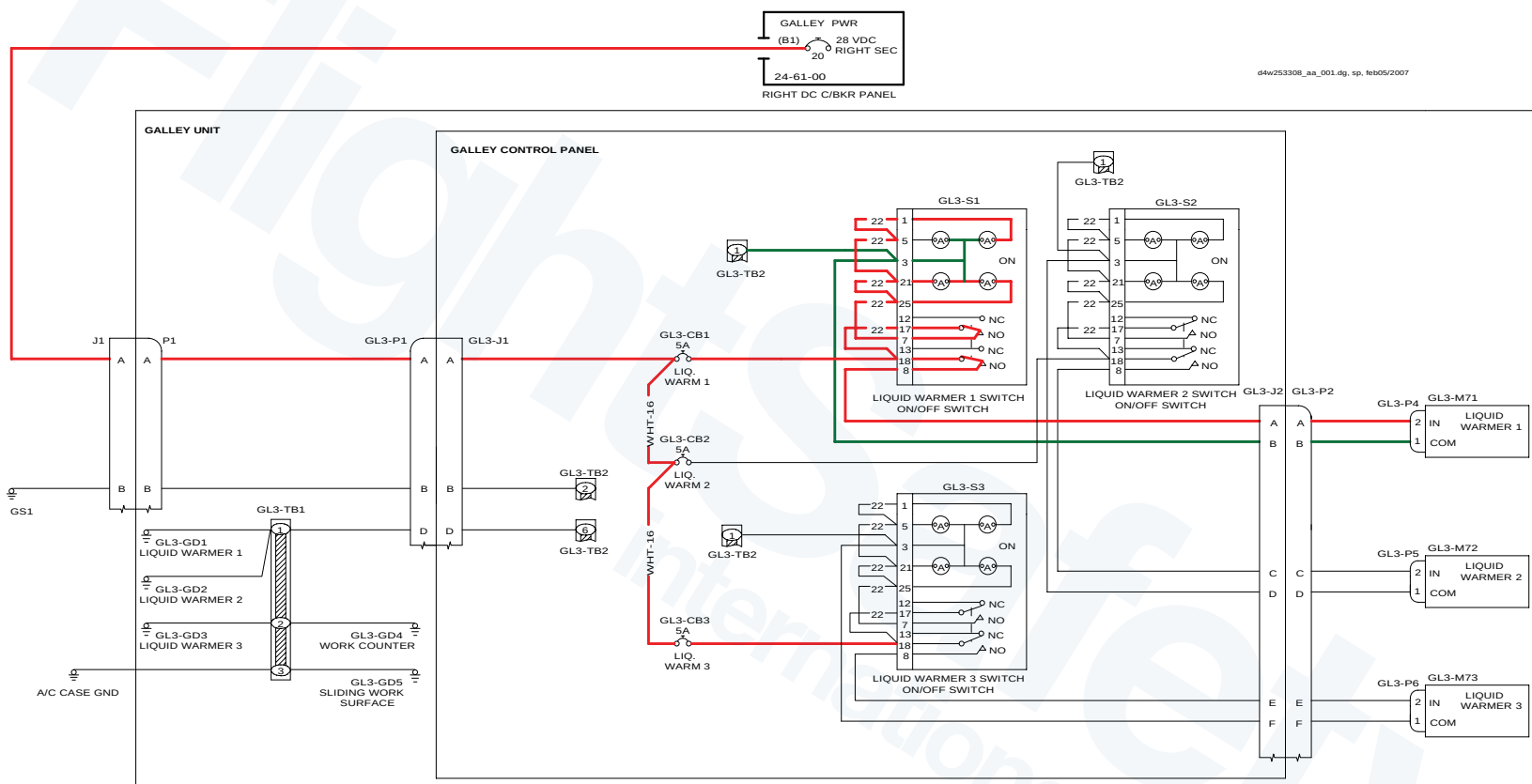
Consists of three liquid warmer switches located on the galley control panel.

The power source C/B for the Hot Jug are located just above the heater switches on the panel.

When selected “ON” the Liquid warmer switches come ON.



Figure 25-33. DC Power Hot Jug Panel



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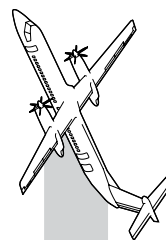
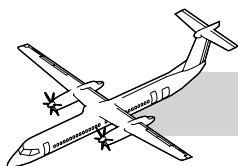


Figure 25-34. Hot Jug Operation Schematic



OPERATION

NOTES

Refer to Figure 25-34. Hot Jug Operation Schematic.

Power from the right hand secondary bus through the galley power circuit breaker is applied to the liquid warmer switches 1, 2 and 3.

Upon selection on any of the switchlight to the ON position 28 VDC through the switch is applied to its respective liquid warmer connector.

When the warmer switchlight is selected ON, 28 VDC is applied to the internal lighting turning the switchlight ON.

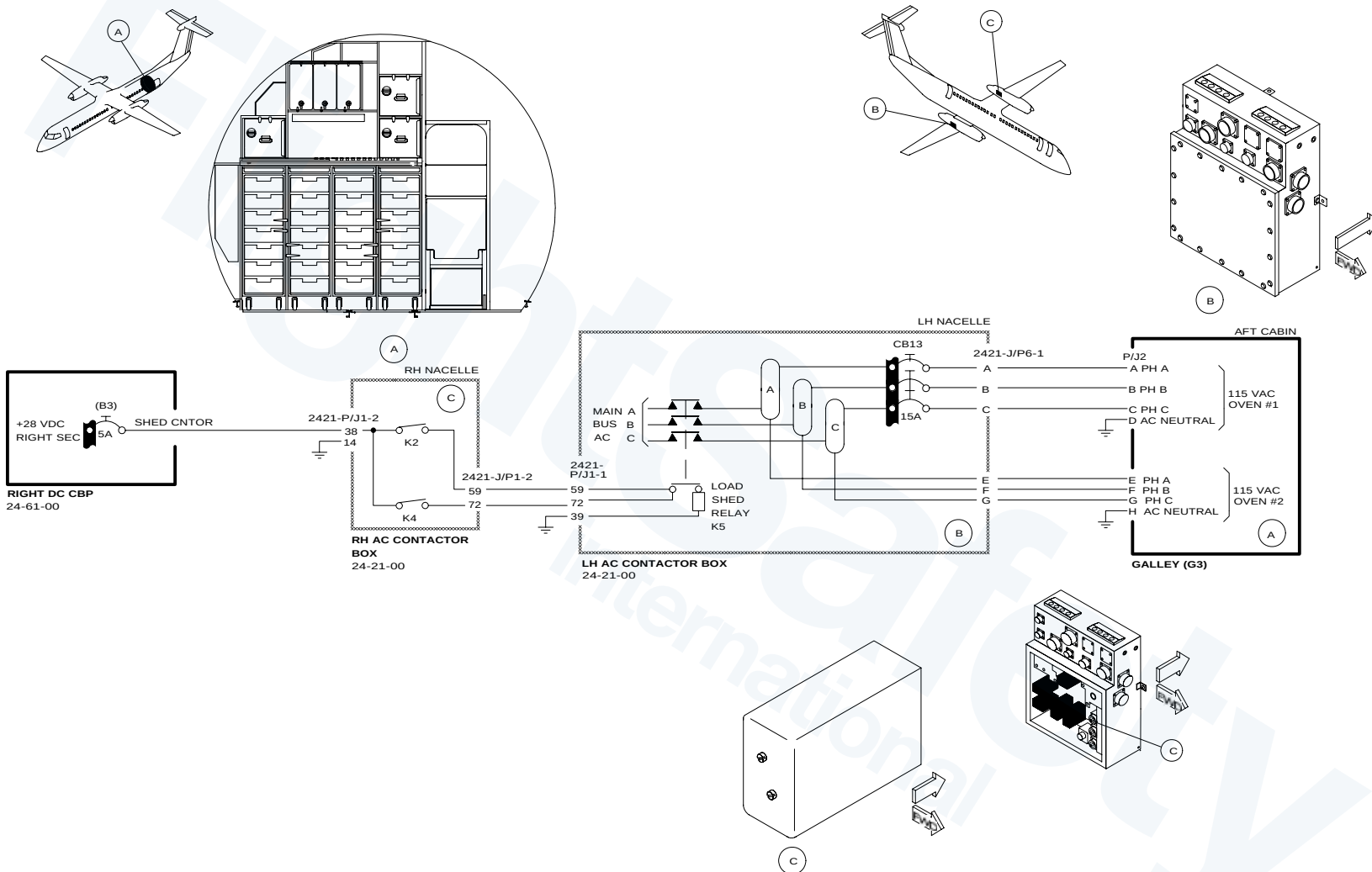
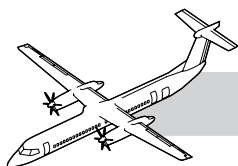


Figure 25-35. Mod G3 Galley with Dual Ovens



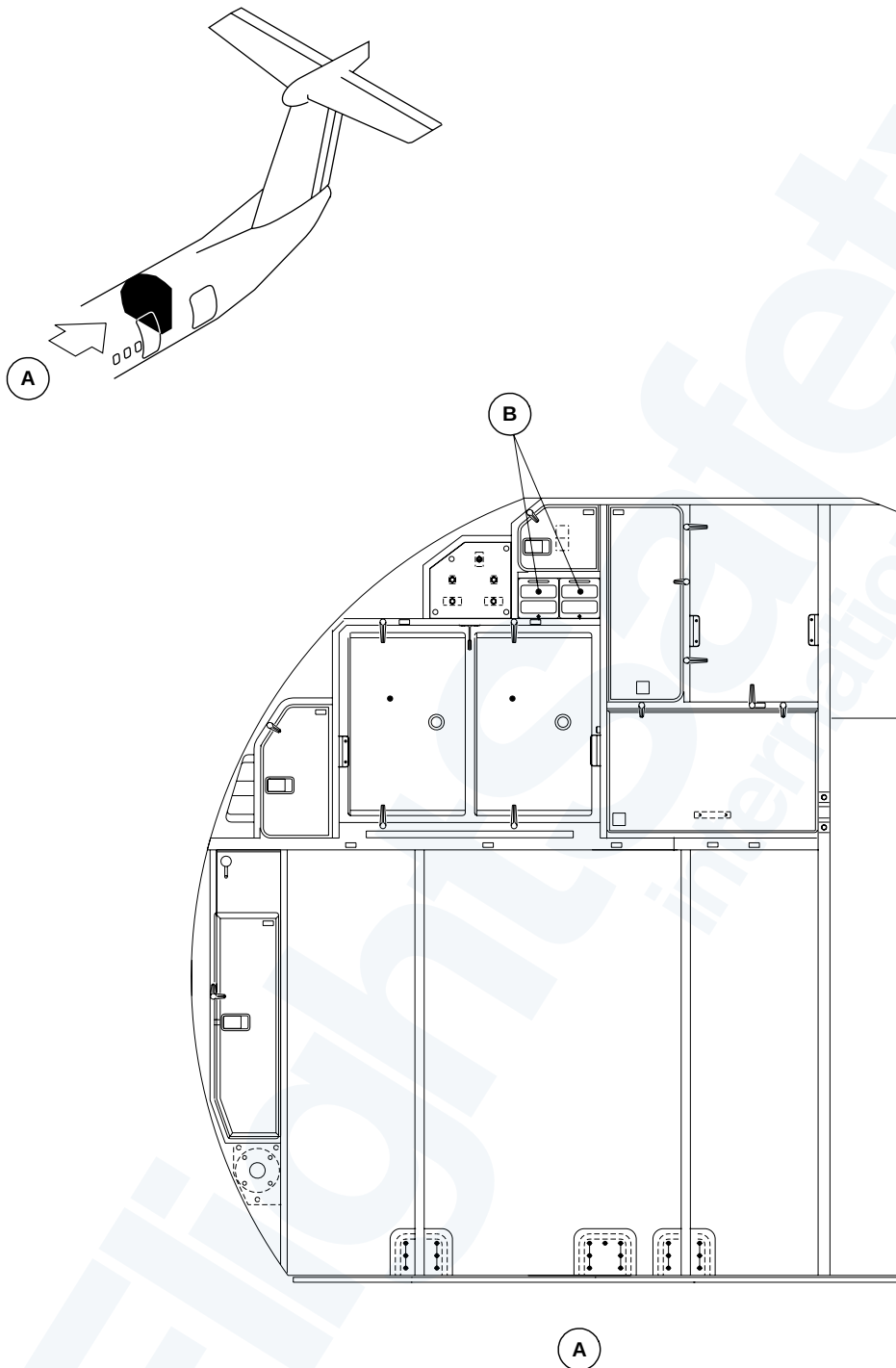
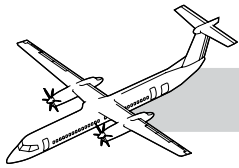
G3 Galley Oven Power Source

NOTES

Refer to Figure 25-35. Mod G3 Galley with Dual Ovens.

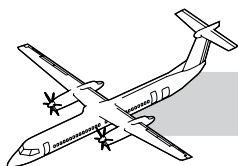
The power for the two galley ovens is provided by the AC variable bus.

In the event of a loss of an AC power source, the load shedding relay K5 will de-activate removing the AC power from the galley.



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Figure 25-36. Galley Oven



G3 Galley with Ovens

General

Refer to:

- Figure 25-36. Galley Oven.
- Figure 25-37. Galley Oven (Controls and Indications).

The G3 galley is also available with 2 ovens to allow the service of hot meals to the passengers while in flight.

System Description

The ovens are operated using an Oven Control Module (OCM). When you push the ON/OFF switch, the ON LED illuminates, the TEMP MED LED illuminates (approximately 180°C (356°F), the HEATING TIME and the SERVING TIME displays show 00.

Turn the TIME SELECTOR Knob to choose how long (up to 99 minutes) you want to heat the food. The HEATING TIME and the SERVING TIME displays show the time in minutes. Push the HEATING TIME SET switch to set the time and the SET LED illuminates.

NOTE

If the serving time is more than two minutes longer than heating time, the fan operates for 30 seconds. The OCM monitors the time and starts the operation of the oven when the serving time equals the heating time.

You can also choose a lower (approximately 140°C (284°F)) or higher (approximately 220°C (428°F)) temperature. You must push the TEMP switch and the LOW or HIGH LED illuminates.

After you have made the selections, push the START switch. The START LED illuminates and the sensing cycle begins. This takes approximately 30 seconds.

NOTE

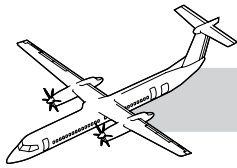
The oven will not start unless you have set at least the heating time. It is not a necessity to set the serving time.

When the heating time only is set, the operating cycle continues to heat the food. When a heating time and a serving time are set, the sensing cycle stops and the serving-time count-down begins. You can see this on the SERVING TIME display. When the serving time equals the heating time, the operation cycle starts and heats the food.

It is possible, after the start, to see how hot the food is at that moment. Press the TEMP switch and the two displays combined will show the temperature, first in °C and thereafter in °F.

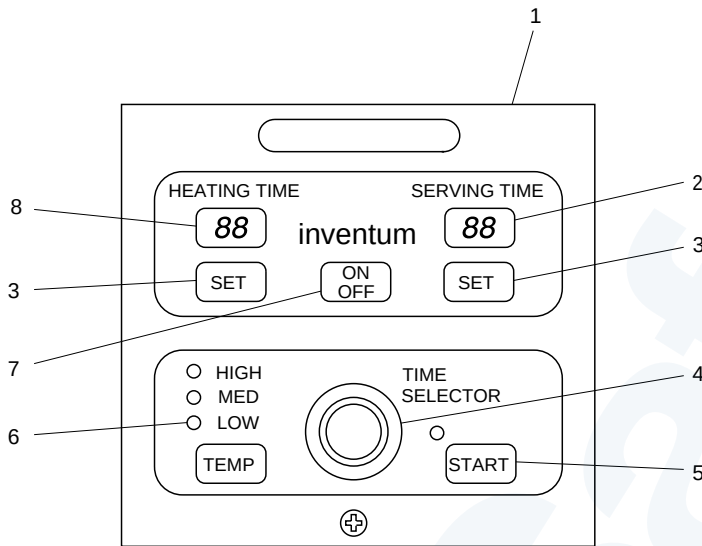
NOTE

If the food is chilled between -2°C and +2°C (28.4°F and 35.6°F), the heating time is automatically increased 25%. If the food is frozen below -2°C (28.4°F), the heating time is automatically increased 50%.



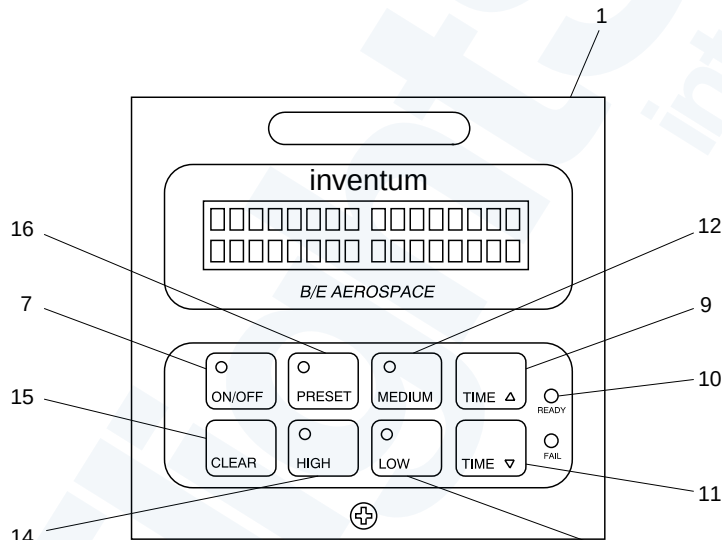
LEGEND

1. Control Panel
2. SERVING TIME Indicator
3. SET Switch
4. TIME Selector Knob
5. START Switch
6. LOW Indication Light
7. ON/OFF Switch Light
8. HEATING TIME Indicator
9. TIME Increase Switch Light
10. READY Indication Light
11. TIME Decrease Switch Light
12. MEDIUM Switch Light
13. LOW Switch Light
14. HIGH Switch Light
15. CLEAR Switch Light
16. PRESET Switch Light



(B)

WITHOUT MODSUM 4-900865

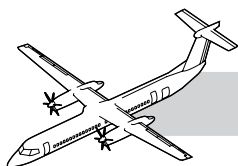


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Figure 25-37. Galley Oven (Controls and Indications)



25-40-00 LAVATORY

GENERAL

Refer to:

- Figure 25-38. Forward Lavatory.
- Figure 25-39. Lavatory Door and Aft Wall.

The lavatory compartment is installed on the right forward side of the passenger compartment, aft of the flight compartment.

The lavatory compartment contains these items:

- Externally serviced toilet unit
- Wall mirror
- Ashtray (Optional)
- Two towel dispensers
- Wet wipe dispenser
- Toilet paper dispenser
- Waste disposal
- Smoke detector
- Fire Extinguisher.

On aircraft with the assist handle (825CH01242): An assist handle in the lavatory is attached to a panel above the toilet roll holder to help elderly and disabled passengers.

The lavatory compartment has an entrance door. A LAVATORY OCCUPIED sign is installed in the passenger compartment. When the lavatory compartment door is locked, an OCCUPIED sign on the door and the LAVATORY OCCUPIED sign in the passenger compartment come on.

Any announcements made over the Passenger Address (PA) system will be heard from the PA speaker installed in the ceiling. A lavatory compartment RETURN TO SEAT sign and an attendant call button are installed on the upper amenity unit.

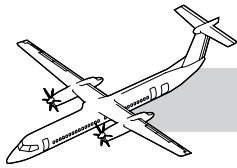
The lavatory compartment RETURN TO SEAT sign illuminates with the passenger compartment fasten seat belt signs. The lavatory call button can be used if assistance is necessary.

The air conditioning system supplies conditioned air to the lavatory compartment through a gasper outlet in the upper amenity unit (refer to ATA 21-00-00).

A metal waste disposal container, with a door assembly spring-loaded to the closed position, is designed to contain fires. A fire extinguisher automatically operates when the temperature in the waste disposal container reaches a set point. If there is smoke from a fire or other source in the lavatory compartment, the smoke detector will cause an audible tone to sound and the repeater lights in the passenger compartment ceiling flash (ATA 26-00-00).

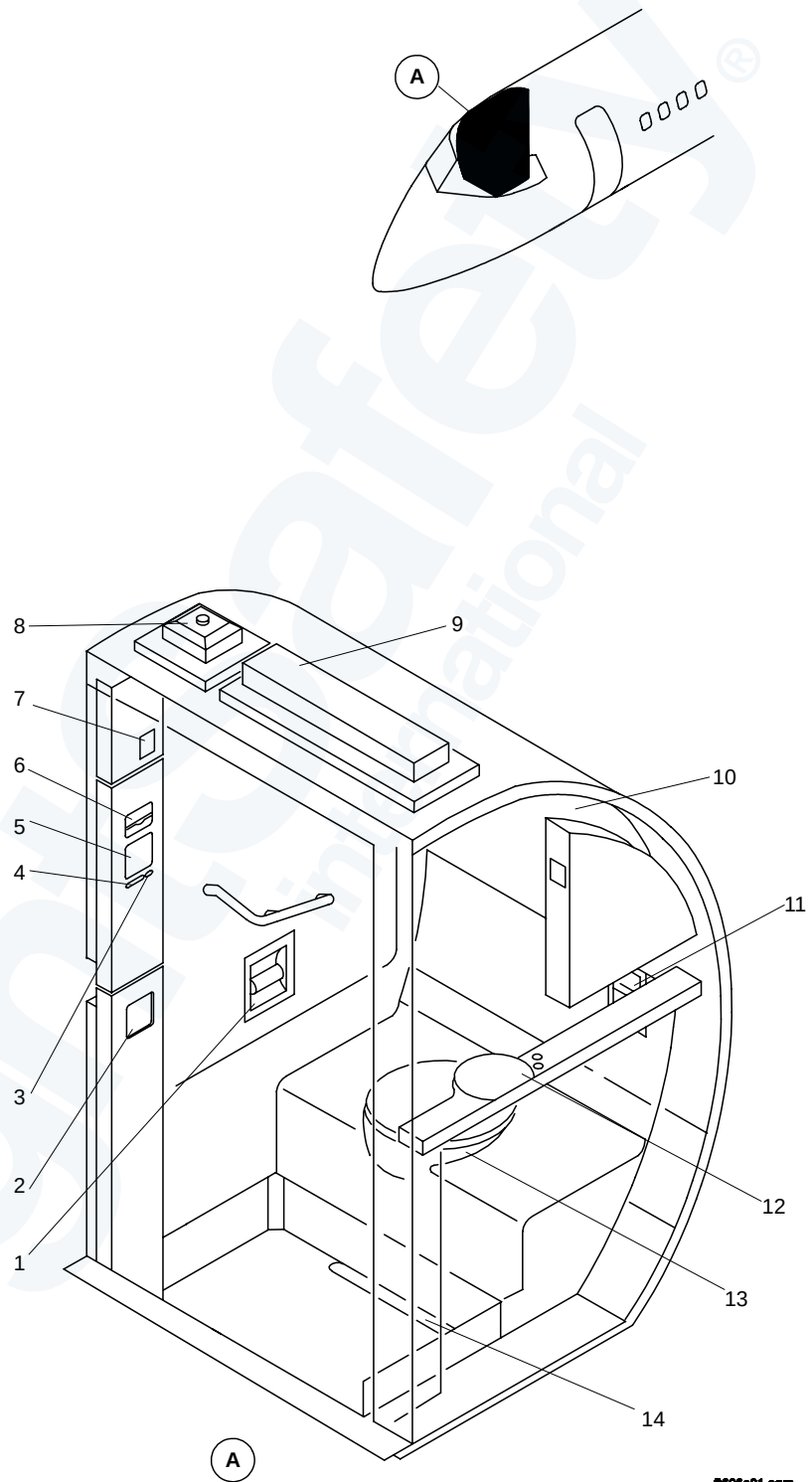
Lighting of the Lavatory is covered in ATA 33-00-00.

Operation of the water and waste is covered in ATA 38-00-00.



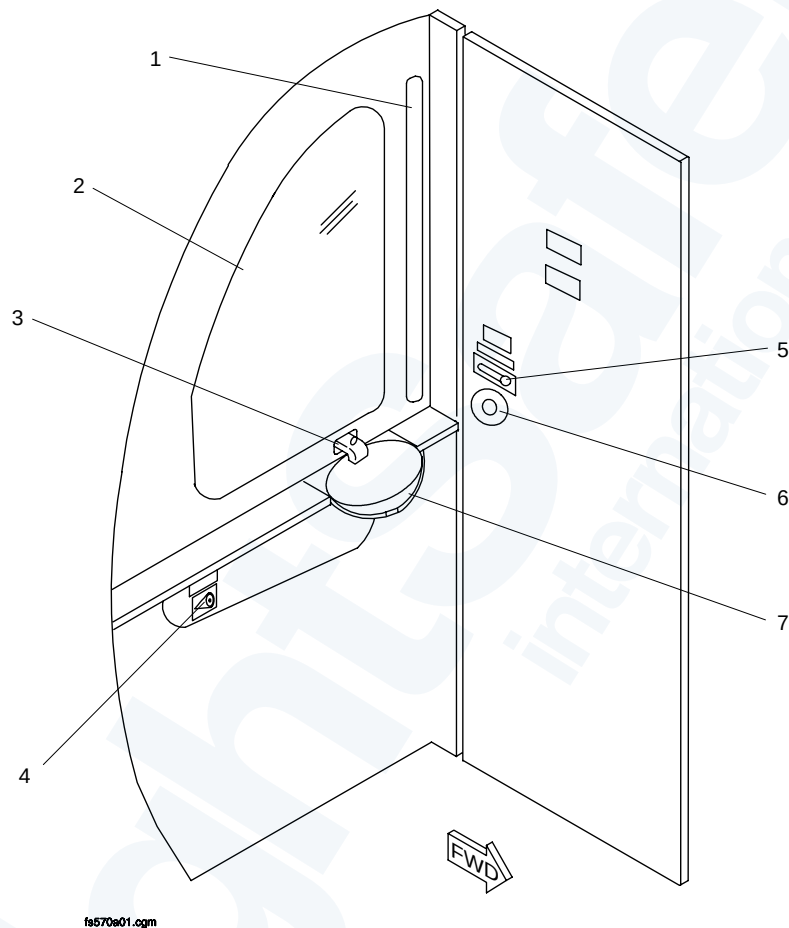
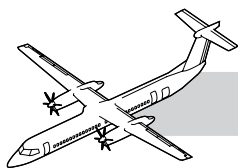
LEGEND

1. Toilet Paper Dispenser
2. Waste Disposal
3. Flight Attendant Call Button
4. Return to Seat Sign
5. Sanity Napkin
6. Wet Wipe Dispenser
7. Gasper Outlet
8. Smoke Detector
9. Overhead Light
10. Mirror (Rear Wall)
11. Ash Try
12. Sink
13. Toilet Unit
14. Floor Drain.



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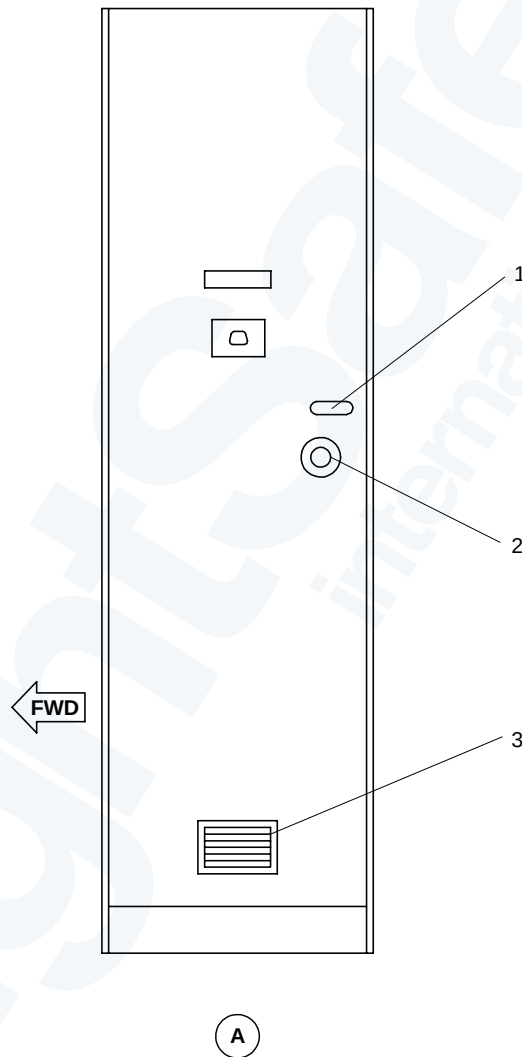
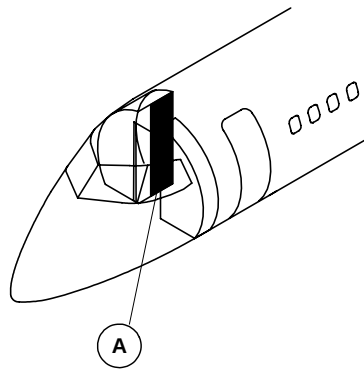
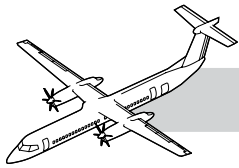
Figure 25-38. Forward Lavatory



LEGEND

- 1. Light Unit
- 2. Wall Mirror
- 3. Faucet
- 4. Push to Flush Button
- 5. Dual Action Lock
- 6. Internal Handle
- 7. Sink.

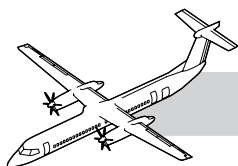
Figure 25-39. Lavatory Door and Aft Wall



- LEGEND**
- 1. OCCUPIED Sign
 - 2. External Door Handle
 - 3. Grille.

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Figure 25-40. Lavatory Entry Door



COMPONENT DESCRIPTION

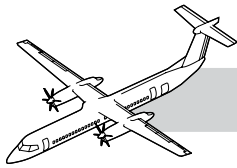
NOTES

Lavatory Entry Door

Refer to Figure 25-40. Lavatory Entry Door.

The lavatory compartment has an entrance door with an external handle, an internal handle and dual action lock. If necessary, the door can be unlocked from outside the compartment. An opaque OCCUPIED sign is installed on the door exterior and a LAVATORY OCCUPIED sign is installed in the passenger compartment.

The lavatory door gives access to the lavatory and has a kick strip on both sides. The door is held in the closed position by a latch, which is unlatched with the door handle. Grilles on both sides of the door let air move between the lavatory and the passenger compartments.



Refer to Figure 25-41. Lavatory Interior.

Toilet Shroud

The toilet shroud covers the toilet tank and is sealed to the lavatory compartment structure during assembly.

Upper Amenity Assembly

The upper amenity assembly contains a gasper to supply conditioned air, dispensers for wet wipe towels and napkins, a RETURN TO SEAT sign and a call button.

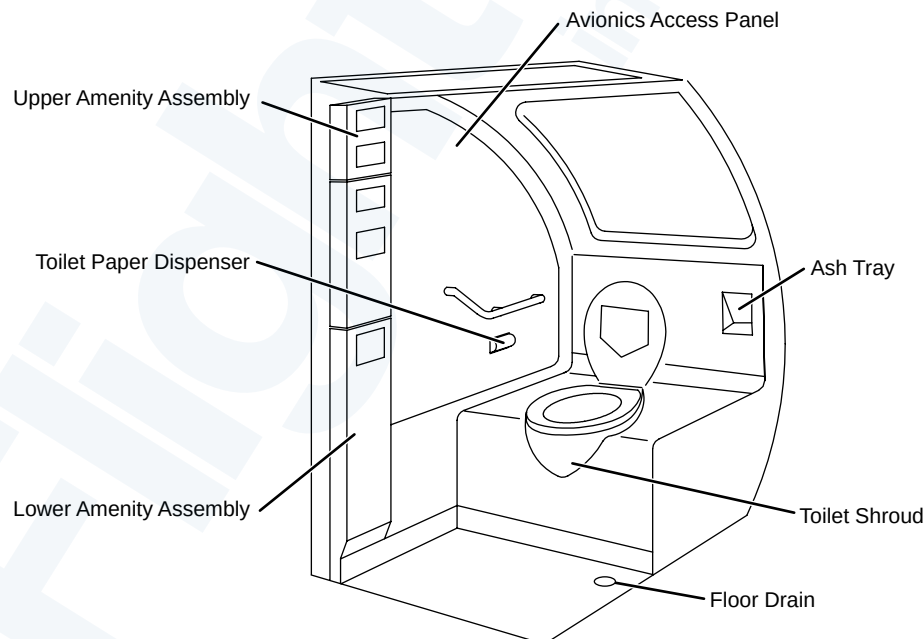
Lower Amenity Assembly

The lower amenity assembly has doors that enclose a towel box and a waste disposal container. The lavatory waste disposal fire extinguisher is installed between the towel box and waste container. The extinguisher discharge tube passes through a shelf just below the fire extinguisher bottle and is directed towards the waste container. Any space between the tube and the shelf is sealed with silicon compound.

The main panel of the assembly is hinged and held closed with latches so that it can be opened for servicing. This panel also has silicon seals on the bottom and lower vertical edges for waste disposal fire containment. The waste disposal door is spring-loaded to the closed position and has flanges which are held against the main panel when the door is in the closed position.

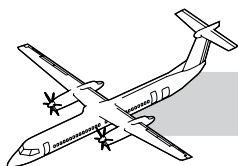
Avionics Access Panel

A part of the forward lavatory compartment wall has an avionics access panel for the rack, immediately aft of the co-pilot's seat. This panel has a support handle which can be used by lavatory compartment occupants. A toilet paper dispenser is also installed in a recess in this panel.



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Figure 25-41. Lavatory Interior



Lavatory Floor Pan

NOTES

The lavatory floor pan is attached to the toilet assembly above the floor. Any crevices are filled with silicon sealant.

Service Access Door

A service door is installed below the sink, or counter top in the “Dry Lav” configuration. In the “Dry Lav” configuration only the aft face has two containers attached to it which can be used for servicing purposes.

Floor Drain

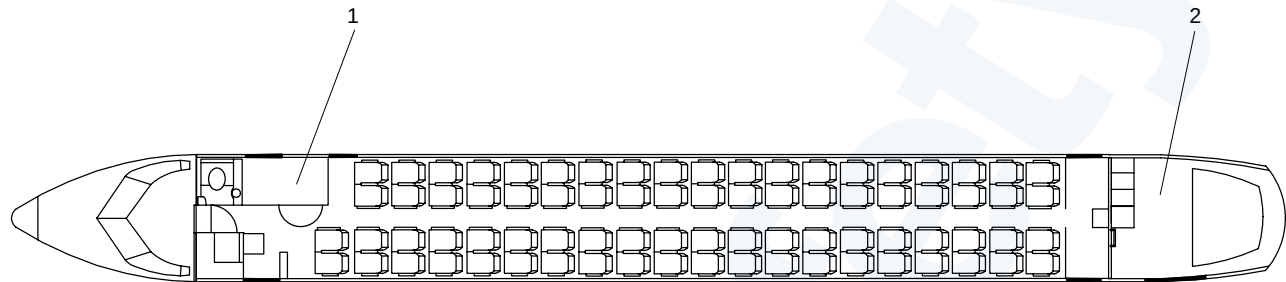
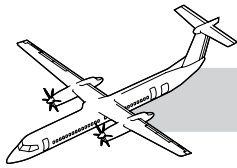
A drain installed on the floor allows fluid spilled on the floor to be directed overboard through the FWD drain mast.

MAINTENANCE PRACTICES

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Detailed Visual Inspection of the Lavatory Waste Compartment, Waste Compartment Door and Sealing at the Lower Amenity Panel

Give a detailed visual inspection of the lavatory waste compartment, waste compartment door, and sealing at the lower amenity panel.

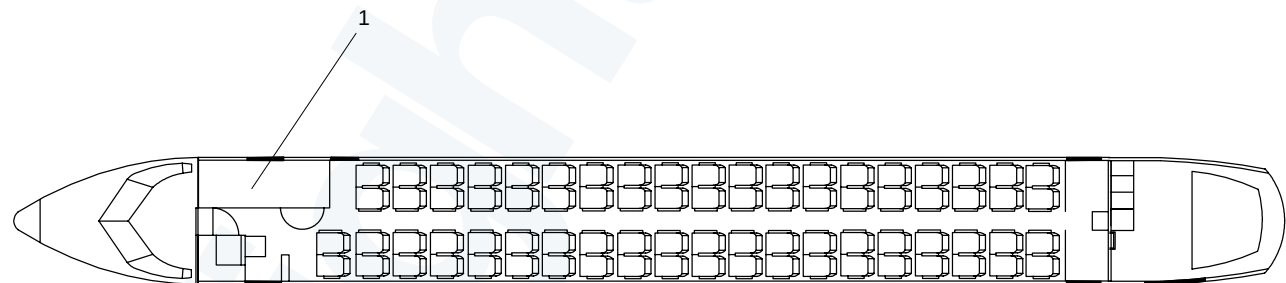


LEGEND

- 1. Forward Baggage Compartment
- 2. Aft Baggage Compartment.

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Figure 25-42. Forward Baggage Compartment

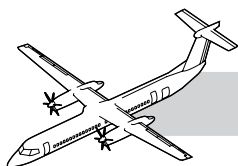


LEGEND

- 1. Forward Baggage Compartment with Lavatory Removed.

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Figure 25-43. Forward Baggage Compartment (Lavatory Removed)



25-50-00 BAGGAGE COMPARTMENTS

INTRODUCTION

The forward and aft baggage compartments have sufficient space for the safe storage of the passengers' baggage and cargo.

GENERAL

The forward and aft baggage compartments are Class C compartments.

They have the components and properties that follow:

- Smoke detectors, one in the forward baggage compartment and two in the aft baggage compartment (refer to ATA 26)
- A built-in fire extinguishing system which is controllable by the crew (refer to ATA 26)
- Sufficient sealing to ensure minimum leakage of smoke and fire extinguishing agent into the passenger compartment.

The aft compartment has dedicated air supply. It has an inflow valve and an outflow valve, which are automatically closed when fire or smoke is detected (ATA 21 and ATA 26).

Baggage loading for both compartments is accomplished through the related external doors. The forward compartment also has an internal door which allows access from the passenger compartment (refer to ATA 52). There is no internal access to the aft baggage compartment.

COMPONENT DESCRIPTION

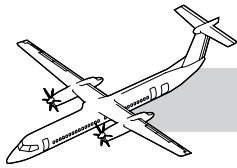
Forward Baggage Compartment

Refer to:

- Figure 25-42. Forward Baggage Compartment.
- Figure 25-43. Forward Baggage Compartment (Lavatory Removed).

The forward baggage compartment is located on the right side of the aircraft in front of the forward emergency exit. For the baseline interior configuration, the baggage compartment is located between the compartment rear bulkhead (Stn. 70) and the aft wall of the lavatory. This configuration gives the forward baggage compartment a volume of approximately 91 ft³ (2.58 m³).

If the lavatory is not installed, the compartment extends forward to the flight compartment bulkhead (Stn.-33) for a total volume of 135 ft³ (3.82 m³).



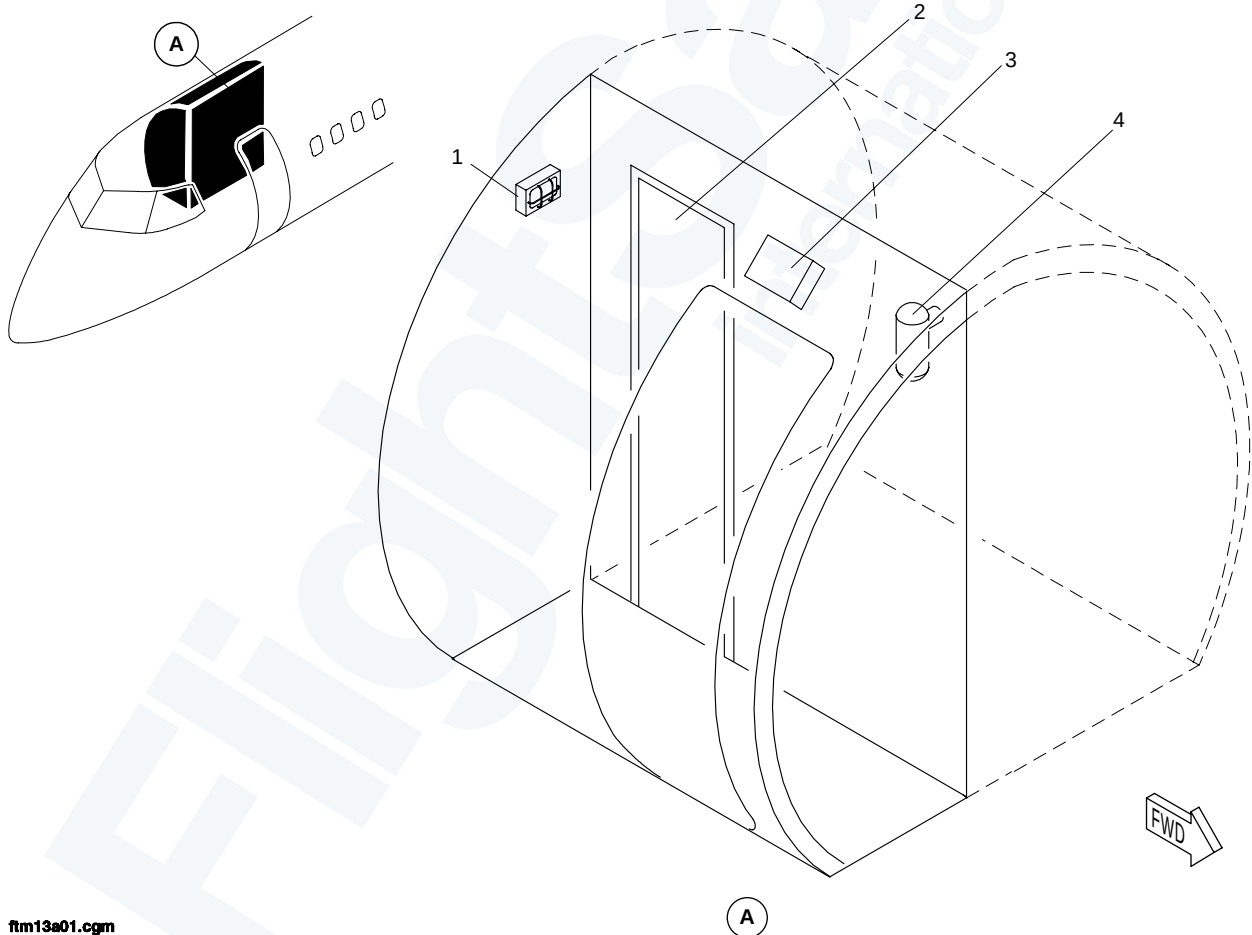
Refer to Figure 25-44. Forward Baggage Compartment - Detail.

If the aircraft is equipped with a G6 galley (refer to ATA 25-30), the compartment rear bulkhead is moved forward to Stn. 39 to accommodate the galley. This configuration gives the forward baggage compartment a volume of 51 ft³ (1.44 m³).

The passengers' baggage is loaded into the forward baggage compartment through the exterior forward baggage door. Door position information is displayed on the doors page on the Multi Function Display (MFD) (refer to ATA 52).

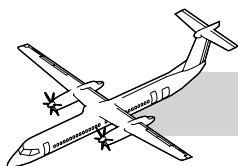
The smoke detector is protected by a metal grille. Fire extinguishing for the forward baggage compartment is supplied by a High Rate Discharge (HRD) extinguisher bottle installed in the compartment forward roof area. The forward baggage compartment also shares a Low Rate Discharge (LRD) extinguisher bottle with the aft baggage compartment (refer to ATA 26).

The forward baggage compartment has a light, with protection from possible fire and from close contact with baggage. The light illuminates when the forward baggage compartment external door is unlocked. The light also illuminates when the interior door is opened.



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Figure 25-44. Forward Baggage Compartment - Detail



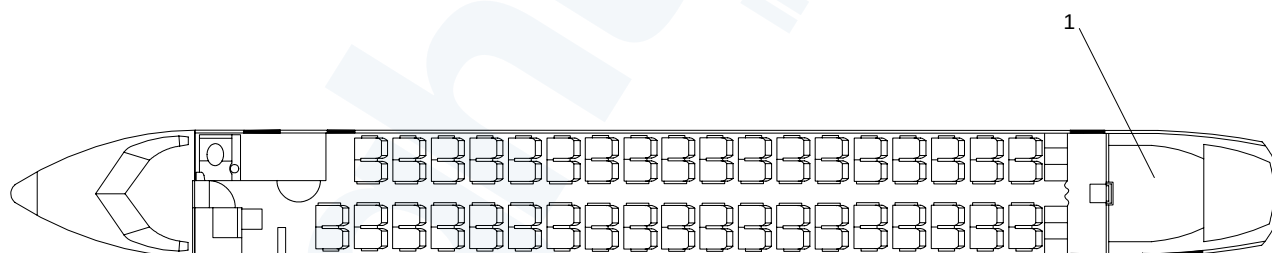
Aft Baggage Compartment

Refer to Figure 25-45. Aft Baggage Compartment.

The aft baggage compartment is located at the rear of the pressurized part of the fuselage immediately forward of the aft pressure dome. The forward bulkhead of the compartment is located at Stn. 701 and the aft (canted) bulkhead is located at Stn. 775. The total baggage compartment volume is 411 ft³ (11.64 m³). If the G3 galley is installed (refer to 25-30-00), the total volume is reduced to 365 ft³ (10.33 m³).

The floor panels are a sandwich construction with unidirectional S-glass/epoxy surfaces and an aramid honeycomb core. The floor panels from the forward bulkhead to the step have a load capacity of 125 lb/ft² (6.0 kN/m²). The panels from the step to the aft (canted) bulkhead have a load capacity of 75 lb/ft² (3.6 kN/m²).

The aft bulkhead of the passenger compartment serves as a divider between the passenger compartment and the baggage compartment. It has a service access panel and decompression blow out grilles, located on the left side of the aft baggage bulkhead. This bulkhead is designed to restrain up to 3,500 lb (1587.6 kg) of loose baggage.

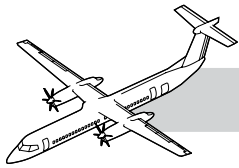


LEGEND

1. Aft Baggage Compartment.

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Figure 25-45. Aft Baggage Compartment



Canted Bulkhead

Refer to Figure 25-46. Canted Bulkhead.

The canted bulkhead (rear wall of the aft baggage compartment) is located at Stn. 775. It is made from three detachable panels which are attached to the fuselage. The panels are made from regular aramid fiber.

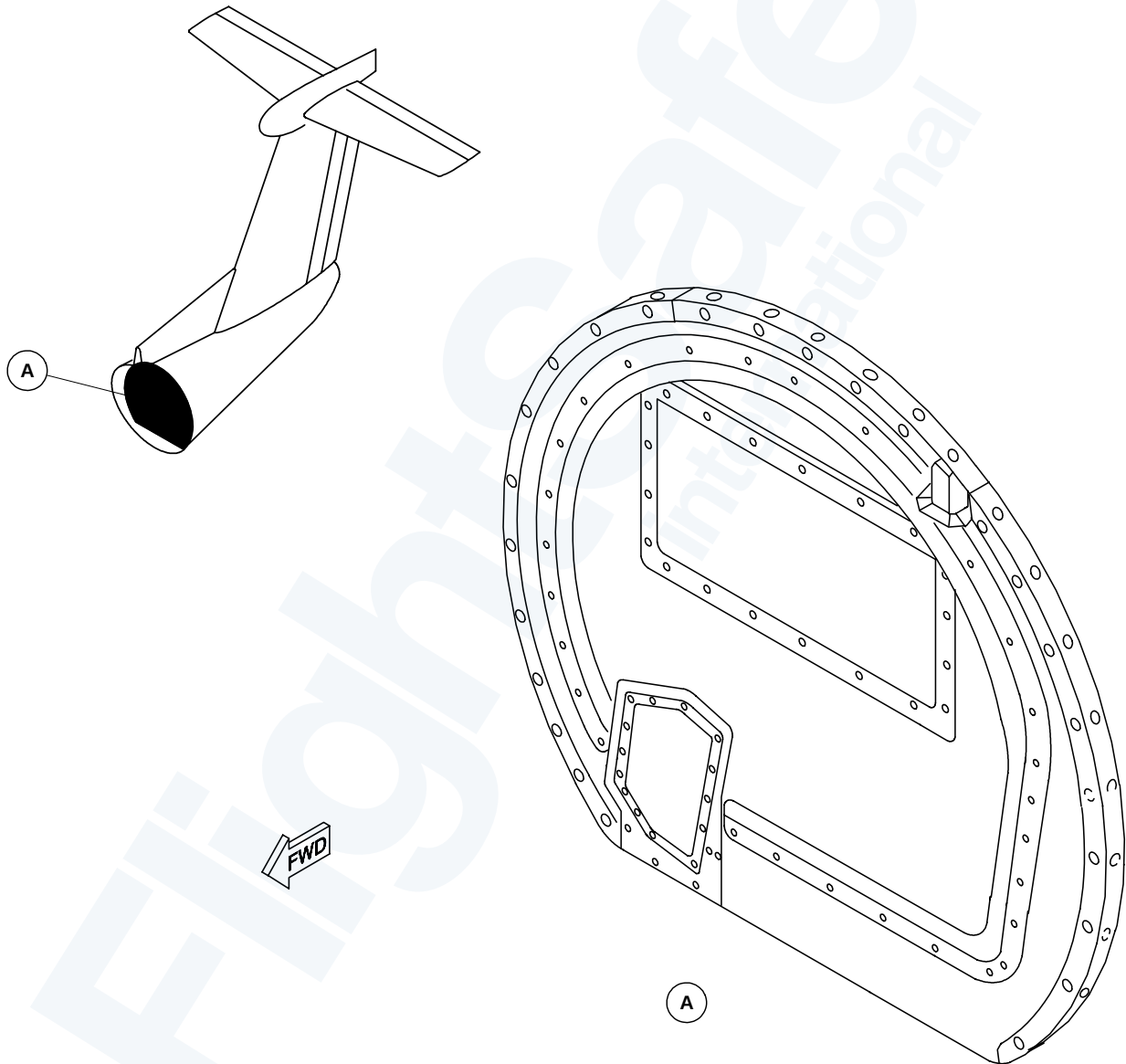
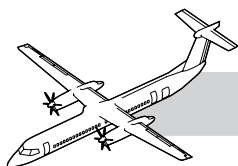


Figure 25-46. Canted Bulkhead



G3 Galley Bulkhead

Refer to Figure 25-47. G3 Galley Bulkhead/Aft Baggage Compartment.

The G3 galley bulkhead (forward wall of the aft baggage compartment) is located at Stn. 701. The bulkhead is made from regular aramid fiber panels. At floor level, the panels are attached to the inboard and outboard seat tracks by fitting pins. At the top, the panels are attached to a bearing by spigots. The bearing is installed in an adjustable housing which is attached to the fuselage structure.

G3 Galley Bulkhead Blowout Panel

A blowout (decompression) panel is installed in the G3 galley bulkhead to the left of the flight attendant's seat. The panel is designed to detach into the baggage compartment, if the pressure differential between passenger and baggage compartment becomes too great. The panel will be caught by a containment cage if the panel is detached.

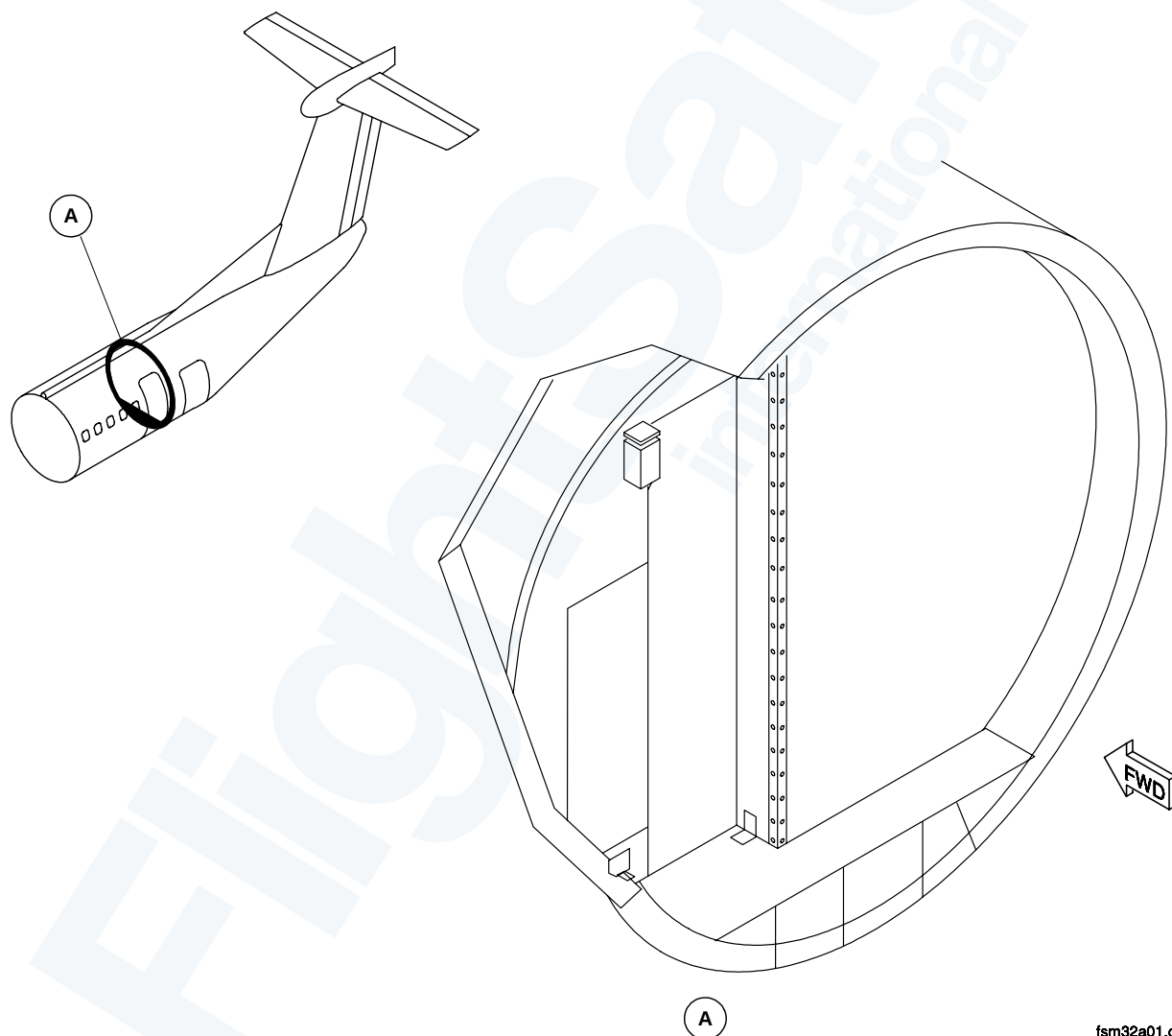
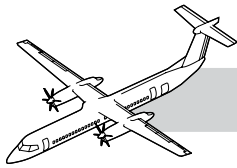


Figure 25-47. G3 Galley Bulkhead/Aft Baggage Compartment



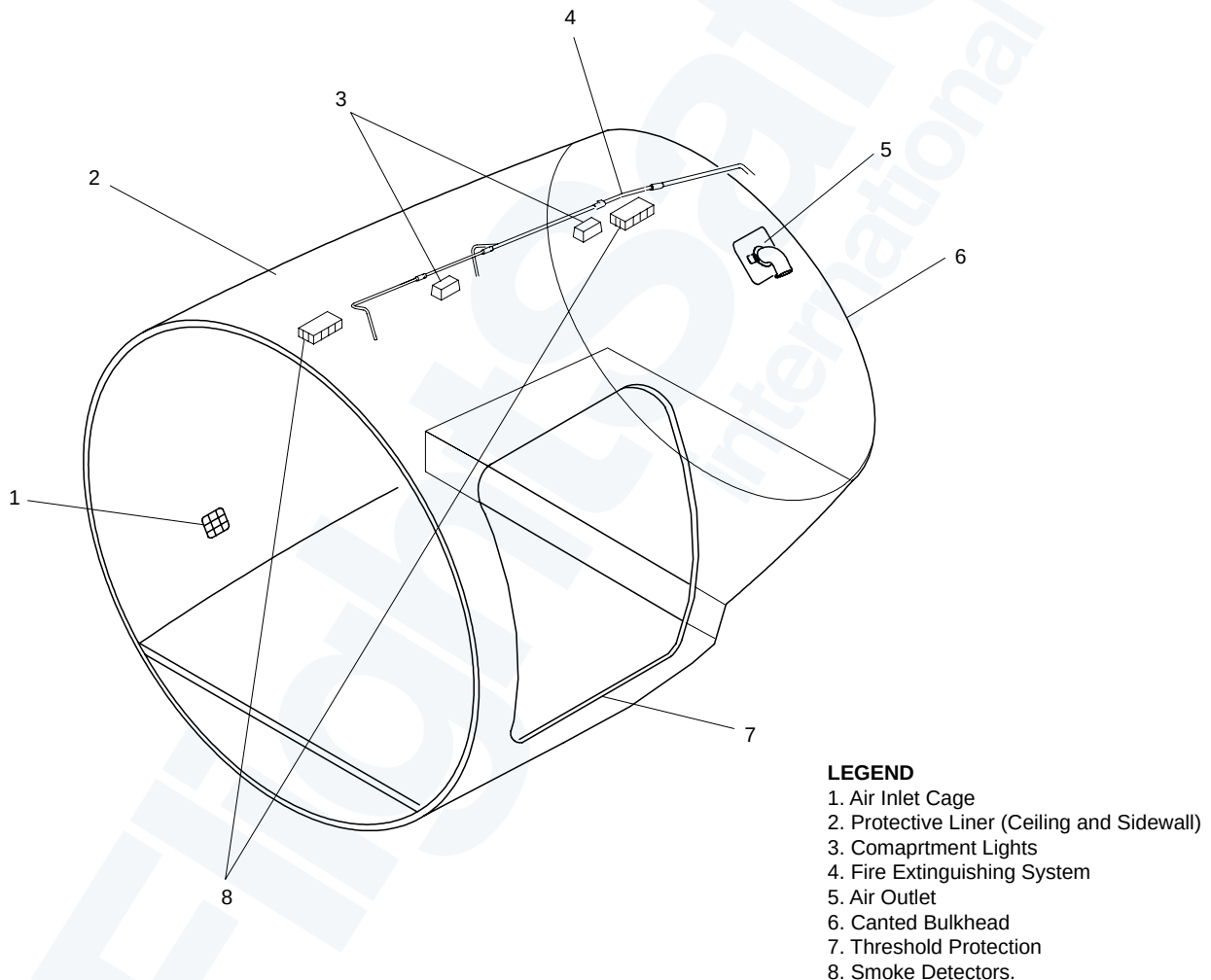
Aft Baggage Compartment

Refer to Figure 25-48. Aft Baggage Compartment - Detail.

The aft baggage compartment is provided with two lamps, a smoke detector and a built-in fire extinguishing system. The lamps are recessed. The smoke detectors, as well as the lamps, are protected from damage by metal grilles.

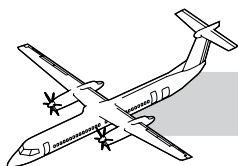
Aft Baggage Compartment Liners and Surrounds

The compartment liner consists of fiberglass/phenolic resin laminate. The edges of the panels at the floor line are metal reinforced to prevent damage to the panels from the baggage. An easily removed covering panel protects the aft pressure bulkhead. A large label, indicating the loading limit and load distribution, is attached to the left side of the aft face of the passenger cabin/cargo compartment bulkhead.

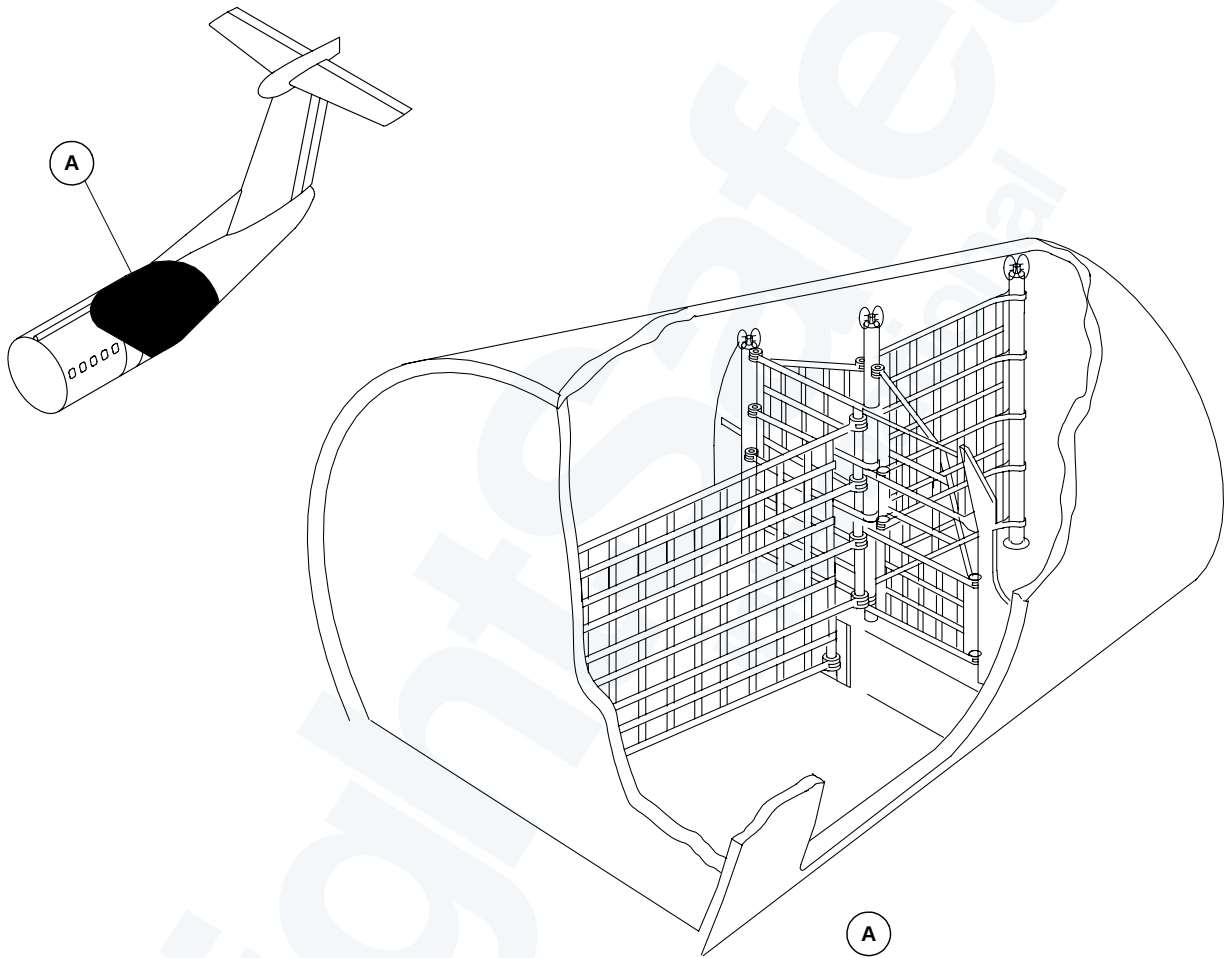
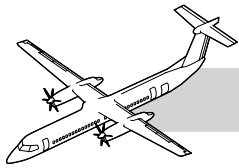


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Figure 25-48. Aft Baggage Compartment - Detail

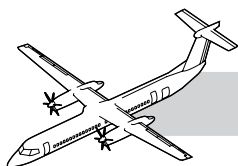


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Figure 25-49. Baggage Divider - Detail



Baggage Restraint

NOTES

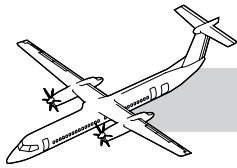
Refer to Figure 25-49. Baggage Divider - Detail.

The baggage restraint system has tie-down rings, spring-loaded tubes, an upper and lower net, and a threshold protector. It is designed to restrain the baggage in flight and to protect the threshold during baggage loading and unloading.

The nets cover the area of the aft baggage compartment door. The lower net is attached by clips to the threshold protector at the bottom and to a spring-loaded drawbolt tube at the top. The upper net is attached to the airframe structure at the top and to a short tube at the bottom.

When the baggage compartment has been loaded, Ty-clip fasteners attach the short tube to the drawbolt tube which is attached to the airframe structure. The threshold protector is raised up and hangs from the drawbolt.

The threshold protector is a heavy gauge metal plate which is attached longitudinally to the baggage compartment floor by a hinged bracket. Protective strips and bumper pads are attached to the plate. An aluminum 90° extrusion is attached to the edge of the plate opposite the bracket. The drawbolt tube and lower net are suspended from this extrusion during the baggage loading process.

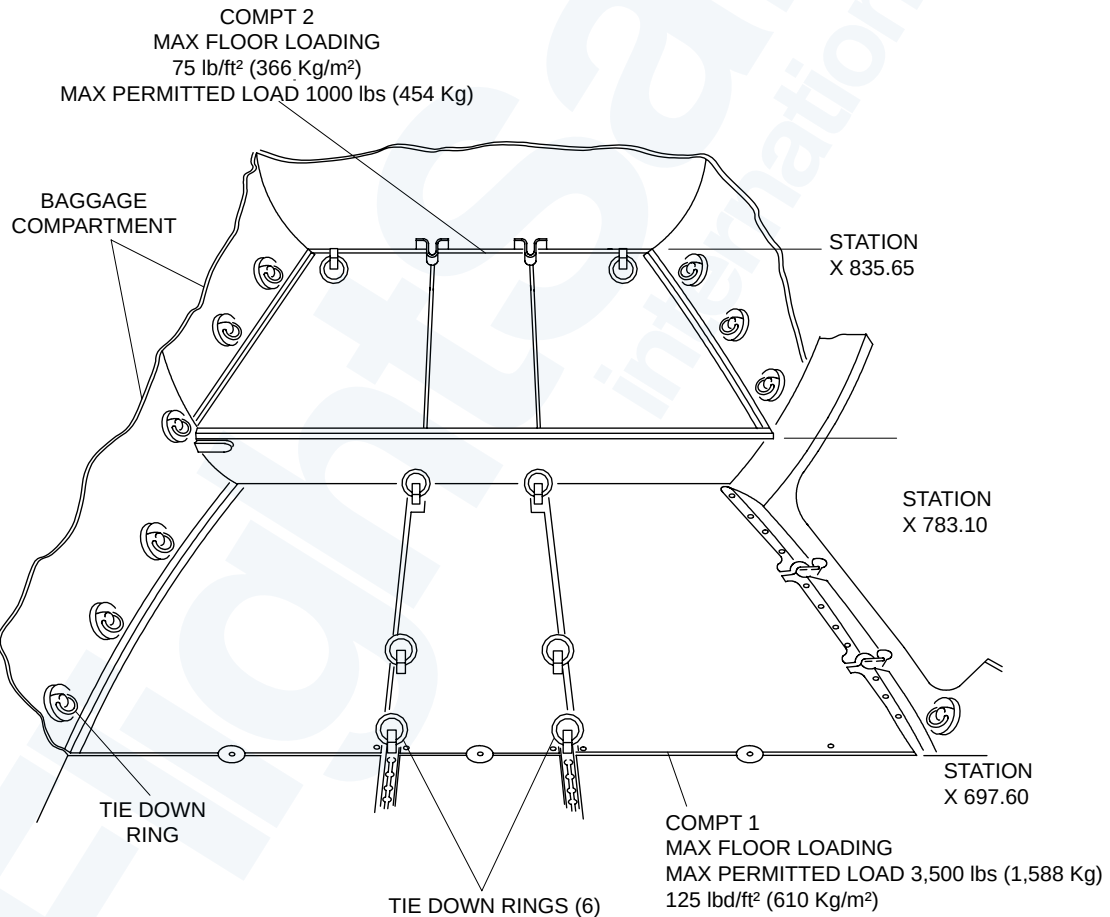


Cargo Tie-Down Fittings

Refer to Figure 25-50. Cargo Tie - Down Fittings.

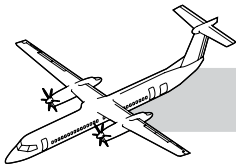
Sixteen 2,000 lb. (907.2 kg) baggage tie-downs are supplied, ten of them located above the cargo floor level, four on the lower floor (behind the aft bulkhead) and two on the step floor.

TOTAL MAXIMUM COMBINED LOAD (COMPT 1 + COMPT 2) = 3500 lbs.



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Figure 25-50. Cargo Tie - Down Fittings



Aisle Cart Restraint

Refer to Figure 25-51. Aisle Cart Restraint.

On aircraft with the aisle cart restraints in the aft baggage compartment (825CH01247): The forward left side of the aft baggage compartment has provisions to hold one or two aisle carts for catering during the return leg.

Four bumpers are attached to the aft baggage compartment bulkhead and two bumpers are attached to the left side of the G3 galley to protect the structure.

Four loops are also attached to the aft baggage compartment bulkhead and two are attached to the left side of the G3 galley. A strap assembly with an adjustable ty-clip at one end and a flat hook at the other end connects to the loops to hold the aisle carts.

A strap stowage loop is also attached to the left side of the G3 galley to hold the strap assemblies when not in use.

NOTE

The aisle carts are part of the baggage load.

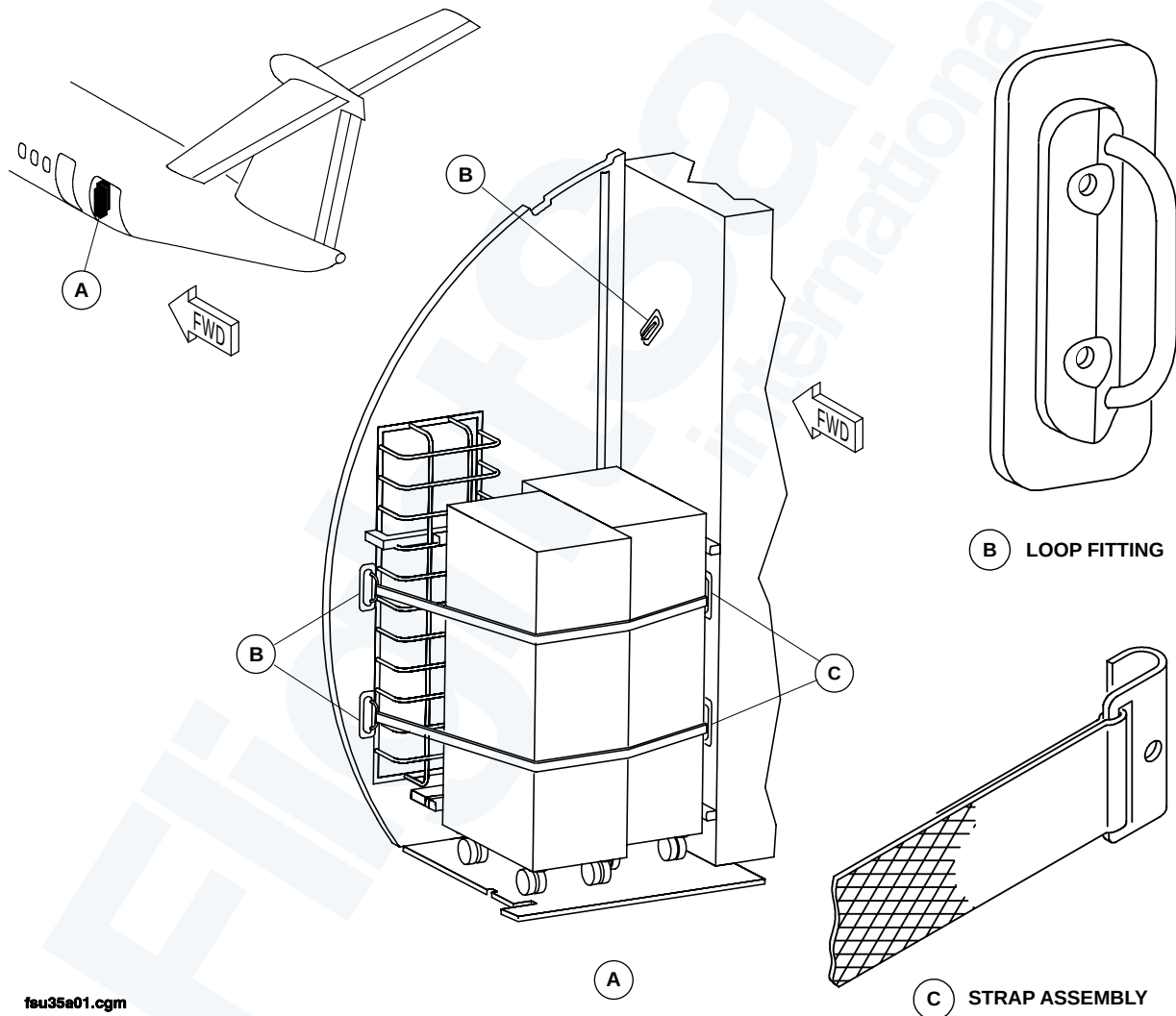
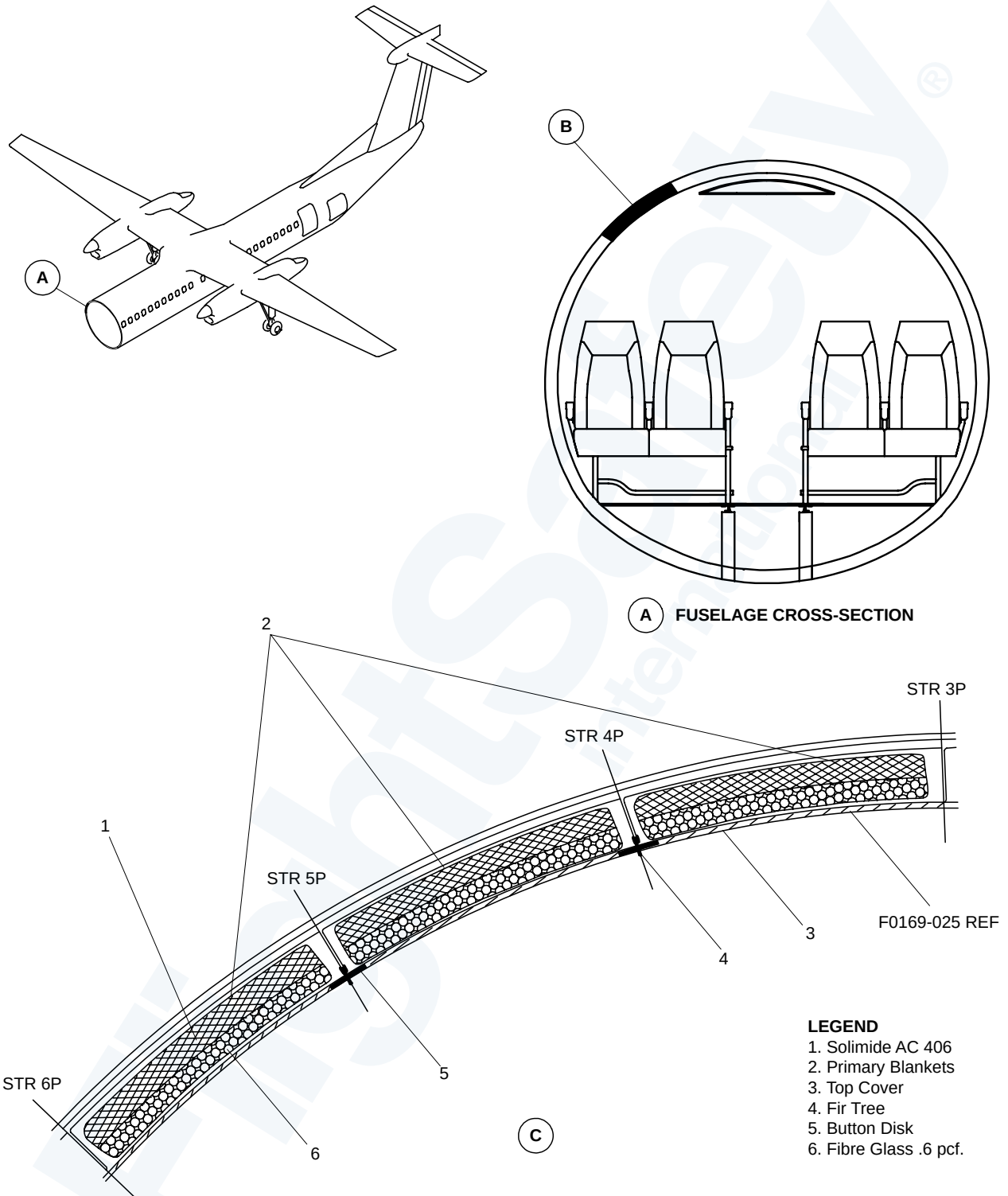
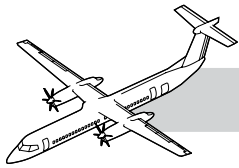
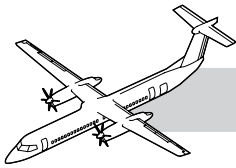


Figure 25-51. Aisle Cart Restraint



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Figure 25-52. Insulation Cross - Section



25-80-00 INSULATION

INTRODUCTION

Insulation is supplied for the flight compartment and cabin for the purposes of acoustic attenuation and thermal energy control. Each blanket is constructed to fit as closely to the structure as possible and require minimal, if any, on-floor modifications. Each blanket is custom fit to a particular location and structural geometry.

GENERAL

Refer to Figure 25-52. Insulation Cross - Section.

The insulation is installed in these areas:

- Flight compartment insulation
- Passenger cabin insulation
- Baggage compartment insulation.

The insulation package consists of primary blankets which fit in the wall cavities and top cover blankets to cover frames. The primary blankets are composed of 1.5 in. of 0.6 lb. fiberglass and 0.5 in. polyimide foam.

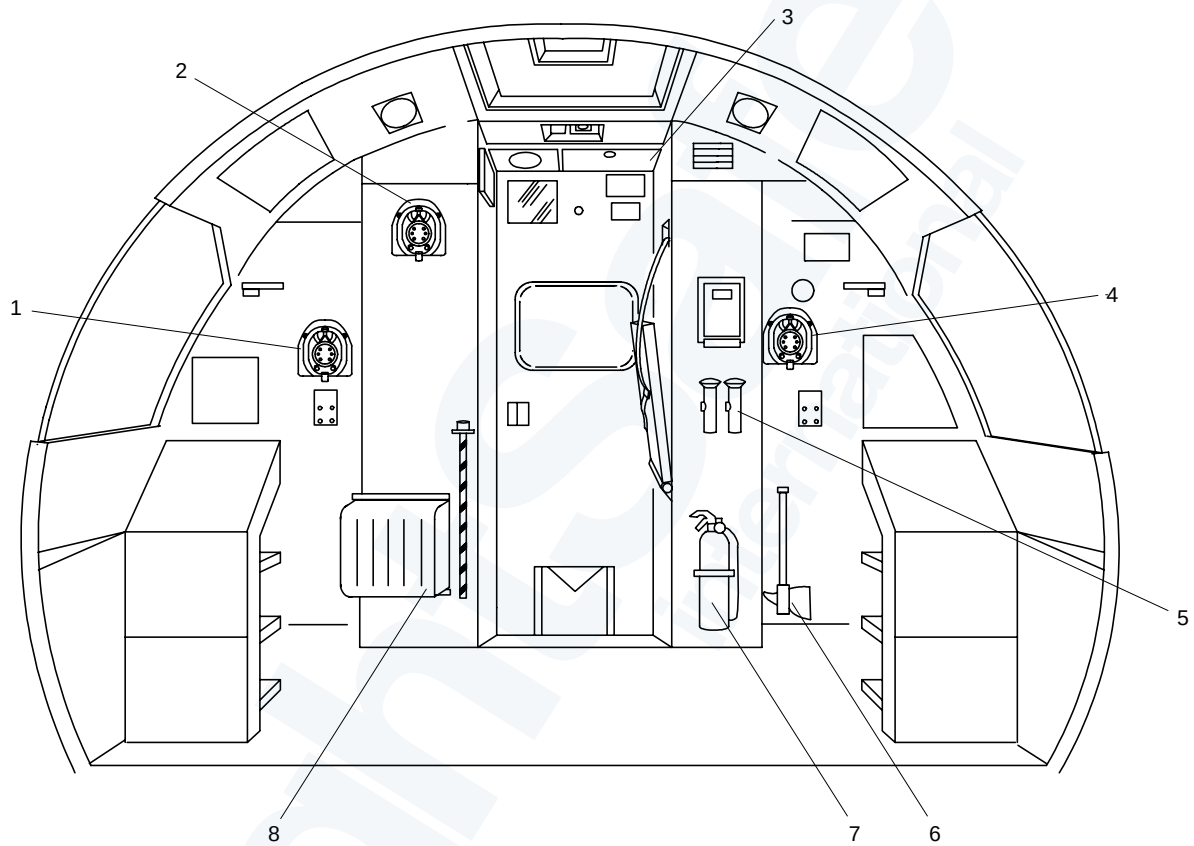
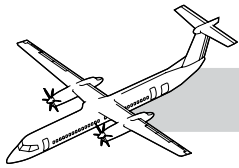
The top covers are made of 0.5 in. polyimide foam. Each insulation bag has a covering of Orcon AN49R vapor barrier film or equivalent. They are heat sealed and constructed with drains that preclude buildup of moisture in the bags and which direct moisture outboard and down the bilges. The drain holes are located on the bottom bevel of each bag.

All blankets are closely tailored to the structure so that noise channels within the wall cavity are eliminated. Each blanket is identified with the precise location by frame station and stringer so that installation time is enhanced. This will also permit for easy removal for zonal inspection requirements on the aircraft systems and structure. The primary blankets are pressed-fit between frame stations and stringers. The top covers are installed

with fir trees through sealed penetrations and secured with plastic tree clips.

For areas around doors and in door structure, individual cavities are filled with 3 in. fiberglass blankets and an over frame blanket provided to cover any bare metal structure.

All cut-outs and channels to accommodate structure and internal hardware are heat sealed with Orcon AN49R film or approved tape.

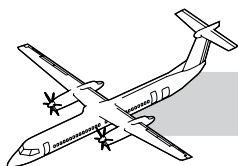


LEGEND

- 1. Co-pilot's Oxygen Mask
- 2. Observer's Oxygen Mask
- 3. Observer's Life Vest Stowage
- 4. Pilot's Oxygen Mask
- 5. Flashlights
- 6. Fire Axe
- 7. Fire Extinguisher
- 8. PBE.

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Figure 25-53. Flight Compartment Emergency Equipment on Aft Bulkhead



25-60-00 EMERGENCY EQUIPMENT

NOTES

INTRODUCTION

The emergency equipment located in the flight compartment and the passenger compartment is used by the flight crew during emergency situations.

GENERAL

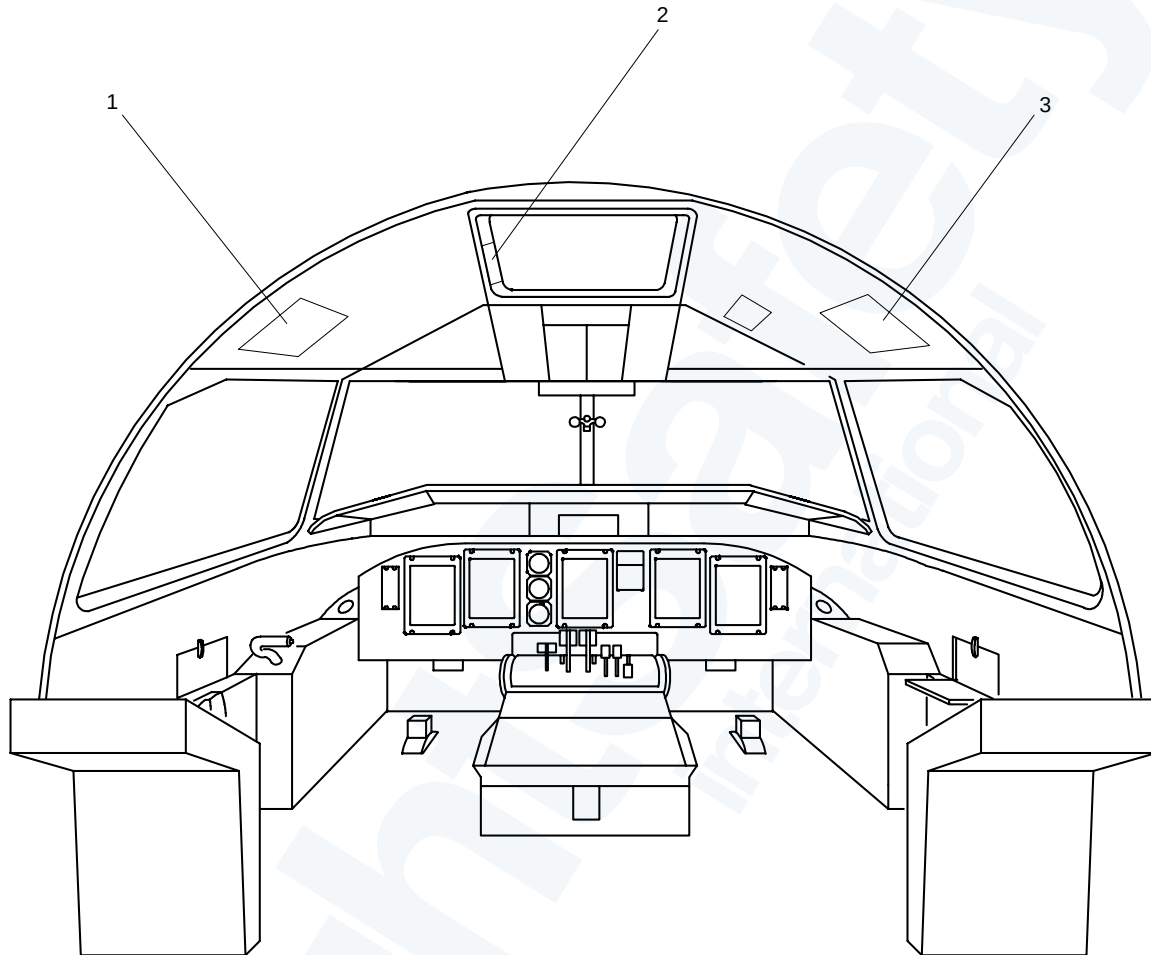
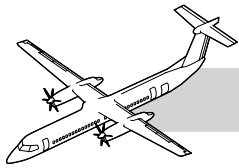
Refer to Figure 25-53. Flight Compartment Emergency Equipment on Aft Bulkhead.

The emergency equipment located in the flight compartment includes these items:

- Fire Axe
- Flashlights
- Life Vest
- Escape Rope
- Smoke Hood.

The emergency equipment in the passenger compartment includes these items:

- Ditching (airstair door) Dam
- Flashlights
- Life vest
- First aid Kit
- Megaphone
- Emergency oxygen
- Medical oxygen.

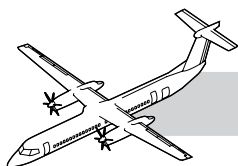


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LEGEND

- 1. Pilot Life Vest Stowage
- 2. Emergency Escape Rope Storage
- 3. Co-pilot Life Vest Stowage.

Figure 25-54. Flight Compartment Emergency Equipment Forward



COMPONENT DESCRIPTION

NOTES

Flight Compartment Emergency Equipment

Refer to Figure 25-54. Flight Compartment Emergency Equipment Forward.

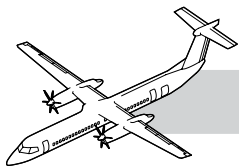
Flight compartment emergency equipment is generally installed on the forward face of the flight compartment bulkhead. Any equipment not visible behind the pilots' seats is clearly marked, at eye level, as to its location. All emergency equipment is located within reach of at least one crew member. The equipment is placarded and is protected from inadvertent damage.

Life Vests

The life vests are used as an aid to flotation for the flight crew in the event of evacuation into water. The life vests are inflated by compressed gas that is connected to a toggle cord activated inflation valve on the jacket. A manual inflation valve and tube are installed, for use in the event the jacket has to be inflated by mouth.

Two life vests are stowed in individual boxes, behind the left and right ceiling panels above the pilot's seat. There is a third life vest stowed on the rear flight compartment bulkhead above the door. The vests are held in the stowage boxes by quick release velcro fasteners.

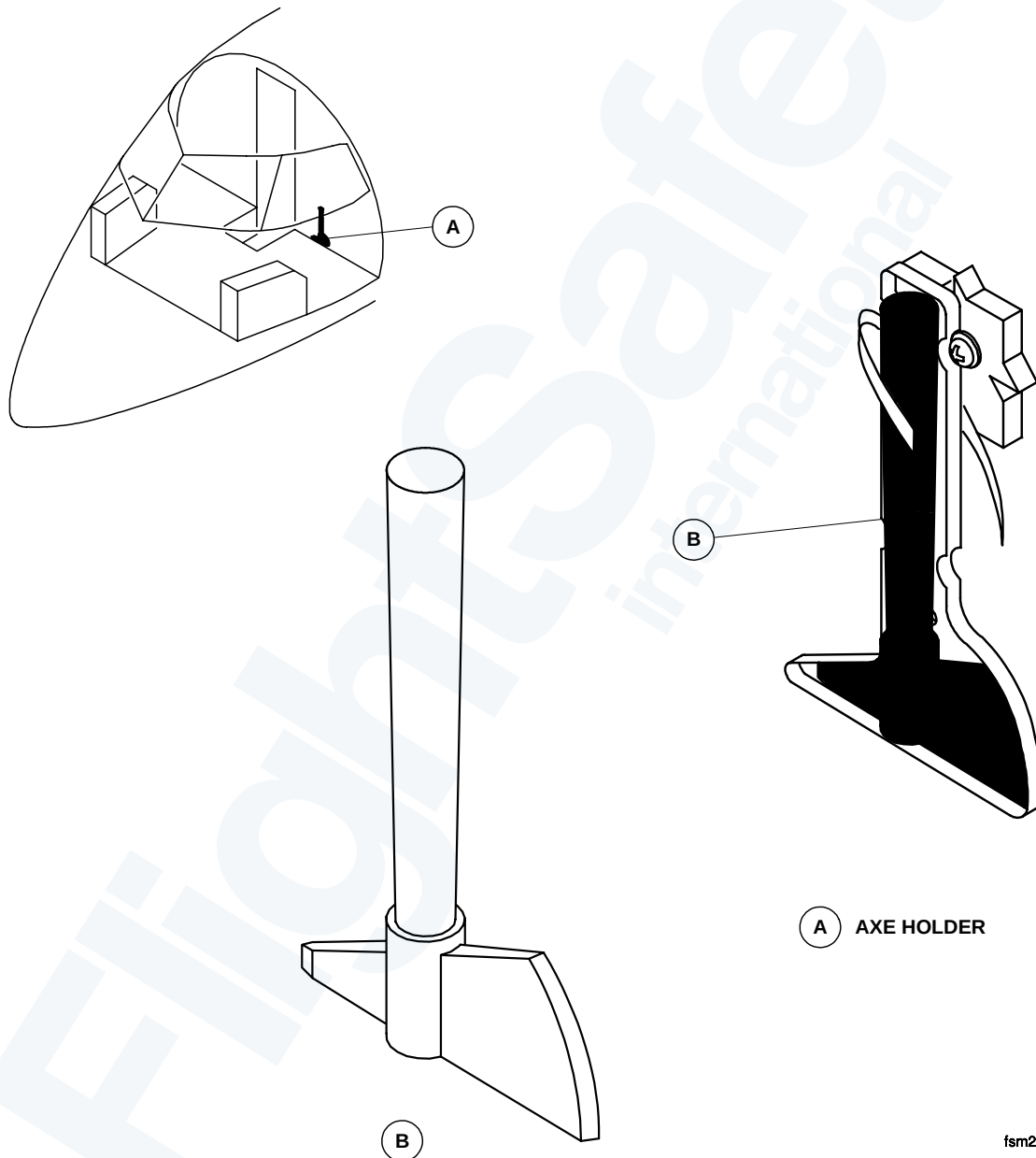
Observers vest location (Refer to Figure 25-6. Observer Seat - Detail.)



Fire Axe

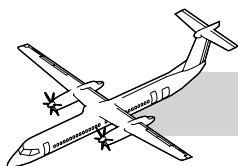
Refer to Figure 25-55. Fire Axe.

The single-bladed fire axe is installed on the flight compartment bulkhead behind the pilot's seat. It is secured to the bulkhead by a Velcro strap.



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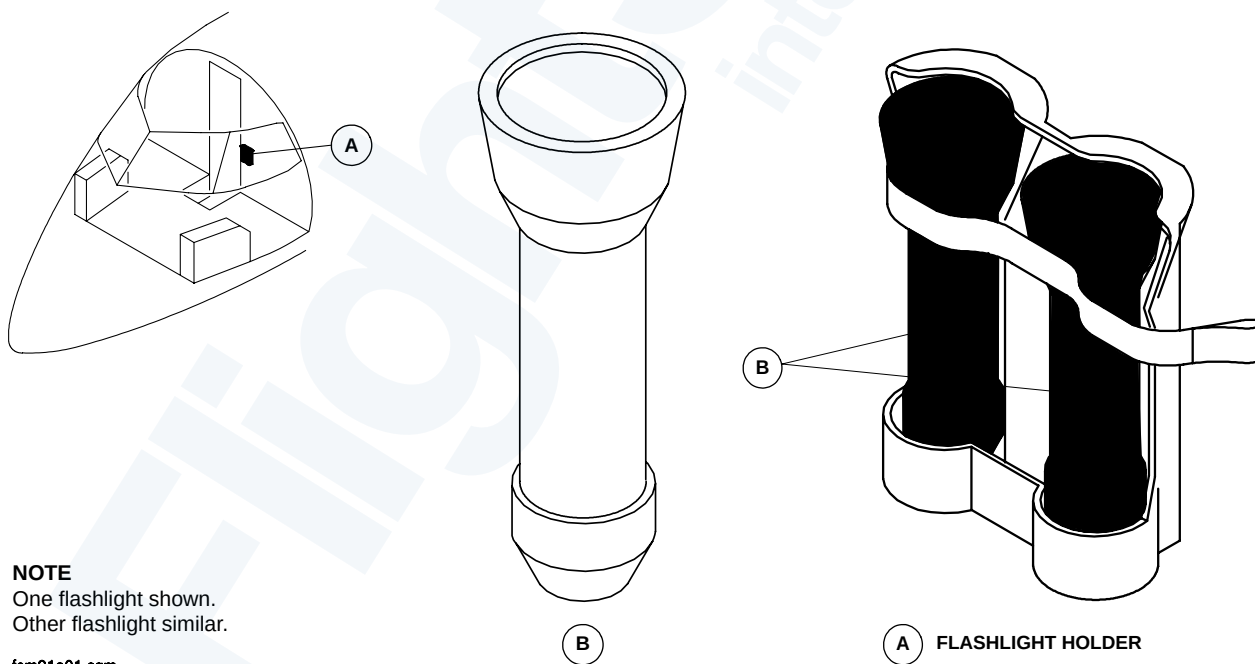
Figure 25-55. Fire Axe



Flashlights

Refer to Figure 25-56. Flashlights.

Two flashlights are located in the passenger compartment. Two are installed on the aft flight compartment bulkhead behind the pilot's seat. They are powered by batteries and secured in their housing by a Velcro strap.

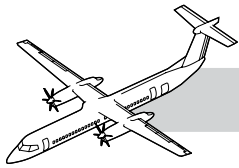


NOTE

One flashlight shown.
Other flashlight similar.

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Figure 25-56. Flashlights



Escape Rope

Refer to Figure 25-57. Escape Rope.

The escape rope is used by the flight crew for evacuating the flight compartment through the emergency escape hatch. It is a 10 feet (3.05 meter) long web strap with 8 knots throughout its length. It is connected to a shoulder bolt installed in the roof structure.

The rope is coiled to fit in a box located at the left side of the escape hatch. The cover of the box is secured by velcro and labeled EMERGENCY ESCAPE ROPE STOWAGE.

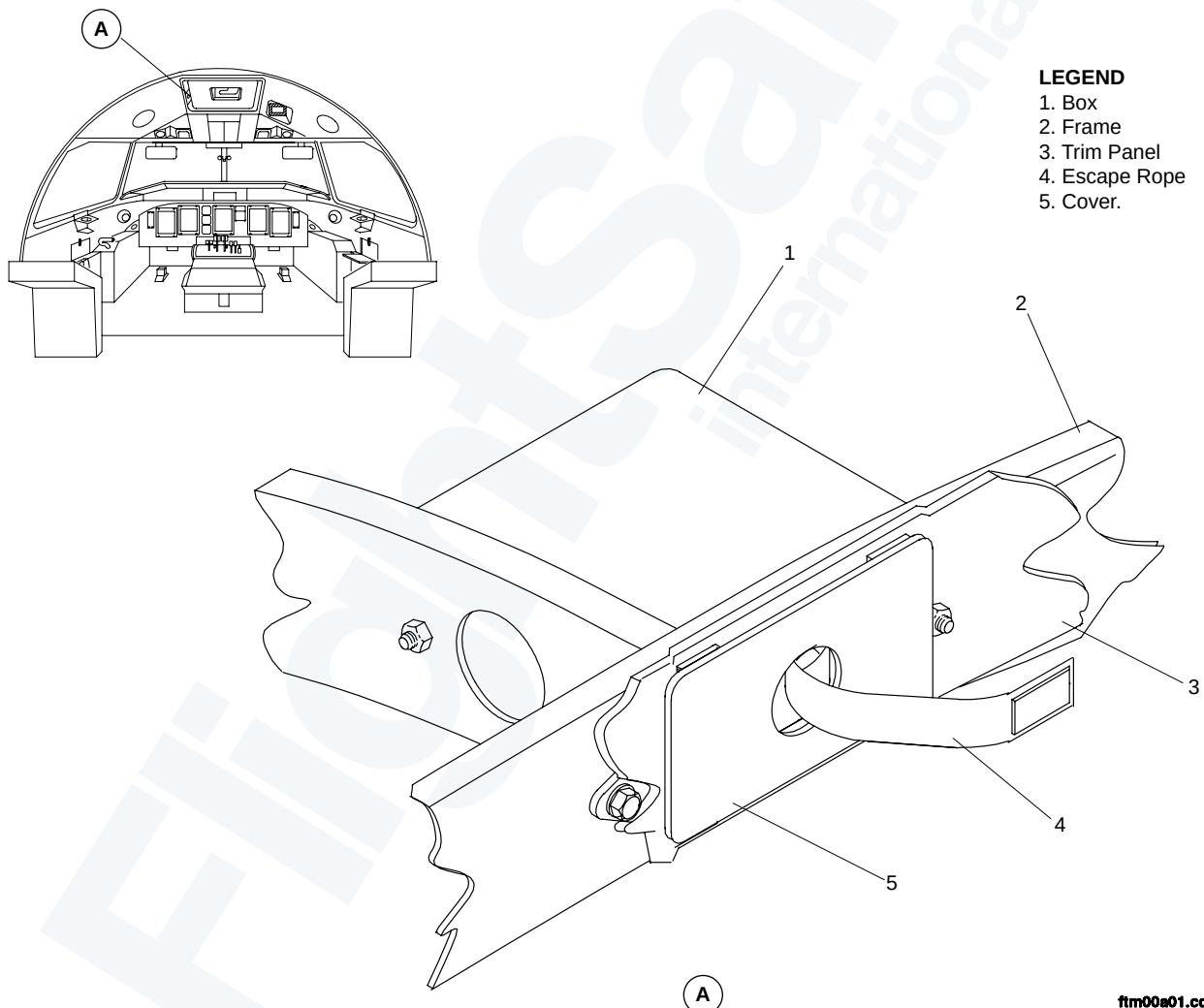
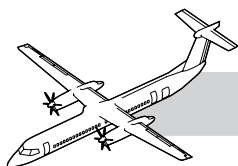


Figure 25-57. Escape Rope

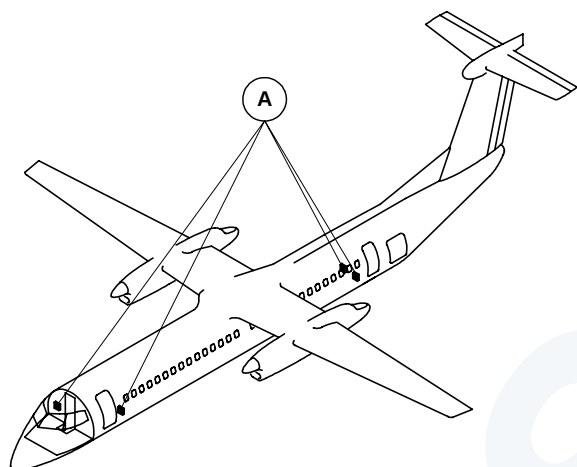
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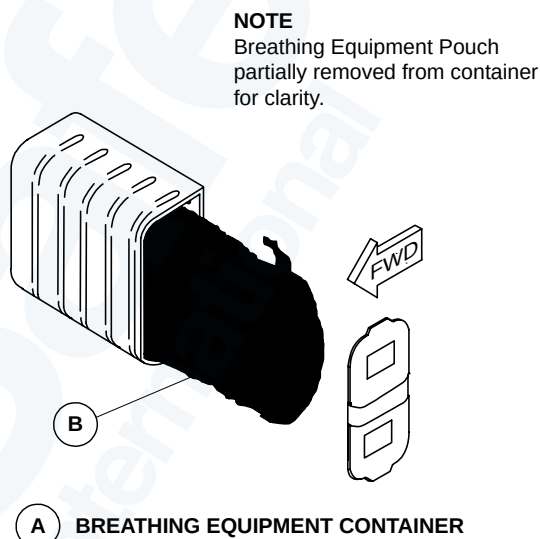
Smoke Hood

Refer to Figure 25-58. Smoke Hood.

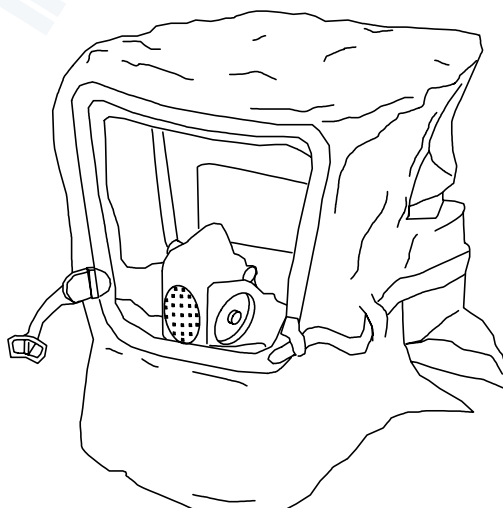
The smoke hood is a self contained, portable, disposable breathing device installed in a quick release stowage on the aft flight compartment bulkhead. The system will supply oxygen for approximately 15 minutes.



NOTE
Breathing Equipment contained within pouch - shown removed.



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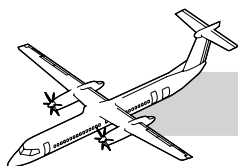


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Figure 25-58. Smoke Hood



Passenger Compartment Emergency Equipment

Refer to Figure 25-59. Passenger Compartment Emergency Equipment - General Configuration.

There are five standard emergency equipment stowage compartments in the passenger cabin.

The locations of the five compartments are:

- One compartment in the forward draft bulkhead, aft of the airstair door
- Two compartments on the forward face of the aft draft bulkhead, on either side of the aisle
- One compartment under the forward flight attendant's seat
- One compartment under the aft flight attendant's seat.

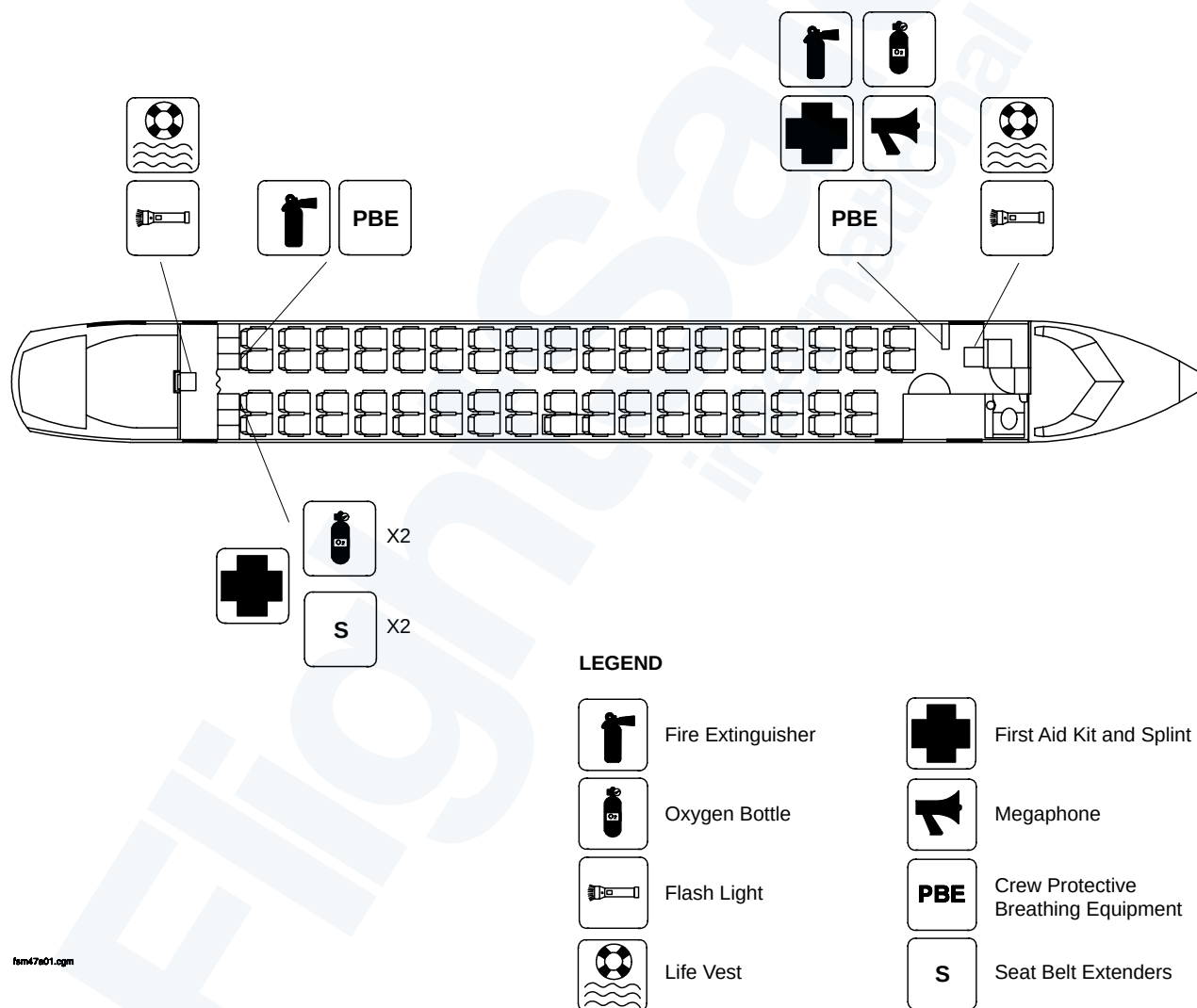
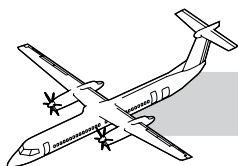
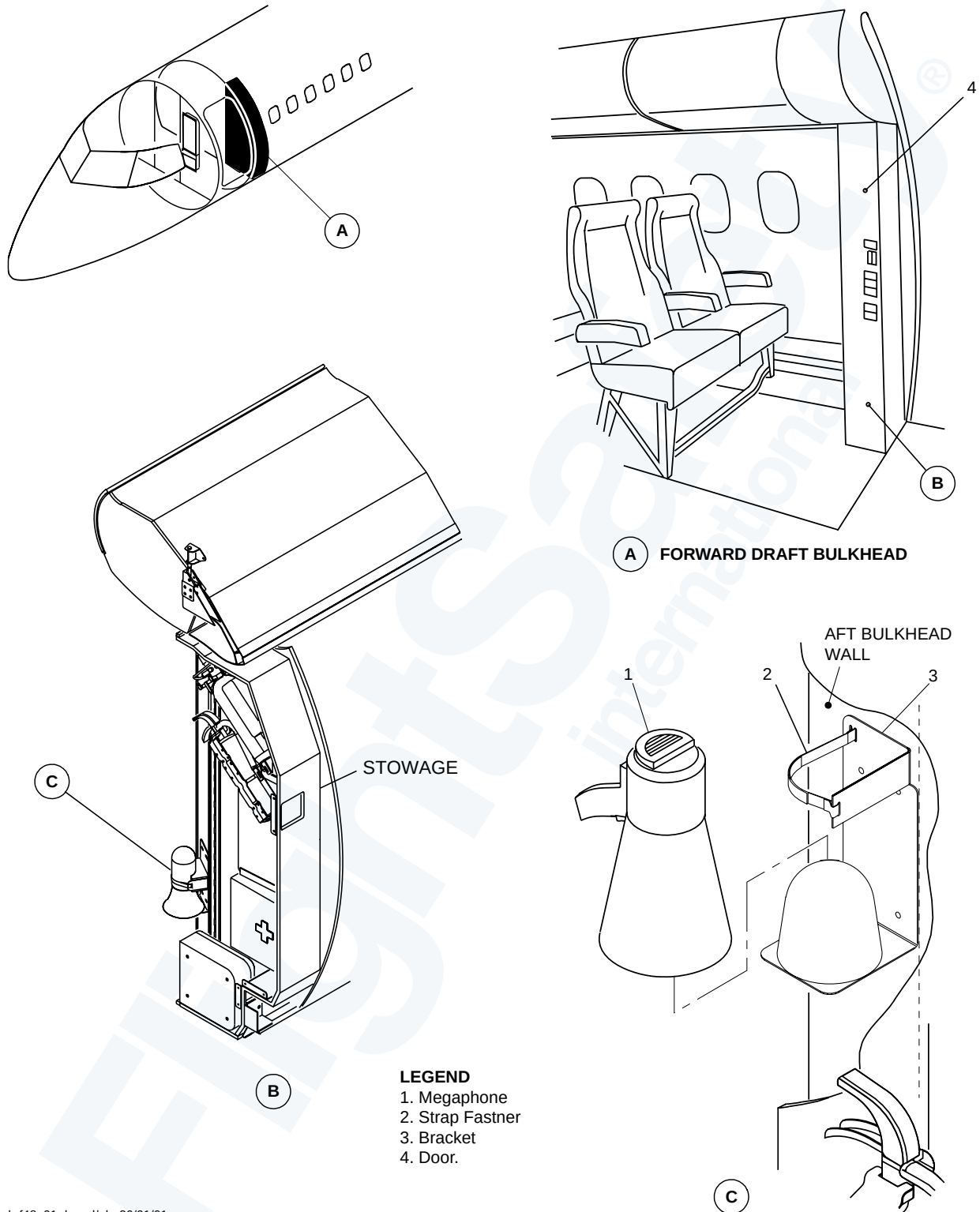
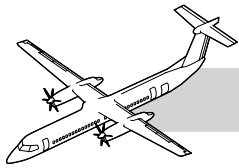


Figure 25-59. Passenger Compartment Emergency Equipment - General Configuration

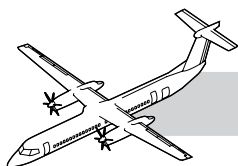


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Figure 25-60. Emergency Equipment - Forward Draft Bulkhead



Refer to Figure 25-60. Emergency Equipment
- Forward Draft Bulkhead.

NOTES

The emergency equipment stowage compartment
in the forward draft bulkhead has:

- One fire extinguisher
- One oxygen bottle with two Puritan
Bennet masks
- One PBE unit
- One first aid kit
- One megaphone.

First Aid Kit

There are two first aid kits located in the
passenger compartment. One kit is located
in the forward emergency equipment
compartment, and the other kit is located in the
right rear emergency equipment compartment.

The first aid kits are sealed to ensure their
completeness and to ensure that they do not
contain unsealed consumable items.

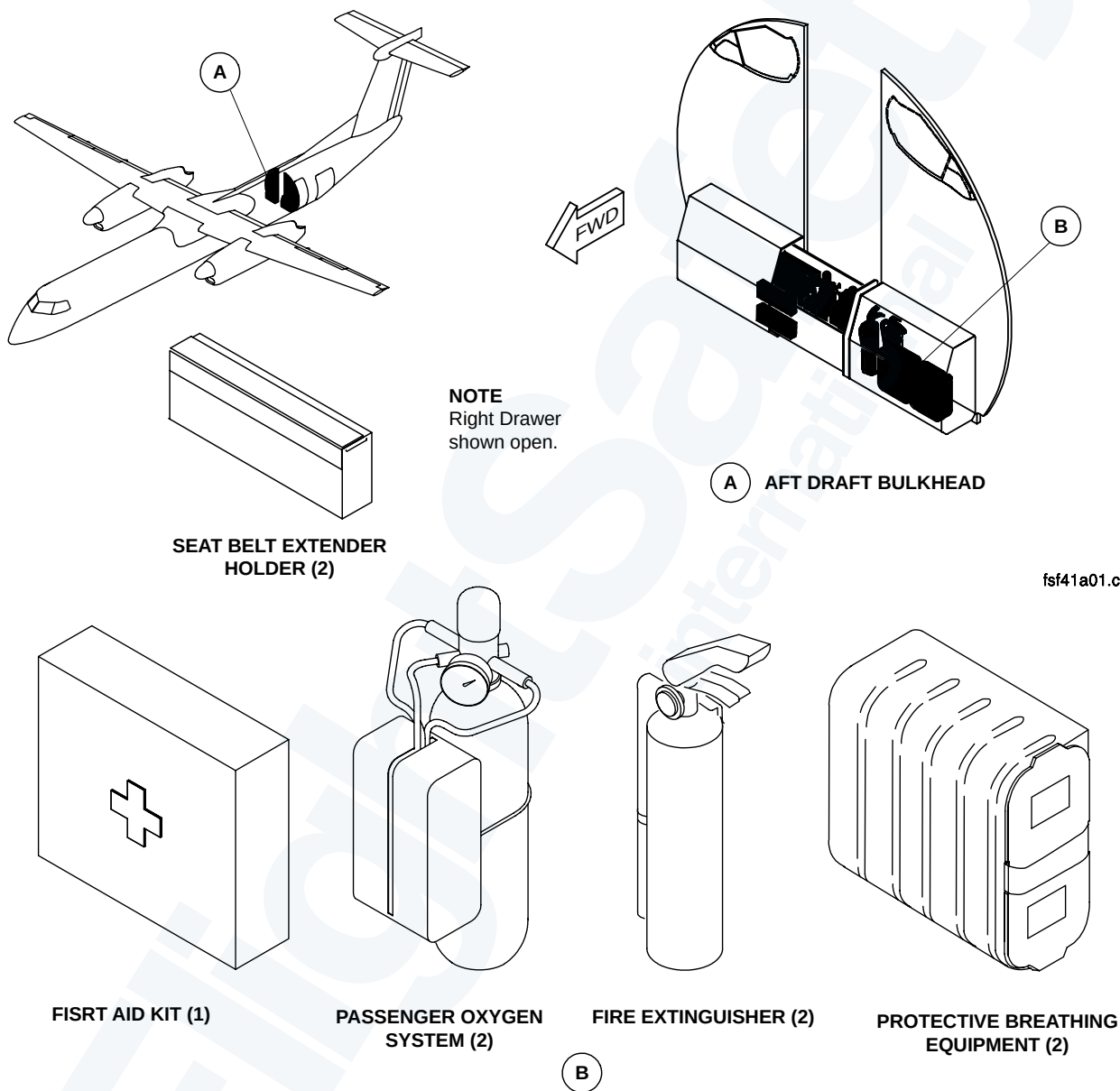
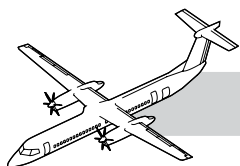
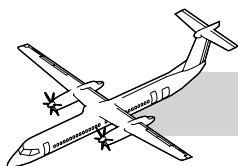


Figure 25-61. Emergency Equipment Aft Draft Bulkhead



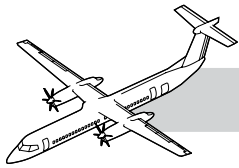
Megaphone

NOTES

The battery operated megaphone is installed in the forward draft bulkhead. The megaphone allows the flight attendant to communicate instructions to the passengers during emergencies when excessive noise may be present.

Refer to Figure 25-61. Emergency Equipment Aft Draft Bulkhead.

The emergency equipment stowage compartment on the left side of the aft draft bulkhead has two fire extinguishers and two PBE units. The right side compartment has two oxygen bottles with Puritan Bennet masks and one first aid kit. Each flight attendant station has a life vest and a flashlight stored under the fold up seat and protected by a quick release panel. All five emergency equipment stowage compartments are clearly placarded as to their contents.



NOTE
Emergency Equipment Compartment
shown in open position.

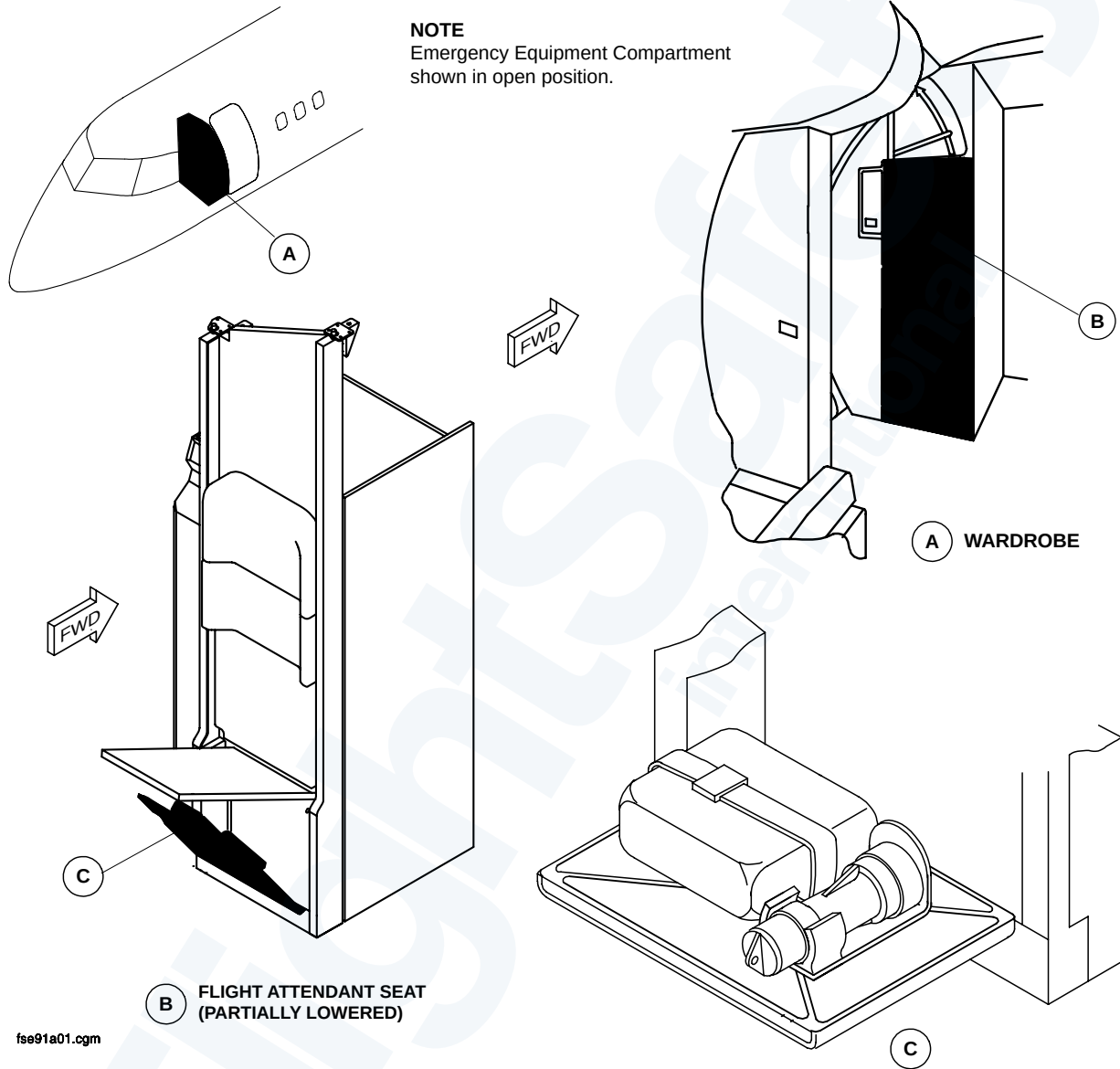
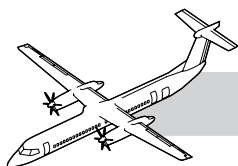


Figure 25-62. Emergency Equipment - Forward Flight Attendant's Seat



Flashlights

NOTES

Refer to Figure 25-62. Emergency Equipment
- Forward Flight Attendant's Seat.

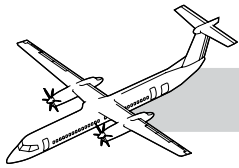
There are two flashlights stowed in compartments located under the flight attendants' seats at the front and rear of the passenger compartment.

They are secured in the stowage by velcro straps and powered by two size D batteries.

MAINTENANCE PRACTICES

Replacement of the Passenger Compartment Flashlight Batteries

- There are two flashlights installed in the flight attendant's forward and aft seat stowage compartments. Each flashlight is installed in a cradle assembly attached by a strap. The flashlights are powered by size 'D' batteries.



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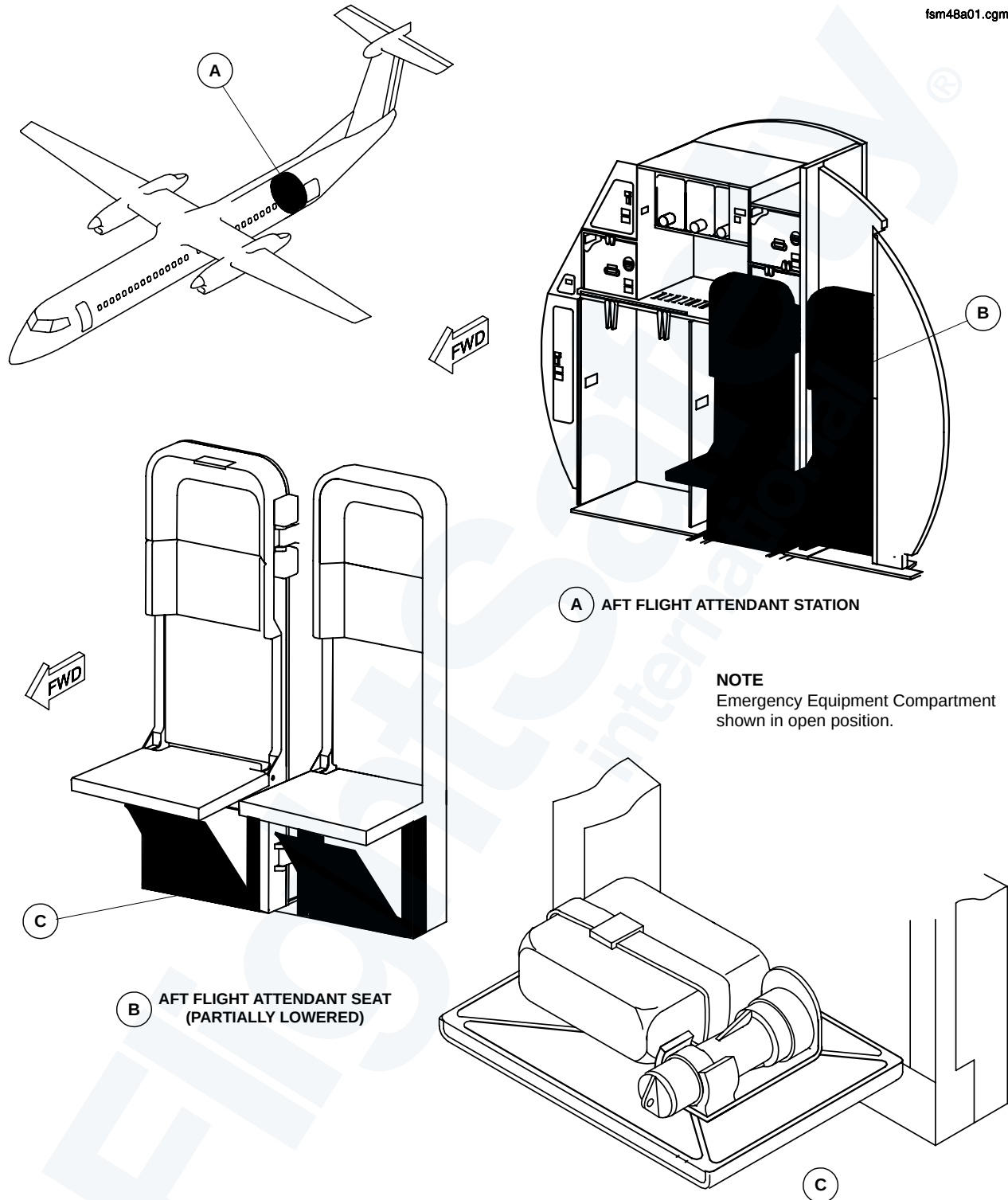
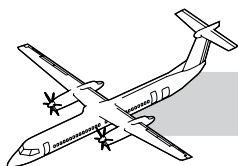


Figure 25-63. Emergency Equipment - Aft Flight Attendant's Seat



Life Vests

There are two life vests, stowed in compartments located under the flight attendants' seats, at the front and rear of the passenger compartment. The vests are identical to those located in the flight compartment. Passenger life jackets are available only as a customer option.

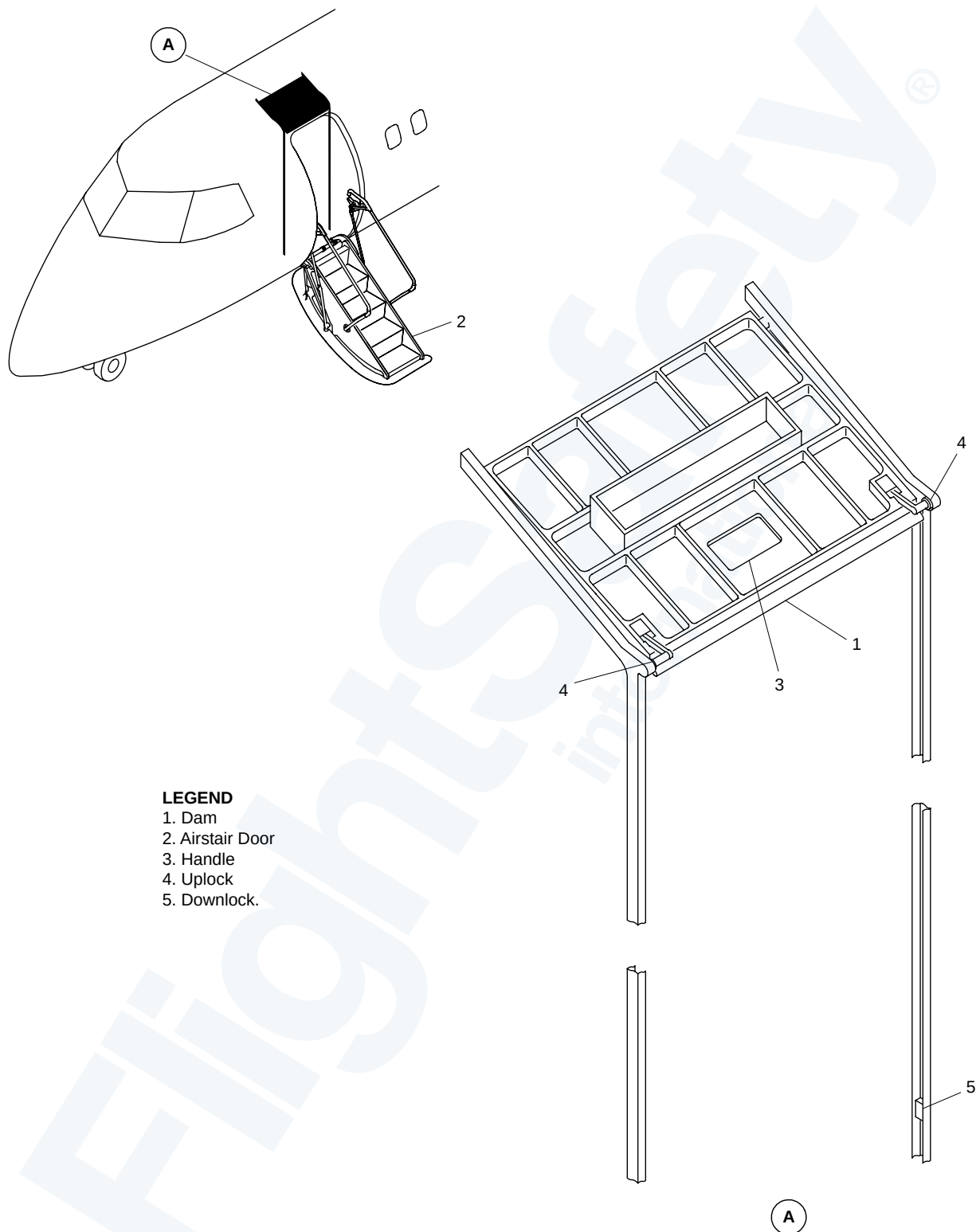
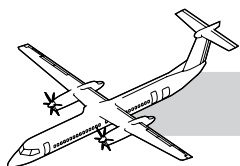
Refer to Figure 25-63. Emergency Equipment - Aft Flight Attendant's Seat.

On aircraft with the SAS interior, the passenger compartment has the emergency equipment that follows:

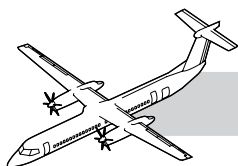
- One megaphone
- Two flight attendant flashlights
- Two flight attendant life vests
- Three Fire extinguishers
- Three flight attendant PBEs
- Three passenger oxygen bottles
- One flight attendant oxygen bottle
- One first aid kit
- Four seat belt extenders.

The passenger compartment emergency equipment is kept in these different areas:

Area	Equipment
Wardrobe	Megaphone (Refer to 25-60-00 Page 8)
Forward flight attendant's seat	Flashlight Life Vest
Aft flight attendant's seat	Flashlight Life Vest
Inboard part of the G6 galley	Fire extinguisher PBE
Compartment in the aft part of the G6 galley	2 Passenger oxygen bottles FA oxygen bottle Seat belt extenders
Aft face of left aft draft bulkhead	Fire extinguisher
Drawer in aft face of left aft draft bulkhead	Fire extinguisher PBE Passenger oxygen bottle Seat belt extenders
Forward left part of aft baggage bulkhead	PBE



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Airstair Door Ditching Dam

NOTES

Refer to Figure 25-64. Airstair Door Ditching Dam.

The ditching dam prevents water from flowing into the cabin, when the airstair door is opened after an aircraft water ditching. With the aircraft floating left wing down, the airstair door exit threshold will be several inches underwater. In this case, the ditching dam will raise the freeboard. The ditching dam is a composite panel with rollers installed on each corner. The dam is located above the forward airstair door and is latched in place. The dam, when not in use, forms the ceiling panel above the door and is retained in tracks forward and aft of the entryway.

The rollers installed on the dam fit in the tracks along each side of the door. When required, the dam is released from its locking clips and pulled down the tracks to rest on the floor, where it is latched in place. This creates a 16 in. (406 mm) high barrier, 13 in. (330 mm) inboard of the airstair door threshold.

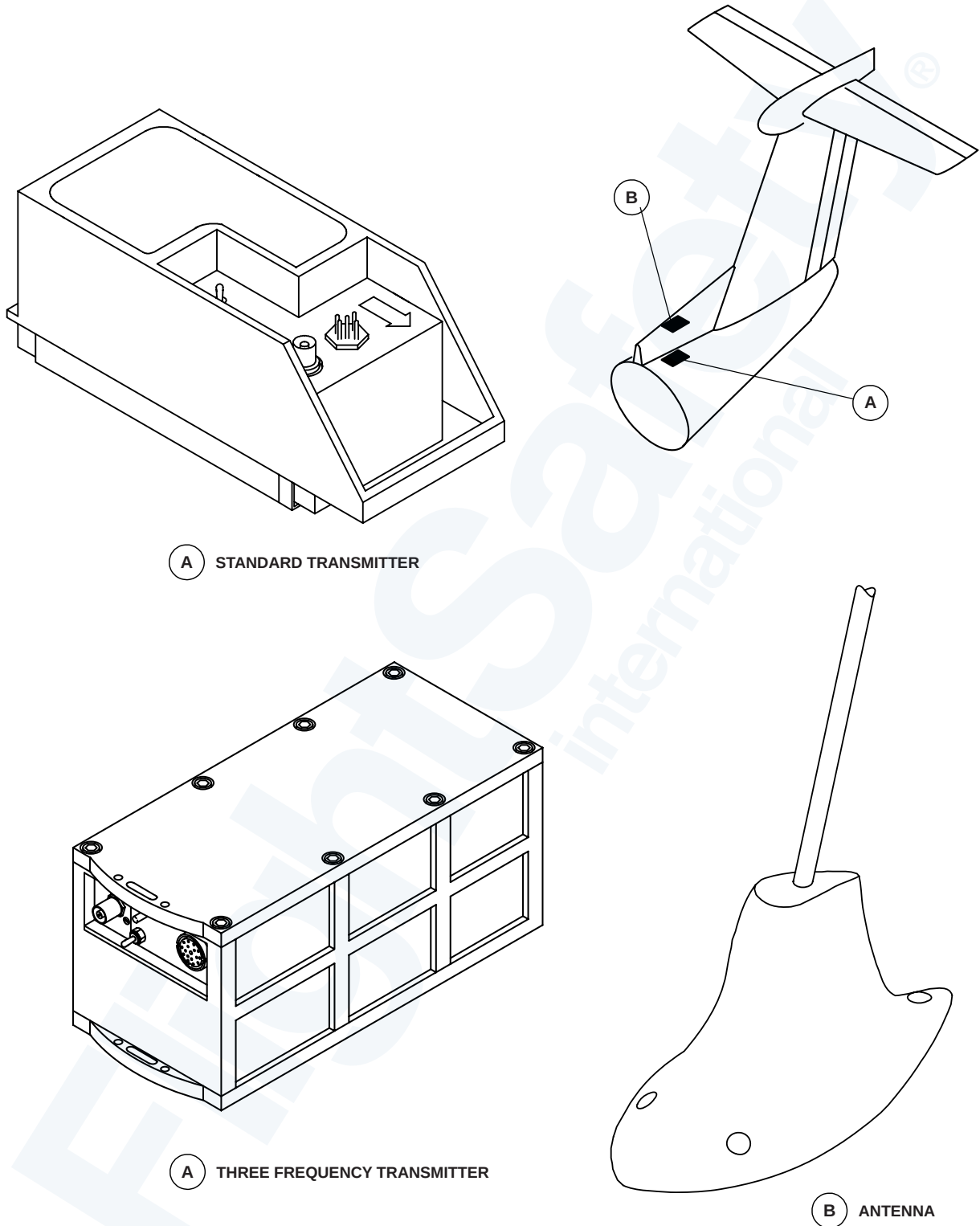
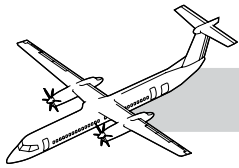
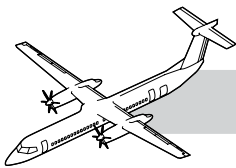


Figure 25-65. ELT Locators



25-63-00 EMERGENCY LOCATOR TRANSMITTER (ELT)

INTRODUCTION

The Emergency Locator Transmitter (ELT) system provides the aircraft with an independent and automatically latched continuous distress signal transmission. An inertial force, similar to those experienced in a crash, activates the transmitter.

GENERAL

Refer to Figure 25-65. ELT Locators.

The ELT system also supplies the flight crew with these functions:

- Visual annunciation of operation
- System reset capability when operating due to an automatically-latched initiation
- Manual system operation.

The ELT system has these components:

- ELT
- Antenna and Cable Assembly
- Remote Control Switch.

The KANNAD 406 AF/AF-H is an ELT installed on aircraft to transmit a distress signal on three frequencies:

- 406 MHz (COSPAS-SARSAT frequency) for precise pinpointing and identification of the aircraft in distress.
- 121.5 and 243 MHz used for homing in the final stages of the rescue operations.

It is certified as an Automatic Fixed (AF) ELT and only works with a three frequency outside antenna.

The electrical power for the system is provided by a battery pack P/N S1820506-XX composed of 3 LiMnO₂ D cells. For replacement of batteries the unit must be sent for manufacture.

The autonomy of the 121.5/243 transmission is close to 100 hours at -20°C with new batteries.

The 406 MHz transmission is deliberately stopped after 24 hours to extend the 121.5/243 transmission for as long as possible. During the worse conditions possible, a distress is pinpointed two hours maximum after the ELT activation and the position is subsequently updated every two hours.

The transmitter battery expiry date is fixed at 6 years after manufacturing. If no activation of the ELT occurs during the battery lifetime, it shall be replaced every 6 years.

The Remote Control Panel (when connected) is also energized by this battery pack.

When activated, the ELT may be switched off by momentarily placing the remote switch in the RESET position.

The ELT system normally operates in an inactive state in order to maximize reliability. It does not supply specific failure indications or annunciation. Periodic operational tests give the operational status of the ELT system.

For normal operation, the three-position remote switch in the flight compartment and the master switch located on the ELT are set to AUTO.

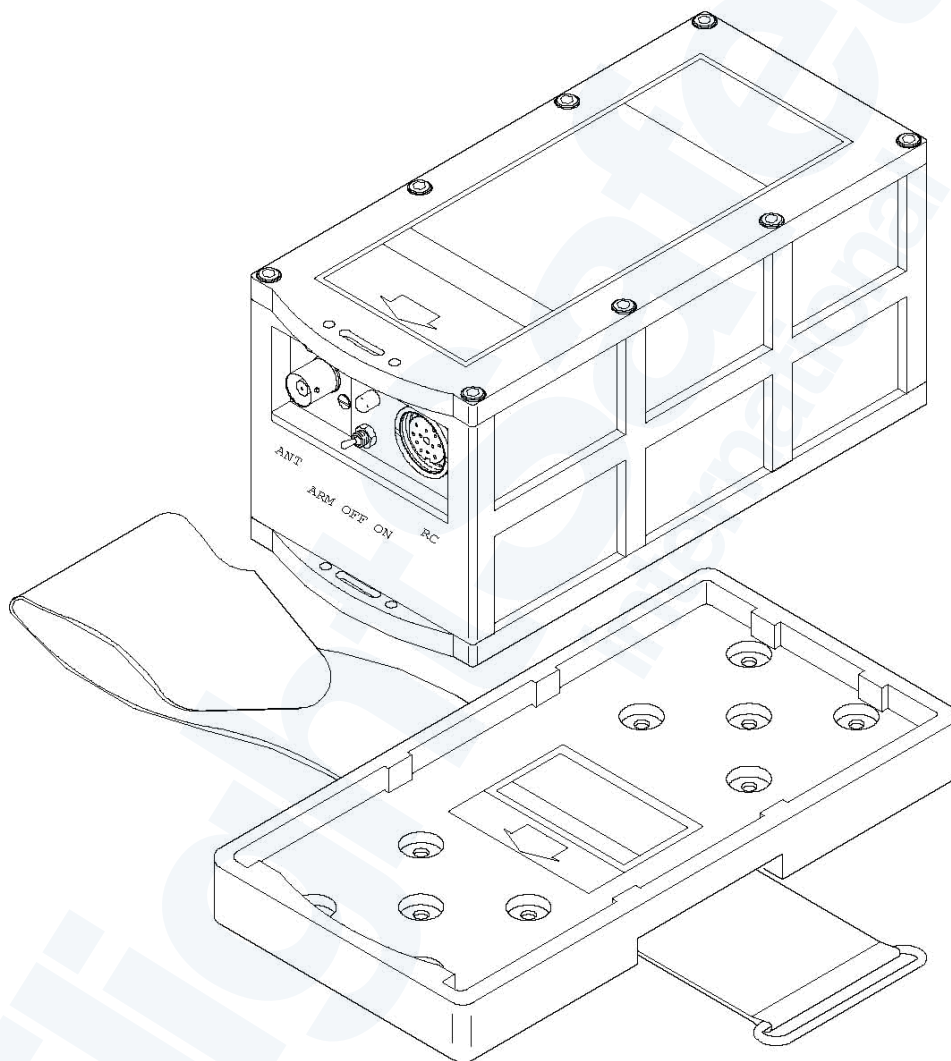
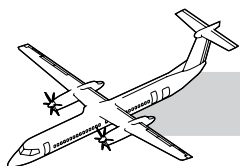
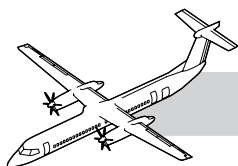


Figure 25-66. KANNAD 405 AF/AF-H and Mounting Bracket



COMPONENT DESCRIPTION

NOTES

KANNAD 406 AF/AF-H Transmitter

Refer to Figure 25-66. KANNAD 405 AF/AF-H and Mounting Bracket.

The KANNAD 406 AF/AF-H transmitter is an integrated design on a single Printed Circuit Board.

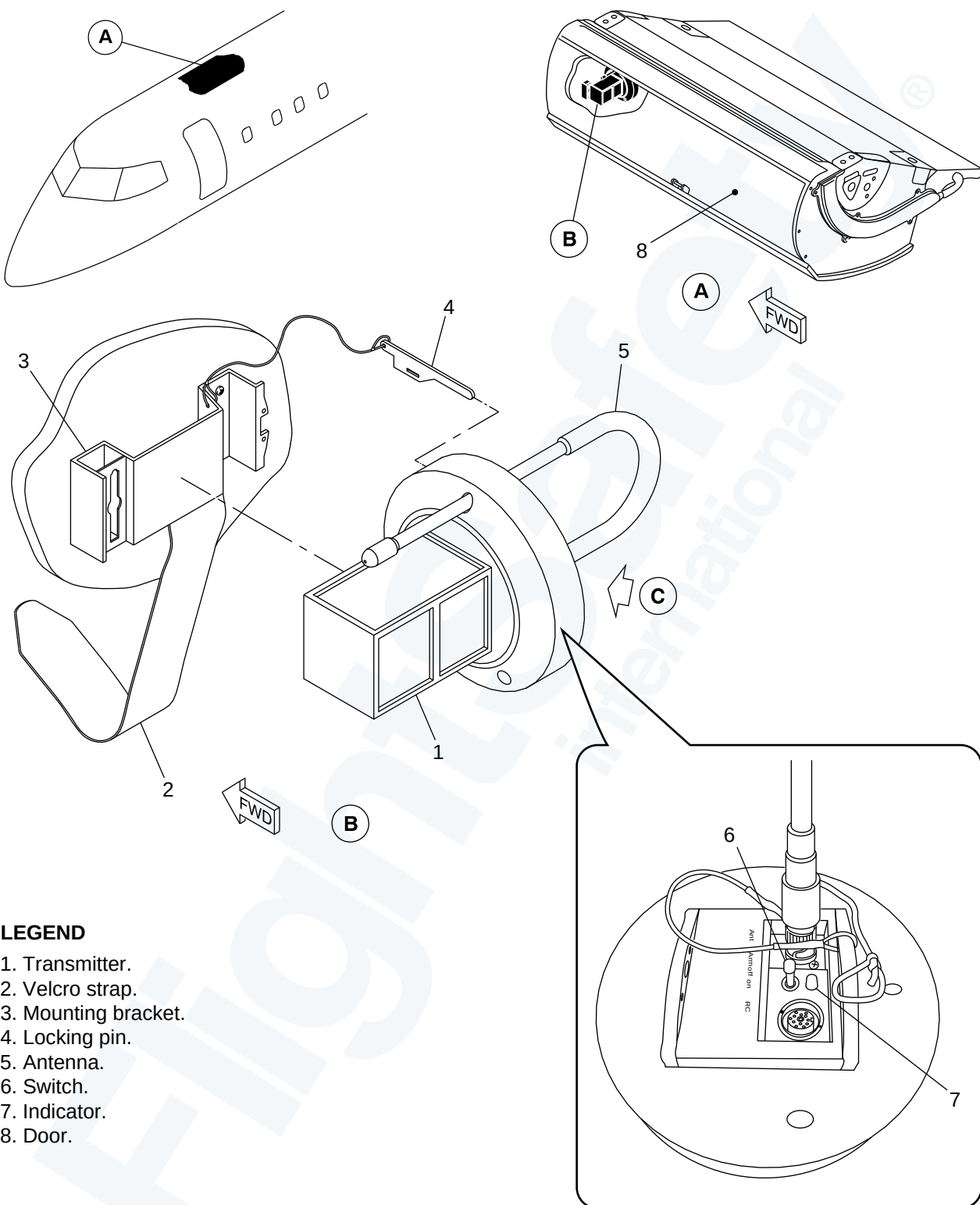
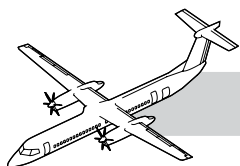
Once activated, the transmitter operates continuously on 121.5 and 243 MHz. The output power is 100 mW on each frequency.

During the 24 first hours of operation, a signal is transmitted on 406 MHz every 50 seconds to the COSPAS-SARSAT satellites. The output power on 406 MHz is 5W (50 times more powerful than the VHF signal).

Phase modulation at 400 BPS enables transmission of all the relevant identification information to the COSPAS-SARSAT satellites in less than half a second.

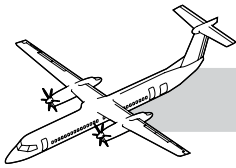
An internal inertia switch is connected to the transmitter printed circuit board. It is activated by longitudinal inertia forces between 5 and 7 G (gravity) acceleration. A label attached to the side of the ELT shows the expiry date of the battery pack currently installed in the ELT.

The ELT is housed within a high-impact, fire retardant waterproof orange colored case. The case is designed to withstand forced landing and crash environmental conditions in an operable condition.



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Figure 25-67. Portable Emergency Locator Transmitter - Removal/Installation



PORTABLE EMERGENCY LOCATOR TRANSMITTER

The portable ELT is attached with a Velcro strap tightly to the mounting bracket installed in the RH overhead bin. The portable ELT is installed with a floating collar to enable the ELT to float if used in water. This floating collar can be removed if the ELT is attached to a life raft or buoyant part. The portable ELT is not installed with a G-Switch (shock detector).

The portable ELT has the components that follow:

- Emergency Locator Transmitter with Antenna
- Mounting Bracket with Locking Pin
- Switch, Indicator and Buzzer
- Programming Dongle.

The portable ELT operates as follows:

- The ELT starts with the self-test sequence.
- The 121.5/243.0 MHz transmission starts immediately after the self-test.
- The 406.0 MHz transmission starts after 50 seconds.
- During transmission, buzzer operates and visual indicator (red) flashes periodically.

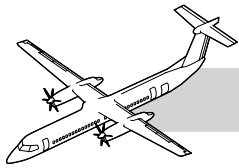
Once activated, the portable ELT can be de-activated manually by selecting the switch to OFF (center) position.

The indicator in the front panel of the portable ELT gives the indication on the working mode of the ELT as follows:

- After the self-test, long flash indicates that the ELT is operational and has no error.
- After the self-test, the series of short flashes indicate that the self test has failed.
- During the operation of the portable ELT, periodic flashes of the indicator indicate that the ELT is transmitting at 121.5/243.0 MHz frequencies. One long flash (1 in every 50 seconds) indicates 406.0 MHz transmission by the ELT.

The buzzer gives audio information on the portable ELT working mode as follows:

- The buzzer produces continuous tone during the self-test.
- The buzzer produces 2 beeps per second during 121.5/243.0 MHz transmission and remains silent when ELT transmits 406.0 MHz signal.



Antenna

Refer to Figure 25-68. ELT Antenna Location.

Connected to the ELT with a 50 Ohm coaxial cable ended with a BNC connector.

The ELT Antenna is a rigid rod antenna weighing 6 ounces.

The ELT antenna fits to the airframe immediately below the outer skin on the dorsal fin. It is forward of the vertical stabilizer at station X+874.7, Y0 and Z+210.8. The antenna protrudes through the aircraft skin and is sealed by a grommet. The antenna connects directly to the ELT through the coaxial antenna cable assembly.

CAUTION

DO NOT OPERATE THE ELT WITHOUT THE ANTENNA. YOU CAN CAUSE DAMAGE TO THE UNIT.

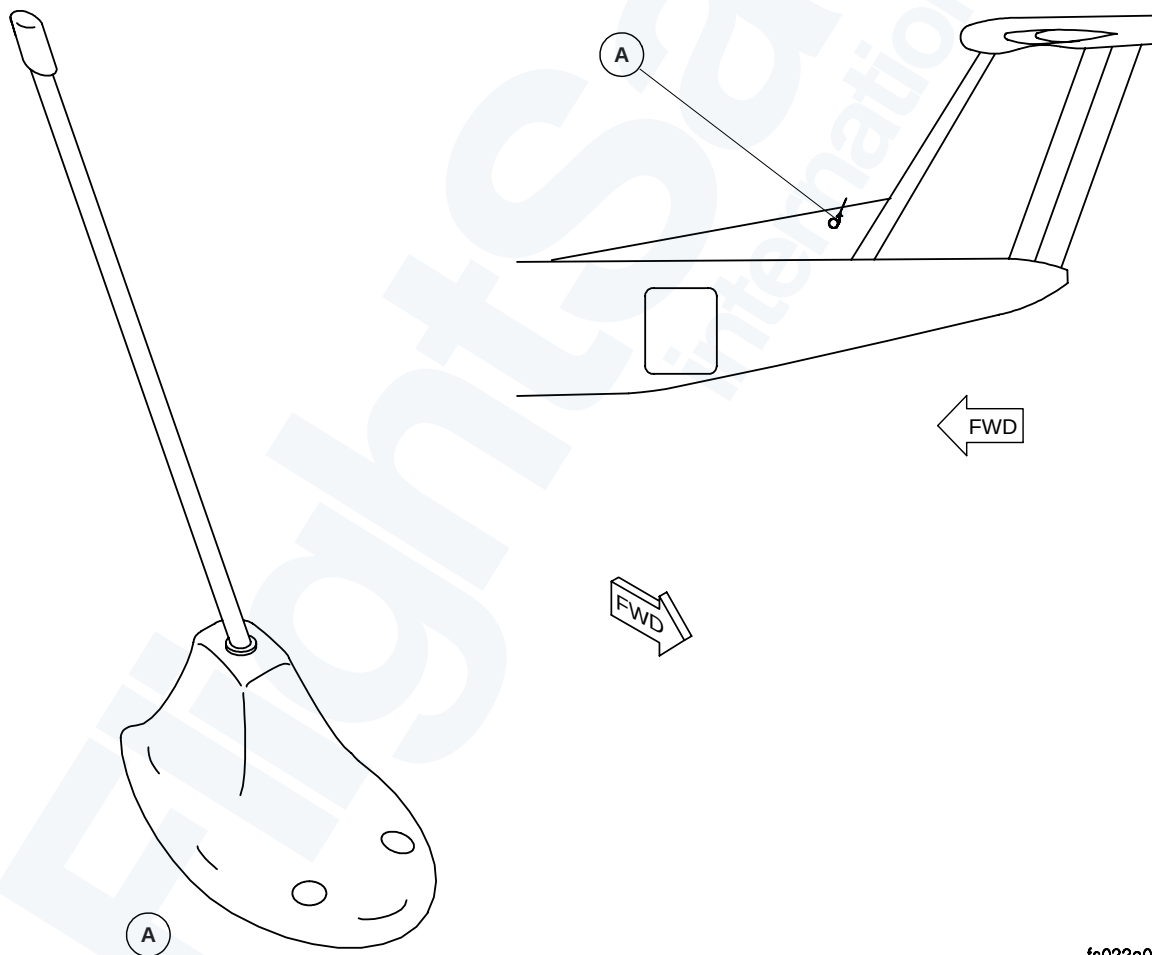
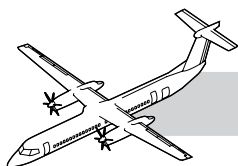


Figure 25-68. ELT Antenna Location

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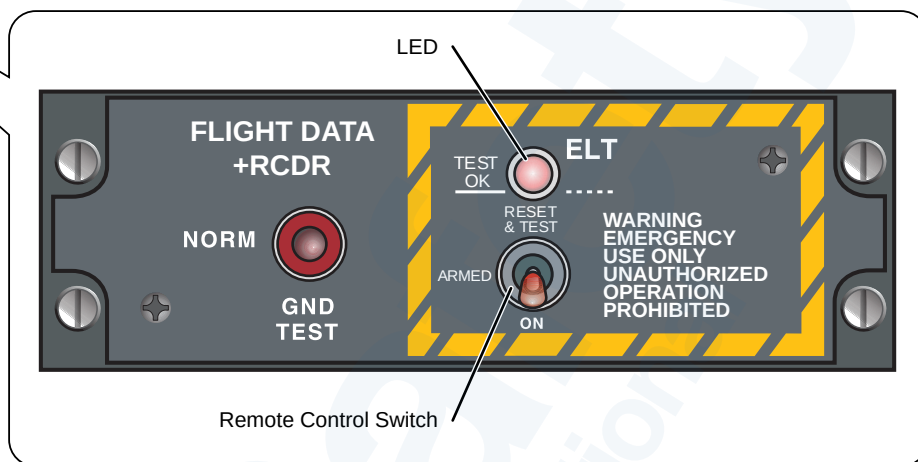
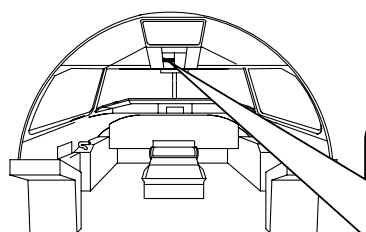
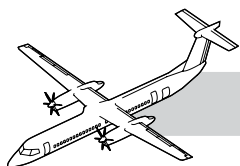


Figure 25-69. ELT Remote Switch

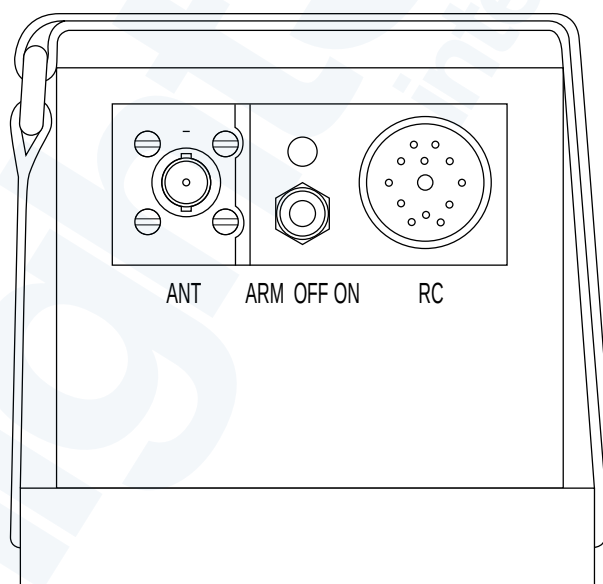
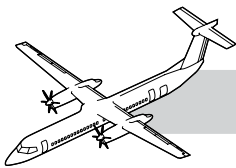


Figure 25-70. Front Panel



CONTROLS AND INDICATIONS

ELT REMOTE SWITCH

Refer to Figure 25-69. ELT Remote Switch.

The remote switch is used to select these ELT operating modes:

- Remote AUTO
- Remote ON
- Remote RESET.

Remote AUTO Mode: The ELT operates in the automatic mode when the master and the remote switches are selected to the AUTO position. The ELT will transmit when the inertia switch above the red ELT monitor light located in the remote switch will come on.

Remote ON Mode: The ELT can be manually activated in the event of an emergency by selecting the remote switch to the ON position. The ON selection overrides the automatic inertia switch. The monitor light located within the remote switch illuminates.

The RESET/Test switch is spring loaded to return the switch to the AUTO position. For test purposes, the ELT may be momentarily activated by placing the remote switch to the ON position. Returning the remote switch to the AUTO position terminates ELT operation. The ELT monitor light goes out.

Remote RESET Mode: The remote RESET mode lets a reset of an inadvertent ELT activation. Momentarily placing the remote switch in the RESET position and then to the AUTO position re-arms the ELT and the ELT monitor light goes out.

The ELT may be tested also by placing the remote switch to the RESET/Test position. In this position the operator can retrieve information about the status of the ELT.

Refer to Figure 25-70. Front Panel.

The following controls are found on the ELT transmitter front panel (from left to right):

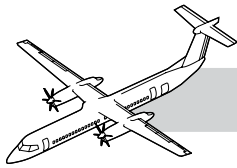
- BNC connector for the antenna
- 3 position switch ARM/OFF/ON
- Red light (LED)
- DIN 12 connector for Remote Control Panel, GPS interface and programming equipment connection.

The red light gives an indication on the working mode of the beacon after the test.

In operating mode, periodic flashes occur during 121.5/243 transmission and one long flash during every 406 burst (1 every 50 seconds).

A buzzer gives aural indication on the working mode of the beacon:

- Continuous beep during test
- 2 beeps per second during 121.5/243 transmission, silence during 406 transmission.



TROUBLESHOOTING

CAUTION

DO THE ELT TEST DURING THE GIVEN TIME THAT IS APPROVED BY THE NATIONAL REGULATORY AUTHORITY. IF YOU DO NOT DO THIS, YOU CAN CAUSE A SEARCH AND RESCUE OPERATION WHICH IS NOT NECESSARY.

Operational Check of the 2-Frequency Emergency Locator Transmitter (ELT) with Inertia

Switch (G-Switch)

- The operational check of the two frequency emergency locator transmitter (ELT) will test the operation of the G-switch and the transmission of the ELT Signal.

Operational Check of the 3-Frequency Emergency Locator Transmitter (ELT) with Inertia

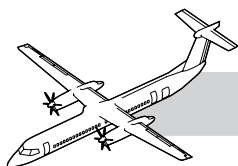
The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Switch (G-Switch)

- The operational check of the 3-frequency (KANNAD 406MHz) emergency locator transmitter (ELT) will test the operation of the G-switch and the transmission of the ELT Signal. There is also an internal check that can be done that will give an indication of the internal function.
- On the overhead console, set and hold the remote control switch to the ON position. You will hear an audible buzzer from the ELT, you will hear a

warbling tone on the loudspeaker or headset, and then the light emitting diode (LED) will flash.

- To test the ELT with the COSPAS/SARSAT receiver, set up the COSPAS/SARSAT receiver (hand-held tester). On the overhead console, set and hold the remote control switch to the RESET & TEST position.
 - Make sure that you hear an audible buzzer from the ELT during the test.
 - Make sure that after approximately three seconds, the (LED) shows the test results that follow:
 - One long flash shows that the test is satisfactory.
 - A sequence of short flashes as follows shows that the test is unsatisfactory:
 - 3+1= Low battery voltage
 - 3+2= Low RF power
 - 3+3= Faulty VCO lock
 - 3+4= No programmed aircraft identification number
 - At the same time, read the data on the hand-held tester and check for 406 MHz Pass information.



PIN PROGRAMMING OPTION

The KANNAD 406 AF/AF-H offers pin-programming capabilities to facilitate maintenance operations especially in case of removals and/or replacement.

A special DIN 12 connector with a Serial Memory Module (called “Programming Dongle”) is connected to the ELT when installed on board. This Programming Dongle contains the identification information of the aircraft and remains on board the aircraft. When an unprogrammed ELT is installed and connected to this Programming Dongle and the “ELT” is switched to “ARM”, it automatically updates its own memory with the identification data contained in the Programming Dongle memory.

When the ELT is removed from the aircraft, it keeps its identification data.

Refer to Figure 25-71. Maintenance Dongle.

For maintenance purposes, it is possible to “reset” the identification information of the ELT by connecting a “Maintenance Dongle” to the ELT. Any accidental transmission with this “maintenance dongle” will not involve SAR operation as the identification code transmitted is recognized by COSPAS-SARSAT as “not on board”.

If the pin programming option is selected by the operator, the following equipment is required :

- One “Programming Dongle” on each aircraft,
- One “Maintenance Dongle” on each spare.



Figure 25-71. Maintenance Dongle

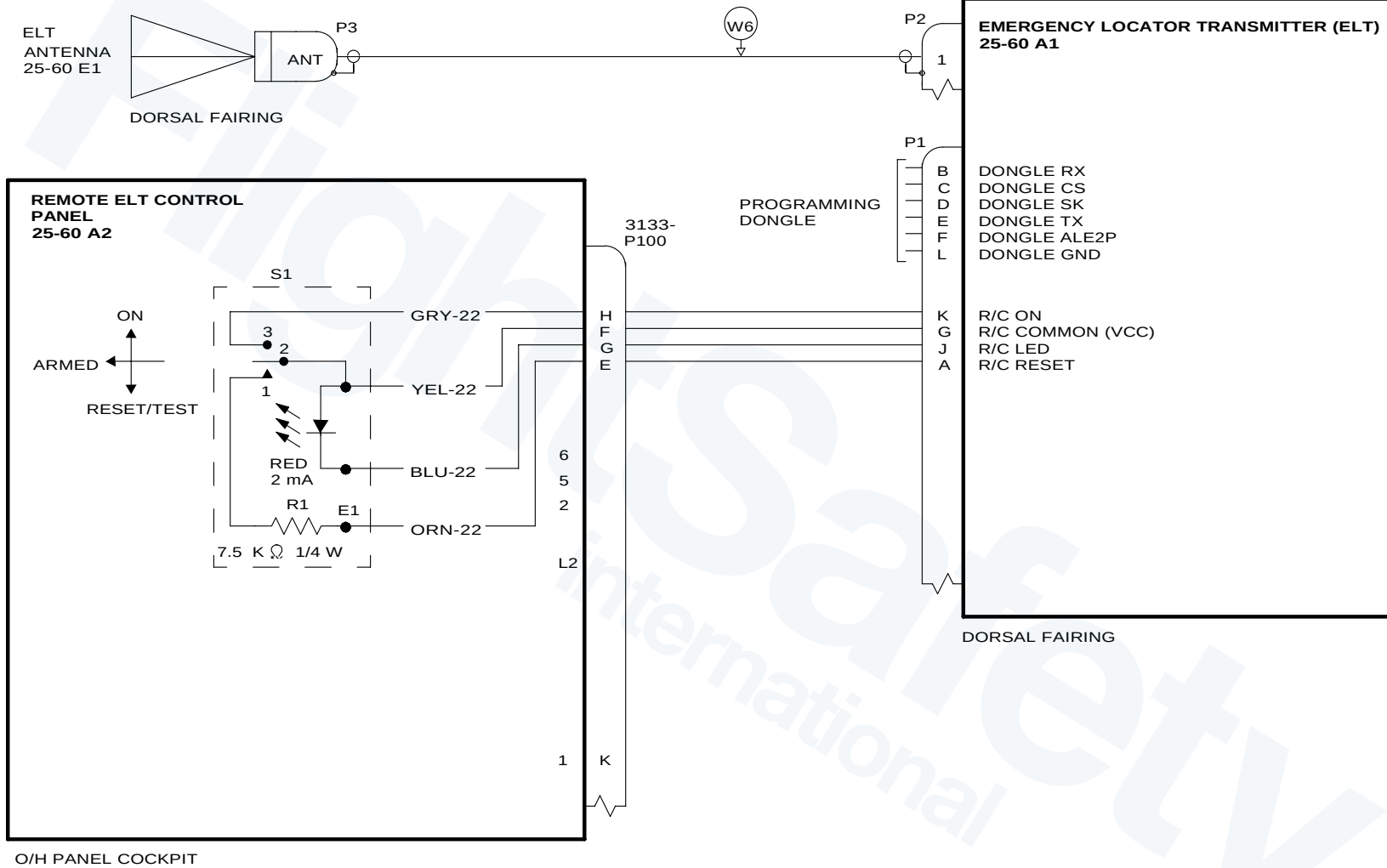
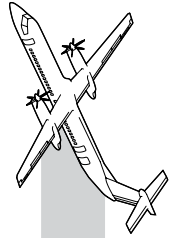
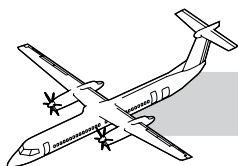


Figure 25-72. Emergency Locator Transmitter (3-Frequency)





OPERATION

Refer to Figure 25-72. Emergency Locator Transmitter (3-Frequency).

1. ELT Modes of Operation

- A. The ELT has two modes of operation:
- Inactive mode - ELT armed
 - Active mode - ELT transmitting the emergency signals.

2. Automatic (normal) Operation

- A. The ELT remote switch on the ELT control panel in the flight compartment is normally set to the ARM position. In this position the ELT is armed and ready for the active mode.
- B. The ELT automatically goes into the active mode when the inertia G switch in the transmitter unit is closed due to an impact.

NOTE

When the ELT transmits the emergency signals, the red LED comes on and flashes continuously, and the buzzer gives out a distinct audible sound.

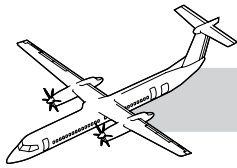
- C. The ELT transmits the signals on the emergency frequencies of 121.5 MHz, 243.0 MHz and 406 MHz to help the search and rescue personnel to locate the aircraft.
- D. The third signal (406 MHz) is transmitted every 50 seconds to a satellite. This signal is encoded with a digital message.
- E. The ELT stops transmission of the emergency signals when it is reset.
- F. After the ELT is reset with the remote switch, the red LED stops flashing, stays illuminated for approximately one second and then goes out.
- G. When the ELT system senses a problem, the LED flashes a coded signal after the one second pulse (refer to the Operational Test of the ELT System).

3. Manual Operation

- A. Manual operation of the ELT system is usually done for the test purposes or to activate the ELT in an emergency.
- B. The ELT can be manually operated with the remote switch in the flight compartment or with the transmitter unit ON/OFF switch in the tail section.
- C. The ELT operates when the remote switch in the flight compartment is put to the ON position.
- D. After 50 seconds of normal ELT operation, the third signal (406 MHz) is transmitted to the satellite, which is considered to be a valid distress signal.
- E. To stop the ELT from transmission of the emergency signals, the ELT must be reset. Refer to Reset the ELT.
- F. The standard emergency frequency of 121.5 MHz and 243.0 MHz can be heard on the VHF and UHF communication systems that are in range and tuned to the emergency frequencies. The third signal (406 MHz) is encoded with a digital message and is sent to a satellite.

4. Reset the ELT

- A. For manual reset, switch to position off on the ELT
- B. To reset using the Remote Control Panel, the switch has to be in position "ARM" on the ELT than press the "RESET & TEST" on the remote control.



25-00-00 SPECIAL TOOL & TEST EQUIPMENT

- HOBART 75 KVA AC Ground Power Cart (or equivalent)
- GSB2000001 Borescope - 115 Volt, 60 Hz
- GSB2000015 Borescope - 220 Volt, 50 Hz
- SD20113 or equivalent Latch Tool
- P.A.-1 Polypropylene Sweep Tool
- T7134010 Perforating Tool
- GSB2525002 Wheel
- GSB2525001 Tool - PSU Mounting Hook
- GSB2400007 Kit - Connect Pins and Sockets